

Universidad Complutense de Madrid

FACULTAD DE CIENCIAS FÍSICAS
DEPARTAMENTO DE FÍSICA ATÓMICA, MOLECULAR Y NUCLEAR



Time Asymmetry in UHE Cosmic Ray Showers. Mass Composition Studies in the Pierre Auger Observatory

Memoria para optar al grado de Doctor en Física
por la Universidad Complutense de Madrid
presentada por
Diego García Pinto

Bajo la dirección del profesor
Fernando Arqueros Martínez

Madrid, Marzo 2009

A mi padre.

Estoy seguro de que ahora conoce todas las respuestas.

Y por su puesto a mi madre.

Contents

List of Figures	vii
1 Introduction	1
2 Ultra High Energy Cosmic Rays	5
2.1 Introduction	5
2.2 History	6
2.3 Energy spectrum	7
2.4 Origin and Acceleration mechanisms	10
2.5 Propagation and source identification of UHECRs	12
2.6 Phenomenology of Extensive Air Showers	16
2.6.1 Hadronic processes	19
2.6.2 Electromagnetic component	21
2.6.3 Muon component	22
2.6.4 Heitler model	23
2.7 Composition of cosmic rays	27
2.7.1 The shower maximum depth X_{max}	27
2.7.2 Fluctuations in X_{max}	28
2.7.3 Muon content	29
2.7.4 Rise time $t_{1/2}$	31
2.7.5 Azimuthal asymmetry in $t_{1/2}$	34
2.7.6 Lateral Distribution Function	34
2.7.7 Relative number of muons and electrons	35



CONTENTS

2.7.8	Radius of curvature R_c	36
3	The Pierre Auger Observatory	37
3.1	Introduction	37
3.2	Motivation	38
3.3	Fluorescence Detector	41
3.3.1	Data acquisition and trigger conditions	42
3.3.2	Detector Calibration and Atmospheric Monitoring	43
3.3.3	Event Reconstruction using the FD	44
3.4	Surface Detector	48
3.4.1	Calibration of the Surface Detector	50
3.4.2	SD Trigger System	52
3.4.3	Event Reconstruction using the SD	57
3.5	Offline Software Framework	63
3.6	Asymmetry in Extensive Air Showers	64
3.6.1	Sources of azimuthal asymmetry	67
3.6.2	Effects of asymmetry in shower reconstruction parameters	79
4	Mass dependence of the time Asymmetry	83
4.1	Introduction	83
4.2	Asymmetry in the time structure as an indicator of shower evolution	87
4.2.1	Slant depth dependence of the asymmetry in the time structure	88
4.3	The risetime	90
4.4	Monte Carlo simulated data	91
4.5	Asymmetry parameters for mass analysis	95
4.5.1	The longitudinal asymmetry development	96
4.5.2	Energy dependency and hadronic models	98
4.5.3	Relationship of the asymmetry parameters with X_{max}	102
4.6	Cut optimization from sensitivity and efficiency	102
4.6.1	Sensitivity	102
4.7	Extracting the mean primary mass	105

5	Determination of Composition	111
5.1	Introduction	111
5.2	Asymmetry analysis on data	112
5.2.1	Event Selection	112
5.2.2	Asymmetry longitudinal development and asymmetry parameters	113
5.3	Comparison of Data and MC	115
5.3.1	Muon Test	119
5.3.2	Effect of the measurement uncertainties in $t_{1/2}$	122
5.4	Inferring X_{max}	125
5.4.1	Data samples and event selection	125
5.4.2	Relationship between X_{max} and the asymmetry parameters . . .	128
5.4.3	The elongation rate method using the XAsymMax parameter . .	134
5.5	Mass composition	135
5.5.1	Uncertainties in the determination of the mass composition . . .	138
6	Conclusions	143
	References	147

List of Figures

2.1	Energy spectrum of cosmic rays.	8
2.2	Energy spectrum of cosmic rays measured by different experiments. In this figure the flux has been multiplied by $E^{2.5}$ to evidence the spectral features. The achievable energy by some terrestrial accelerators are also shown for comparison. Figure taken from [16].	9
2.3	Hillas plot showing the size and magnetic field strengths of possible astrophysical candidates to accelerate UHECRs. Objects below the corresponding lines are not able to accelerate protons or iron nuclei to an energy of 10^{20} eV. Figure taken from [25].	11
2.4	Left: upper limits at 95% CL on the fraction of photons for different energies obtained from Auger data (black arrows) along with those obtained from AGASA (A, A2), Haverah Park (HP), Yakutsk (Y), AGASA-Yakutsk (AY) and Auger hybrid data (FD). Predictions of the photon fraction from different top-down models (SHDM, TD and Z-burst) are also shown as well as predictions for GZK photons. Right: Limits at 90% CL for a diffuse flux of μ_ν from Auger data. Limits from other experiments are also shown.	13
2.5	Energy of a proton as a function of the propagation distance through the CMBR for three different initial energies. Figure taken from [41].	14
2.6	Energy loss length for the propagation of extragalactic cosmic rays as a function of nucleon energy. The hadron production corresponds to the GZK effect. Figure taken from [16].	15



LIST OF FIGURES

2.7	Energy spectrum obtained from data taken at the Pierre Auger Observatory. The first part of the spectrum (up to 40 EeV) is fitted to a power-law function which is extrapolated to the highest energies to evidence the flux suppression in this region.	16
2.8	Projected view of 20 trajectories for protons of several energies emanating from a point source. The magnetic field is described by cells of size 1 Mpc with mean value 1 nG oriented randomly in each cell. Figure taken from [46].	17
2.9	Projection of the celestial sphere in galactic coordinates with circles of 3.2° centred at the arrival directions of the 27 highest-energy events detected by the Pierre Auger Observatory. The positions of AGNs with redshift $z \leq 0.017$ are indicated by red asterisks. The solid line represents the border of the field of view (for zenith angles smaller than 60°) for the southern site of the Observatory. The dashed line is the super-galactic plane. Darker blue color indicates larger relative exposure.	18
2.10	Schematic diagram of the principal EAS processes.	19
2.11	Schematic view of an electromagnetic shower (a) and hadronic shower (b). Figure taken from [69].	24
2.12	Variation of X_{max} with energy measured by different experiments as well as predictions from different models. Figure taken from [72].	29
2.13	Variation of $\langle X_{max} \rangle$ as a function of the energy measured with the Pierre Auger Observatory. Data are compared to predictions from different hadronic interaction models.	30
2.14	Fluctuations of X_{max} as a function of the energy. Data are compared with the predictions the QGSJETII model. Above energies of 10^{18} eV a trend to increasing masses is observed. Figure taken from [77].	31
2.15	Distribution of jumps for events at a given energy, zenith angle and distance to the core range. Lines corresponds to the different fitted functions of muon and electromagnetic signal. Figure taken from [80].	32
2.16	Risetime at 1000 m against energy for Monte Carlo predictions (lines) and data (black points) at a fixed zenith angle (38°). Blue = proton, red = iron, solid = QGSJET01, dashed = Sibyll2.1. A dashed black line is the result of a line fitted to the data. Figure taken from [84].	33

LIST OF FIGURES

2.17	Left: Average value of $\langle \Delta_i \rangle$ derived from a single event. The blue line represents the benchmark function and the points the individual measurements of $t_{1/2}$. Right: The average value of Δ_i as a function of X_{max} for the selected hybrid events. A correlation is seen. Figures taken from [86].	34
3.1	The cosmic ray spectrum measured by HiReS and AGASA experiments. The straight lines segments are the results of a fit to a triple power law. Figure taken from [105]. .	40
3.2	Schematic view of a FD telescope. From left to right: shutters, aperture system (including filter and correction ring), PMT camera and mirror.	42
3.3	Scheme of the geometry of a cosmic ray induced shower. The SDP and R_p , χ_0 and T_0 parameters are shown as well as pixel information $(\bar{n}_i, t_i, \chi_i)$. Figure taken from [115].	45
3.4	Example of a typical event reconstructed using only the FD information (red line) and using also timing information provided by the SD (blue line). Fitted parameters are shown to emphasize the improvement of the hybrid technique. Figure taken from [115].	46
3.5	Typical longitudinal profile of an event with an energy around 25 EeV observed using the Auger fluorescence telescopes. The red solid line is a fit to a Gaisser-Hillas function.	47
3.6	Schematics of the surface array. Surface detectors are represented as purple points. The light blue zone marks the status at 13 th June of 2008. A few holes remain. . . .	48
3.7	Picture of a Surface Detector unit.	49
3.8	Charge and pulse height histograms from an SD station with the signal from all 3 PMTs summed. The dashed histogram is produced by an external muon telescope providing the trigger to select only vertical and central muons. The first peak in the black histogram is caused by low energy particles. The second peak is due to vertical through-going atmospheric muons. Figure taken from [130].	52
3.9	Traces of the 3 PMT signals of a tank. a) Satisfying the T1 threshold condition. b) Satisfying the T2 ToT condition. c) Satisfying the T2 threshold condition.	54
3.10	The hexagon geometry around the central station used for T3 trigger decisions. Figure taken from [134].	55
3.11	The two possible 3ToTC1 configurations. All symmetry transformations of the triangular grid are also valid. Figure taken from [135].	56
3.12	The three (minimal) 4C1 configurations. All symmetry transformations of the triangular grid are also valid. Figure taken from [135].	56



LIST OF FIGURES

3.13	Shower geometry including parameters used in the SD reconstruction.	58
3.14	Station signals as a function of distance (dots) for a simulated event and the reconstructed LDFs using different values of β (dashed lines). Figure taken from [139]. . .	60
3.15	Calibration curve showing the correlation between $\log E_{FD}$ and $\log S_{38^\circ}$. Figure taken from [45].	62
3.16	Schematic view of the shower geometry. The incoming direction of the primary particle defines two regions, “early” ($-\frac{\pi}{2} < \zeta < \frac{\pi}{2}$) and “late” region ($\frac{\pi}{2} < \zeta < -\frac{\pi}{2}$).	65
3.17	Event 14899272 recorded by the Haverah Park array with reconstructed energy $E \sim 3 \times 10^{18}$ eV and $\theta = 43^\circ$. Left: Data in the shower plane and axis position (cross). Right: Fit result to the equation 3.13.	66
3.18	Schematic view of a shower with zenith angle θ . An early tank (1) will see a greater difference in paths lengths (P_1P_2) than a late one (2), (P_3P_4).	67
3.19	Particles going away from the shower axis with symmetric angles β . When projecting the ground densities on the shower plane, a purely geometric asymmetry appears. Figure taken from [152].	69
3.20	Comparison of the ground asymmetry (electromagnetic energy density as a function of ζ angle) observed in simulated inclined shower (solid lines) to the prediction of a vertical shower at the same depth, after projection (dashed line). Figure taken from [152].	71
3.21	The geometrical forward-backward asymmetry for photons and electrons.	72
3.22	The azimuthal dependence of the integrated signal at $\theta = 25, 36, 53, 66$ (from top to bottom) at different distances from the shower axis, for normal-shaped tanks (flat-shaped) (open squares) and high-shaped tanks (filled triangles). The average signal is normalized to the same value. Figure taken from [153].	73
3.23	Difference in the path Δt traveled by the particles at ground, for two points at the same distance R from the shower axis measured at the shower plane. Their corresponding distances in the detector plane are r_1 and r_2	75
3.24	Maximum difference of path travelled by particles Δt , as a function of the distance to the shower core in the shower plane, for three different zenith angles.	76
3.25	Intersection of a cone with the ground plane.	77
3.26	The azimuthal dependence of the integrated signal, at different distances from shower axis, for normal-shaped tanks and high-shaped tanks.	78

LIST OF FIGURES

3.27	Scheme of a shower with zenith angle θ . The true core position is located in A. The signal in the early tank is greater than that in the late one due to the attenuation of shower particles. The circular assumption shifts the core to position B.	79
3.28	Difference of estimated core location from SD and Hybrid analysis on the shower plane. The origin of the scatter plot is the estimated core location from Hybrid analysis and black dots indicate that from SD. Red circle in the scatter plot shows a 1σ of 2-dimensional-Gaussian fit. Effect of the asymmetry can be seen in bottom panel. Figure taken from [159].	80
3.29	Asymmetry in the LDF as a function of the core distance, for different primaries using QGSJET01 (top) and SIBYLL2.1 (bottom) for hadronic interactions. The curves are the results of the fit with $\tan^{-1}$ function. Figure taken from [161].	81
4.1	Integrated detector signals in the shower frame of event #184599. The size of the circles are proportional to $\log(S/[VEM])$ and the numbers are given as inset accordingly. Green indicates early arrival times while red corresponds to late arriving particles in the detectors. The open cross marks the position of the shower core.	84
4.2	Calibrated FADC traces of the SD stations of the event #184599. The colour code represents the different PMTs: PMT1 = red, PMT2 = green, PMT3 = black. Stations 31, 33, 34, 49 and 52 corresponds to early stations. Stations 43, 44, 45, 48, 53 and 53 to late stations.	85
4.3	Risetime as a function the azimuth angle ζ for two $\sec\theta$ bins from events recorded with the EA of the Pierre Auger Observatory.	86
4.4	The time structure measured in the surface detector is related with X_{max} . The smaller is the X_{max} value the smaller is the difference in path lengths for particles arriving at the detector, case b). Muons travel in straight line while EM particles are subjected to multiple scattering and therefore the muon contribution dominates the first portion of the signal.	87
4.5	Schematic view of shower development when arriving at three different zenith angles	89
4.6	Footprint of a 10^{19} eV proton shower with $\theta = 45^\circ$. The triggered tanks are represented with circles with area proportional to the signal intensity.	94
4.7	Simulated signal produced by the muon and electromagnetic components in an early station (right) and a late one (left).	95



LIST OF FIGURES

4.8	Risetime vs. ζ for primary energy 10 EeV and different angles ($32^\circ, 45^\circ, 53^\circ$ and 60° , from top to bottom) for protons (left) and iron nuclei (right).	97
4.9	Asymmetry longitudinal development for 10^{19} eV protons. The three asymmetry parameters describing the longitudinal development are shown.	98
4.10	Asymmetry longitudinal development at primary energies 3.16, 10, 17.8, 31.6, 56.3 and 100 EeV (from top to bottom and from left to right). Monte Carlo protons (red dashed line) and iron nuclei (blue solid line) are shown.	99
4.11	XAsymMax vs. $\log(E/\text{eV})$ for both hadronic models; SIBYLL and QGSJETII. The points are the result of the MC analysis for both proton and iron. Lines represent the fit to a linear function.	100
4.12	AsymHeight vs. $\log(E/\text{eV})$ for both hadronic models; SIBYLL and QGSJETII. The points are the result of the MC analysis for both proton and iron. Lines represent the fit to a quadratic function.	100
4.13	XAsymWidth vs. $\log(E/\text{eV})$ for both hadronic models; SIBYLL and QGSJETII. The points are the result of the MC analysis for both proton and iron.	101
4.14	Correlation between $\langle X_{max} \rangle$ and the three parameters of the asymmetry longitudinal development: XAsymMax (upper plot), AsymHeight (middle plot) and XAsymWidth (lower plot) for two different primaries, iron (squares) and proton (solid dots), and for two hadronic models, QGSJETII-03 (green) and SIBYLL 2.1 (pink).	103
4.15	Sensitivity parameter (equation 4.10) for different sets of cuts for 10^{19} eV showers.	104
4.16	Efficiency (equation 4.11) versus zenith angle for different cuts in signal and core distance for 10^{19} eV primaries for both primaries proton (red) and iron (blue). Cuts in signal intensity are $S > 15$ VEM (top) and $S > 10$ VEM (bottom). The core distance intervals correspond to 500-1500 m, 500-1800 m and 500-2000 m (from left to right).	106
4.17	Efficiency (equation 4.11) for different cuts in signal and core distance for 10^{19} eV primaries for protons (left) and iron (right).	107
4.18	Asymmetry development for the different samples with mixed composition, from pure proton (dashed blue) to pure iron (continuous red) in steps of 10% for primary energy 10^{19} eV.	108
4.19	Position of the maximum, width and height of the longitudinal asymmetry development as a function of the mixture fraction X_{Fe} , for a primary energy of 10^{19} eV. $X_{Fe} = 1$ corresponds to 100 % iron and $X_{Fe} = 0$ to 100 % proton.	109

LIST OF FIGURES

4.20	Consistency check of iron fraction with three different samples: 100% iron(upper curve), 50% iron and 50% proton(middle curve), 100% proton(lower curve).	110
5.1	Detection efficiency of protons showers as a function of the energy for the Pierre Auger Observatory surface array using the T5 trigger condition. The effective detection area becomes saturated at 3.16 EeV ($\log(E/\text{eV}) = 18.5$).	113
5.2	$\langle t_{1/2}/r \rangle$ vs ζ for events with reconstructed $\log(E/\text{eV})$ in the bin 18.4 - 18.6 and reconstructed θ in four different bins, $30^\circ - 33^\circ$, $43^\circ - 46^\circ$, $49^\circ - 52^\circ$ and $59^\circ - 62^\circ$, from top to bottom and from left to right.	114
5.3	$\langle t_{1/2}/r \rangle$ vs ζ for events with reconstructed $\log(E/\text{eV})$ in the bin 18.8 - 19.0 and reconstructed θ in four different bins, $30^\circ - 33^\circ$, $43^\circ - 46^\circ$, $49^\circ - 52^\circ$ and $59^\circ - 62^\circ$, from top to bottom and from left to right.	115
5.4	Asymmetry longitudinal development of real data for six $\log(E/\text{eV})$ bins in the range 18.0 - 18.6.	116
5.5	Asymmetry longitudinal development of real data for five $\log(E/\text{eV})$ bins in the range > 18.6	117
5.6	XAsymMax vs $\log(E/\text{eV})$ for real data.	118
5.7	AsymHeight vs $\log(E/\text{eV})$ for real data.	119
5.8	XAsymWidth vs $\log(E/\text{eV})$ for real data.	120
5.9	XAsymMax vs $\log(E/\text{eV})$ for data (points) and Monte Carlo. Monte Carlo predictions are represented by their fitted functions. At $\log(E/\text{eV}) < 18.5$ (dashed vertical line) SD trigger efficiency depends on primary and therefore data can not provide reliable information on mass composition.	121
5.10	AsymHeight vs $\log(E/\text{eV})$ for data (points) and Monte Carlo. Monte Carlo predictions are represented by their fitted functions.	122
5.11	XAsymWidth vs $\log(E/\text{eV})$ for data (black points) and Monte Carlo.	123
5.12	XAsymMax versus $\log(E/\text{eV})$ for MC simulations using the QGSJETII interaction model for both standard and that with enhanced muon content at ground.	124
5.13	AsymHeight versus $\log(E/\text{eV})$ for MC simulations using the QGSJETII interaction model for both standard and that with enhanced muon content at ground.	125
5.14	XAsymWidth versus $\log(E/\text{eV})$ for MC simulations using the QGSJETII interaction model for both standard and that with enhanced muon content at ground.	126



LIST OF FIGURES

5.15	Comparison of XAsymMax between data and QGSJETII simulation with enhanced muon content at ground.	127
5.16	Comparison of AsymHeight between data and QGSJETII simulation with enhanced muon content at ground.	127
5.17	Comparison of XAsymWidth between data and QGSJETII simulation with enhanced muon content at ground.	128
5.18	Example of $\sigma_{t_{1/2}}$ (black dots) and $\sigma_{t_{1/2}}/r$ (red dots) as a function of the core distance.	129
5.19	Comparison of the XAsymMax parameter versus energy obtained weighting risetimes with the measurement uncertainties (dots) and without weights, i.e. arithmetic mean values (triangles), obtained for a reduced data sample.	130
5.20	Correlation with $\langle X_{max} \rangle$ of the three parameters which describe the asymmetry in the shower development: XAsymMax (upper plot), AsymHeight (middle plot) and XAsymWidth (lower plot).	131
5.21	Correlation of XAsymMax with $\langle X_{max} \rangle$ for the different hadronic interaction models and different primaries along with data for comparison (upper plots). In the lower plot the calibration curves for data and two different hadronic interaction models are shown. Values of the parameters obtained from the linear fit to data are also presented.	133
5.22	XAsymMax versus X_{max} (calibration curve) obtained for real data (solid line). Deviations of the calibration curve (dashed lines) obtained from the propagation of uncertainties are shown for 1σ region (cyan) and 2σ region (blue).	134
5.23	Correlation of AsymHeight with $\langle X_{max} \rangle$ for different hadronic interaction models. Linear fits to data are also shown.	135
5.24	$\langle X_{max} \rangle$ vs $\log E$ as inferred from the XAsymMax parameter measured with the SD. Results for $\log E$ lower than 18.4 should be taken with caution since the surface detector is not fully efficient.	136
5.25	$\langle X_{max} \rangle$ vs $\log E$ inferred from the XAsymMax parameter measured with the SD. Predictions for a pure iron or proton composition according to several models as well as results from direct measurement of X_{max} using the FD are also shown for comparison.	136
5.26	XAsymMax parameter vs $\log E$ (left panel) and $\langle X_{max} \rangle$ vs $\log E$ (right panel) inferred from XAsymMax using the presented calibration curve for both data (black symbols) and MC predictions (red lines: proton, blue lines: iron) from QGSJETII-03 (solid lines) and Sibyll 2.1 (dashed lines).	137

LIST OF FIGURES

- 5.27 Mass composotion X_{Fe} obtained by minimizing equation (4.12) using only the term
corresponding of XAsymMax. Predictions from QGSJETII model are used. 138
- 5.28 Mass composotion X_{Fe} obtained by minimizing equation (4.12) using only the term
corresponding of XAsymMax. Predictions from SIBYLL 2.1 model are used. 139
- 5.29 Effect of the assumed primary energy in MC on mass composition. Ordinate represents
the difference in X_{Fe} when the MC input energy, instead of the reconstructed one, is
used. The continuous line represents the average deviation in the full energy range. . 140
- 5.30 Effect of the cuts on mass composition. Ordinate represents the difference in X_{Fe}
when the cut in core distance is restricted to the 600 - 1800 m interval (left) for both
MC and real data samples. Effect for an intensity signal cut of 15 VEM (right). The
continuous lines represents the average deviation in the full energy range. 141

Introduction

Since the discovery of cosmic rays at the beginning of the twentieth century, almost one hundred years ago, some of the most important properties of these energetic particles such as the origin, production mechanisms or mass composition, remain still being a mystery. Specially unsolved are the properties of Ultra High Energy Cosmic Rays (UHECRs), defined as those with an energy above 10^{18} eV, which exceed in several orders of magnitude the maximum energy attainable in the most recent man-made accelerators such as the Large Hadron Collider.

The flux of cosmic rays is strongly dependent on primary energy decreasing up to 1 particle per km^2 and per century for energies around 10^{20} eV. The observation of UHECRs with such low flux is possible through the detection of the so-called Extensive Air Showers (EAS). The interaction of an extremely energetic cosmic ray with an atmospheric nucleus induces the development of a cascade of secondary particles which can be observed from the ground with an appropriate instrument. Because of the extremely low flux, large instrumented surfaces are necessary to study the most energetic cosmic rays. The Pierre Auger Observatory is nowadays the largest detector constructed to study UHECRs.

The Pierre Auger Observatory was conceived as a hybrid detector designed to study with high significance the energy spectrum, arrival direction distribution and mass composition of UHECRs. The hybrid concept, in which the Pierre Auger project is pioneer, implies a combination of the two most successful techniques previously used in the study of high energy cosmic rays. A Surface Detector composed of 1600 Cherenkov stations



1. INTRODUCTION

samples the secondary particles at ground level while a Fluorescence Detector made up of 24 telescopes registers the longitudinal development of the shower by collecting the faint fluorescence light emitted by atmospheric nitrogen molecules previously excited by cascade particles crossing the atmosphere.

The flux suppression above around 10^{19} eV postulated by Greisen, Zatsepin and Kuz'min due to the interaction of UHECRs with the Cosmic Microwave Background Radiation has been recently observed with high significance by the Pierre Auger Collaboration. In addition, a relevant breakthrough reported by the Collaboration has been the observation of correlations between the position of nearby Active Galactic Nuclei and the arrival direction of the highest energy cosmic rays.

Determining the mass composition of UHECRs is crucial for the interpretation of these results, i.e. the energy spectrum and the distribution of arrival directions, as well as for an appropriate understanding of the acceleration mechanisms and the possible sources. In this work a method to extract the mass composition of UHECRs is presented. The technique is based on the observed asymmetry in the time structure of the water Cherenkov signals recorded by the SD of the Pierre Auger Observatory. The amplitude of this time asymmetry varies with the incoming zenith angle θ of the primary cosmic ray in a way which is reminiscent of the longitudinal development of the shower. As will be shown the $\sec\theta$ value for which the asymmetry longitudinal development reaches a maximum is a very useful mass-sensitive parameter well correlated with the shower maximum depth.

This thesis is organized as follows. In chapter 2 a brief review of the current knowledge of UHECRs is presented including a description of the phenomenology of EAS and a summary of the most used techniques developed for primary composition studies. Chapter 3 is focused on the description of the Pierre Auger Observatory. The main properties of both fluorescence and surface detectors are described emphasizing the great advantages of a hybrid design. The tools developed to reconstruct the registered events in order to obtain relevant parameters of the primary are also presented in this chapter. A special section is dedicated to summarize the main features of the asymmetry observed in the total signal measured with the Cherenkov tanks. The novel method develop to study the primary composition of cosmic rays based on the observed azimuthal time asymmetries is explained in detail in chapter 4. The results of the application of this technique to real data of the SD of the Pierre Auger Observatory and

preliminary results on mass composition are presented in [5](#). In chapter [6](#) the most relevant results reported in this work are summarized.

Ultra High Energy Cosmic Rays

2.1 Introduction

Energetic particles which are continuously striking the Earth's atmosphere are called cosmic rays. The incoming rate of these particles is strongly dependent on energy decreasing from several particles per m^2 and per second ($E > 10^{11}$ eV) to 1 particle per km^2 and per century ($E > 10^{20}$ eV). Since the flux of these UHECRs¹ is extremely low the knowledge about the nature, acceleration mechanisms and propagation of these particles, which exceeds in several orders of magnitude the energies achievable in man made accelerators, is nowadays rather limited. Properties of UHECRs such as energy spectrum, arrival direction distribution or mass composition are studied through the large showers of particles they generate in the atmosphere.

In this chapter, an overview of the field of cosmic ray physics with special attention to UHECRs is presented. In the first section, some of the most important landmarks in the field are mentioned. Several important items such as the energy spectrum (section 2.3), origin and acceleration mechanisms (section 2.4) and propagation (section 2.5) are covered. Also the phenomenology of Extensive Air Showers (EAS) is described in section 2.6. Finally a review of the analysis techniques and current experimental data on the composition of UHECRs is presented in section 2.7.

¹Cosmic rays with energies above 10^{18} eV are known as Ultra High Energy Cosmic Rays (UHECRs).



2.2 History

Cosmic rays were discovered by Victor Hess in 1912 [1] through a series of balloon flights when he realised that the ionization this radiation produced on an electroscope increased with altitude, indicating an extraterrestrial origin for the radiation. This result was later supported by Millikan [2] when he probed that the ionization dropped with depth as electroscopes were sunk in lakes. This significant discovery initiated a close and fruitful connection between astronomy and particle physics since cosmic rays, name coined by Millikan due to the belief they were gamma rays, provided particle physicists with high energetic particles until the advent of particle accelerators. The observation of cosmic rays in cloud chambers led to the discovery of yet unknown particles where the positron and the muon were the first of them. In 1932, Anderson observed in his cloud chamber the antiparticle of the electron [3] as a result of cosmic ray interactions, the first evidence of antimatter, whose existence had been recently postulated by Dirac [4]. Hess and Anderson were both awarded with the Nobel prize in 1936. Shortly after this landmark, Street and Stevenson [5] and Anderson and Neddermeyer [6] discovered the muon while looking for evidences of a particle predicted by Yukawa [7] and associated with the strong nuclear force that finally turned out to be the pion, discovered in 1947 [8].

A major breakthrough in this field, previously suspected by Rossi [9], was carried out by Pierre Auger and co-workers [10] which were able to record coincidences in particle counters located at the same altitude but separated large distances. They interpreted the measurements as the result of a shower of secondary particles caused by the collision of high-energy primary particles with air nuclei discovering the so-called Extensive Air Showers (EAS). The confirmation of the phenomenon suspected by Rossi was possible using electronics with microsecond time resolution. Based on the recently developed electromagnetic cascade theory, Auger deduced that some detected EAS were initiated by cosmic rays with energies around 10^{15} eV.

The study of EAS has continued increasing the extension of the array of particle counters. Geiger-Müller counters were soon replaced by scintillation counters since the former were unable to provide information about the shower direction [11]. In 1963, Linsley reported the first event with an energy of about 10^{20} eV using the Volcano Ranch array [12].

A new advance in the detection of EAS generated by UHECRs was placed in 1962 when Suga [13] and Chudakov [14] suggested the possibility of using the Earth's atmosphere as a giant calorimeter where shower particles deposit their energy exciting N_2 molecules which later radiate fluorescence light. This pioneering idea has been subsequently exploited successfully by different air shower experiments.

2.3 Energy spectrum

The cosmic ray energy spectrum (figure 2.1), describing the flux of particles J as a function of energy, extends over 13 decades, from 1 GeV to above 10^{20} eV (or 100 EeV, $1 \text{ EeV} = 10^{18} \text{ eV}$) and 32 decades in flux. The energy spectrum of figure 2.2 has been measured with several instruments. At the lowest energies particles are directly detected before interacting with an atmospheric nucleus allowing a measurement of the flux for different primaries. At higher energies, approximately above 10^{14} eV, these measurements were only possible through the development of more and more large air shower experiments. The energy spectrum follows a power law ($J \propto E^{-\gamma}$) as expected for particles accelerated stochastically in shock fronts (see section 2.4). In figure 2.2 the flux has been multiplied by $E^{2.5}$ to point out the observed deviations from the power-law.

At energies around $4 \cdot 10^{15}$ eV there is a change in the spectral index from $\gamma \sim 2.7$ to $\gamma \sim 3.0$, known as the knee, where the flux turns down. It is generally assumed that cosmic rays with energies below the knee are accelerated through diffusive shock acceleration (see next section) in expanding supernova remnants (SNRs) located within our galaxy. The position of the knee also coincides with an increase of the mean primary mass so the change in the spectral index is probably caused by the galactic cosmic ray accelerators reaching their maximum energy (see equation 2.1), or also due to the limitation of the galactic magnetic fields to confine the nuclei in the Galaxy when they reach a given energy. In both cases the maximum energy is proportional to the charge of the particle Z so heavier particles would be accelerated to higher energies. The KASCADE experiment [15] has studied the knee region obtaining energy spectra for different primary mass groups (proton, helium and CNO) and showing that the position of the individual knees is shifted to higher energies with increasing atomic number.



2. ULTRA HIGH ENERGY COSMIC RAYS

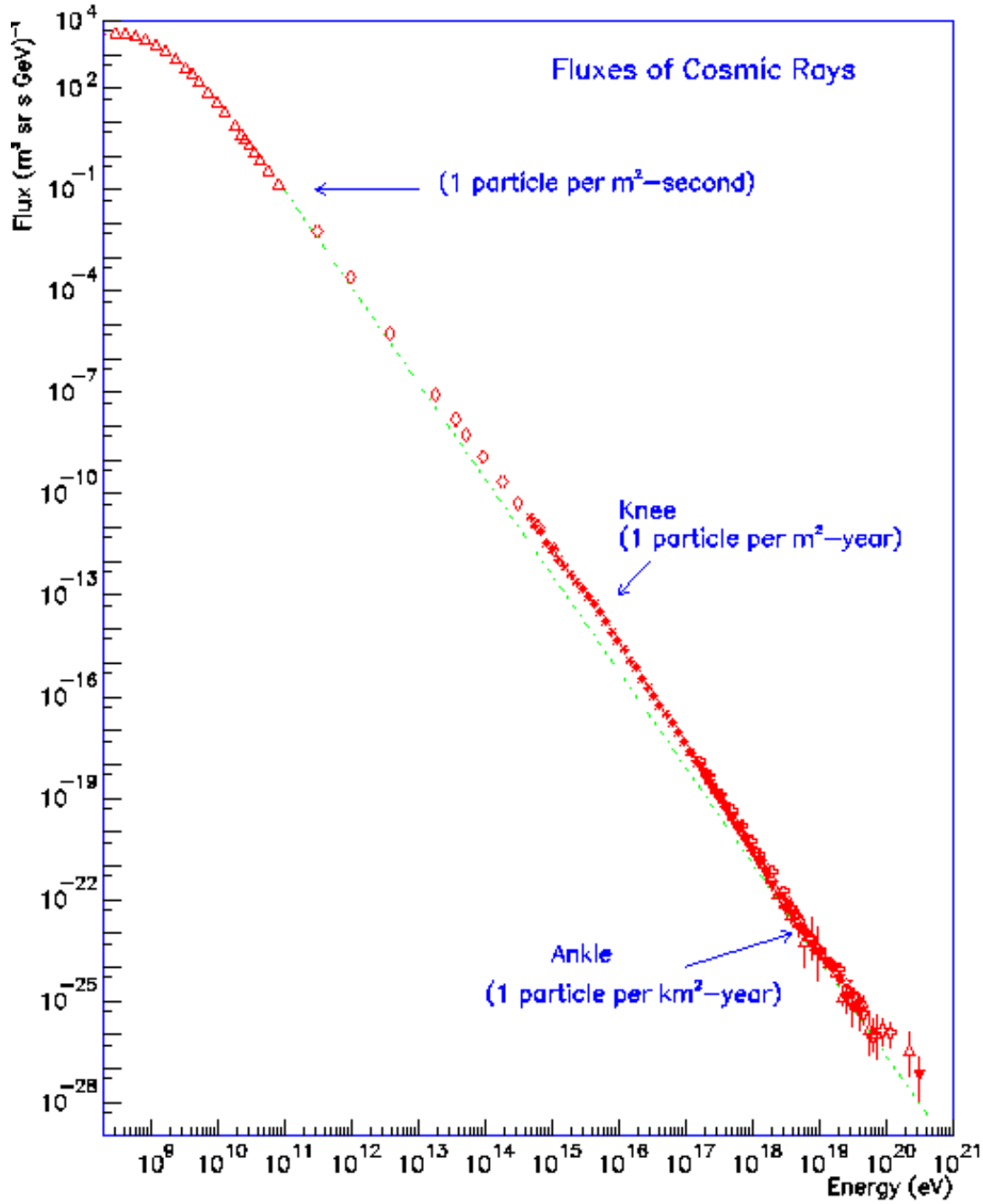


Figure 2.1: Energy spectrum of cosmic rays.

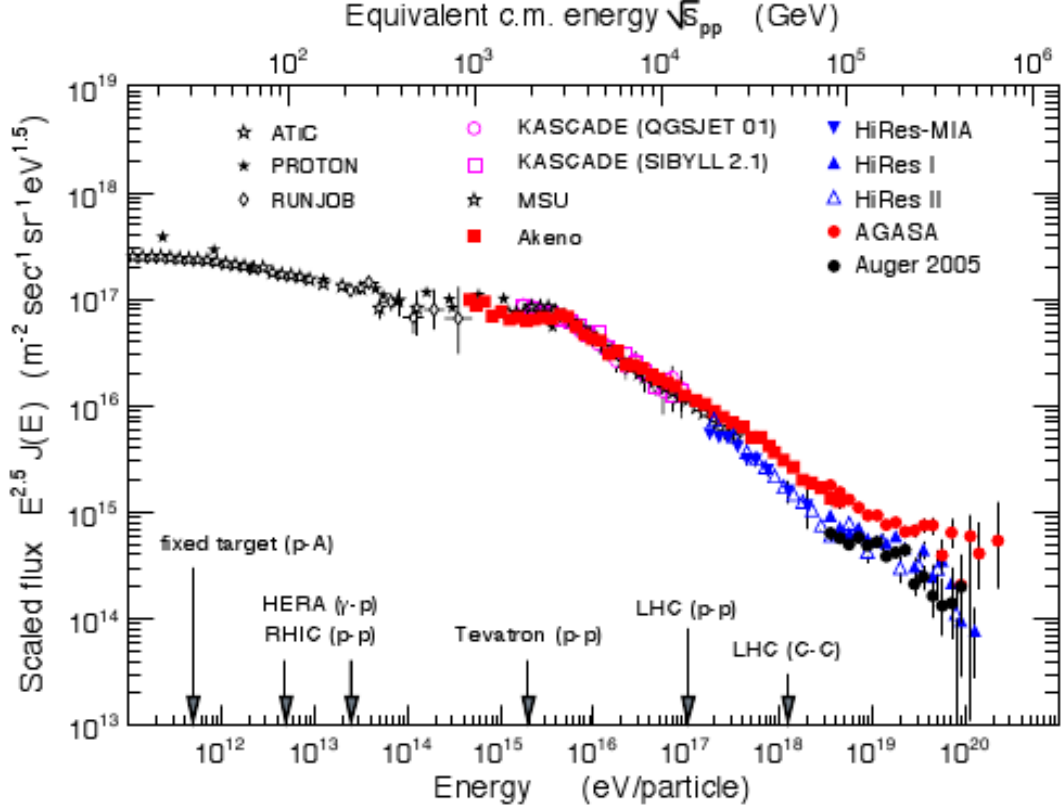


Figure 2.2: Energy spectrum of cosmic rays measured by different experiments. In this figure the flux has been multiplied by $E^{2.5}$ to evidence the spectral features. The achievable energy by some terrestrial accelerators are also shown for comparison. Figure taken from [16].

A second discontinuity has been observed at energies around 0.2 EeV, the so-called second knee, where the flux drops again and the spectral index changes to $\gamma \sim 3.3$. At energies around 4 EeV another break appears, the ankle, where the flux turns up again and the spectral slope returns again to ~ 2.7 . Several theories have been developed to explain the origin of cosmic rays in the region above the knee and the possible acceleration and propagation mechanisms which caused the aforementioned spectral features. Some of these scenarios attribute the spectral shape to a galactic component extending up to the ankle and a pure proton extragalactic component dominating above 10^{19} eV [17]. The ankle would be defined as the point of equal contribution of both galactic and extragalactic components. Other models explain the region between the knee and the ankle as the transition between both components but in this case the source composition of the extragalactic component could be mixed and the ankle



2. ULTRA HIGH ENERGY COSMIC RAYS

might be the point of the end of the transition [18; 19]. A different model [20] suggests that the transition between components takes place at lower energies, close to the second knee, and the change in the slope at the ankle is due to a propagation effect of a dominated proton extragalactic component. Protons propagating through the intergalactic medium interact with the Cosmic Microwave Background Radiation losing energy through pair creation and causing a lack of events at higher energies and a pile up at slightly lower energies. Detailed studies of mass composition in this region would be very useful to discriminate between possible models since they suggest different compositions at the ankle [21; 22].

2.4 Origin and Acceleration mechanisms

The sources of UHECRs as well as the mechanisms involved in the acceleration of these particles are still questions which remain unsolved. Currently, the acceleration of cosmic rays to the energies observed in the spectrum is conventionally explained with the Diffusive Shock Acceleration Mechanism (DSAM) [23], or first-order Fermi mechanism at shocks, based on the original work developed by Enrico Fermi in 1949 [24], nowadays known as Fermi II or second order acceleration. The original theory proposed cosmic rays gain energy stochastically due to encounters with moving magnetic clouds. Although particles may gain or lose energy in each individual encounter, on average cosmic rays gain energy with an efficiency proportional to β^2 ($\beta = u/c$, u = cloud velocity), not enough to accelerate particles to those highest energies observed in the spectrum of cosmic rays.

A more efficient version of Fermi acceleration mechanism is obtained considering encounters of particles with shock fronts. In this mechanism particles are diffusively accelerated while they cross back and forth the shock due to the deflection of charged particles by magnetic anomalies and the average fractional energy gain is of first order in the relative velocity of the shock front. A crucial feature of the DSAM is that particles emerge out with a characteristic power-law spectrum. However, there is a maximum energy attainable through this mechanism depending ultimately on the size of the accelerating object R (R is the linear radius) and the magnetic field strength B :

$$\frac{E_{max}}{\text{EeV}} \simeq \beta Z \frac{B}{\mu\text{G}} \frac{R}{\text{kpc}} \quad (2.1)$$

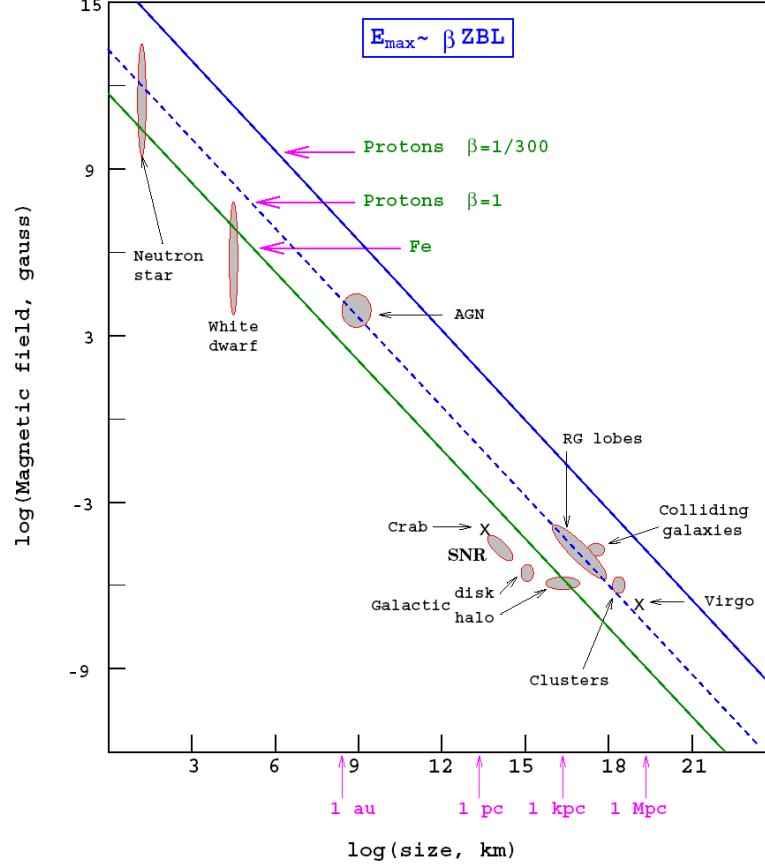


Figure 2.3: Hillas plot showing the size and magnetic field strengths of possible astrophysical candidates to accelerate UHECRs. Objects below the corresponding lines are not able to accelerate protons or iron nuclei to an energy of 10^{20} eV. Figure taken from [25].

where Z is the particle charge and β the velocity of the shock front in units of the speed of light.

Cosmic rays with energies up to some GeV arise predominantly from the Sun since, for example, characteristic night-day variations are presented in the spectrum. From these energies up to a few 10^{15} eV acceleration of cosmic rays can be explained by shock acceleration in galactic SNRs (see [26] for a review of the current knowledge of cosmic ray acceleration in SNRs). Support to this assumption comes from the observation of TeV gamma rays from SNRs [27] even when there is much controversy about if these gamma rays are generated from electrons or, indeed, from protons and nuclei [28]. Some theories point out that cosmic rays could be accelerated in SNRs up to energies of 10^{18} eV [29; 30] but there is not experimental evidences to prove it. From this



2. ULTRA HIGH ENERGY COSMIC RAYS

energies to the highest ones observed in the energy spectrum the theories developed to explain the possible acceleration mechanisms are not conclusive. M. Hillas proposed a simple dimensional argument based on equation 2.1 to demonstrate there are only few astrophysical objects capable to accelerate nuclei and protons to energies of 10^{20} eV. Figure 2.3 illustrates this situation. Furthermore, this plot shows a maximum energy without taking into account possible losses.

Due to the absence of a satisfactory traditional theory to explain the origin of these extreme energetic particles, several scenarios demanding new physics and grouped in the so-called top-down models have been proposed. These models include production of UHECRs due to the decay of exotic particles such as Topological Defects (TD) [31] or superheavy dark matter (SHDM) particles [32; 33; 34], or neutrino interactions with the relic neutrino background (Z-burst model) [35]. A common feature of all these top-down models is that they suggest a large fraction of photons in the flux of UHECRs [36]. The validity of these theories can be studied measuring the flux limits for both photons and neutrinos. The Pierre Auger Collaboration has set upper limits for the fraction of photons at different energies [37] using the large acceptance of the surface detector and has constrained these values to 2.0% for $E > 10$ EeV, 5.1% for $E > 20$ EeV and 31% for $E > 40$ EeV (figure 2.4(left)). These limits put strong constraints on certain top-down models and for example the SHDM scenario does not seem to be an adequate model for the origin of UHECRs. Ultra high energy neutrinos can also arise from these exotic processes but their low cross sections make difficult a measurement of the neutrino flux. The Pierre Auger Collaboration has also estimated an upper limit for the flux of tau neutrinos looking for nearly horizontal showers with a significant electromagnetic component [38]. Results are presented in figure 2.4(right).

2.5 Propagation and source identification of UHECRs

Cosmic rays propagating through space may suffer interactions and their trajectories are affected by both galactic and extragalactic magnetic fields. The study of these effects is crucial since they have important implications in the energy spectrum and distribution of arrival directions observed at Earth. The GZK effect is the most studied interaction process of UHECRs since it is thought that this effect is responsible of/for the shape of the energy spectrum at the highest energies. In 1966, Greisen [39], and

2.5 Propagation and source identification of UHECRs

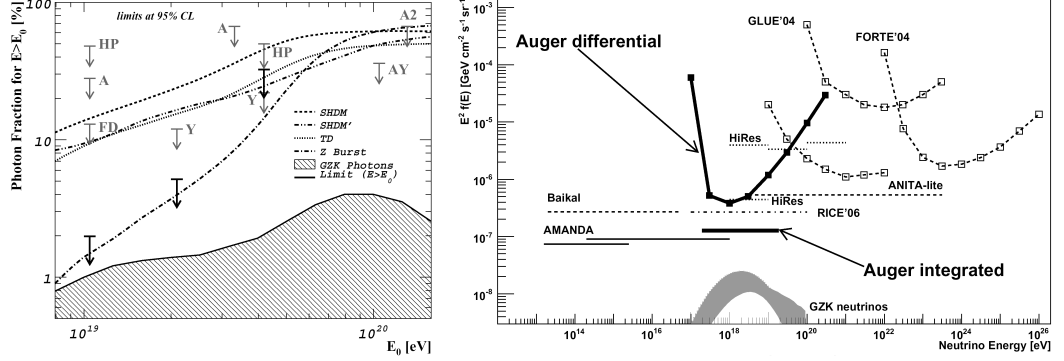


Figure 2.4: Left: upper limits at 95% CL on the fraction of photons for different energies obtained from Auger data (black arrows) along with those obtained from AGASA (A, A2), Haverah Park (HP), Yakutsk (Y), AGASA-Yakutsk (AY) and Auger hybrid data (FD). Predictions of the photon fraction from different top-down models (SHDM, TD and Z-burst) are also shown as well as predictions for GZK photons. Right: Limits at 90% CL for a diffuse flux of μ_ν from Auger data. Limits from other experiments are also shown.

Zatsepin and Kuzmin [40] pointed out that protons with energy above $5 \cdot 10^{19}$ eV lose energy when interact with CMB photons by pion photoproduction via Δ^+ resonance:

$$p + \gamma_{CMB} \rightarrow \Delta^+ \rightarrow p + \pi^0 \quad (2.2)$$

$$p + \gamma_{CMB} \rightarrow \Delta^+ \rightarrow n + \pi^+$$

This interaction would produce a suppression in the energy spectrum for energies around $5 \cdot 10^{19}$ eV, the so-called GZK steepening, and also imposes a limit for the distance at which the sources of these cosmic rays above this energy limit can be located. For example, sources of protons observed at Earth with an energy above 10^{20} eV are not expected to be located at distances larger than 50 Mpc. This phenomena is illustrated in figure 2.5 where the mean energy of protons with three different primary energies is shown as a function of the traversed distance. Protons also interacts with CMBR through pair production with an energy threshold about 10^{18} eV:

$$p + \gamma_{CMB} \rightarrow p + e^+ + e^- \quad (2.3)$$

Although the energy loss for pair production is quite insignificant compared with that released in photo-pion production, this mechanism determines the shape of the energy spectrum since it is thought to be the dominating process in the region between



2. ULTRA HIGH ENERGY COSMIC RAYS

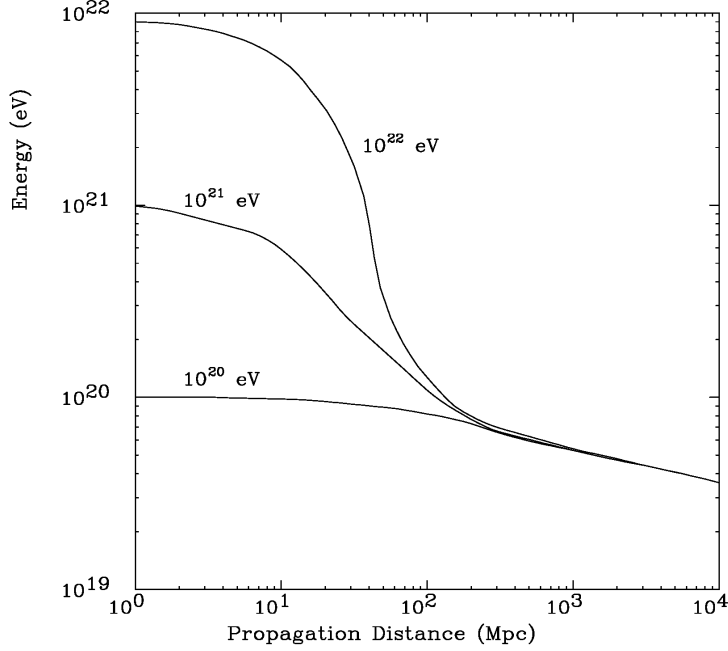
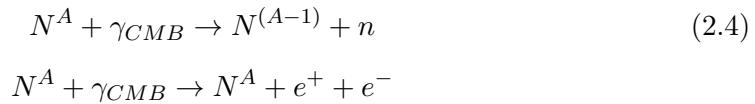


Figure 2.5: Energy of a proton as a function of the propagation distance through the CMBR for three different initial energies. Figure taken from [41].

the second knee and the ankle. In figure 2.6 the mean path lengths for the main mechanisms of energy losses suffered by UHECRs as they propagate to the Earth are presented [42]. Cosmic rays with energies above $5 \cdot 10^{19}$ eV travelling distances larger than 100 Mpc will suffer a significant energy degradation.

Heavy nuclei also interact with CMBR and intergalactic Infrared Background Radiation (IBR) through photodisintegration and pair production with similar energy thresholds [43; 44]:



The Pierre Auger Collaboration has recently presented a measurement of the energy spectrum of cosmic rays in the ultra-high energy region [45] and evidences of the GZK steepening have been found. As shown in figure 2.7 a suppression in the flux is observed for energies above around 40 EeV with a significance of 6σ . If an extrapolation of the power-law fitted at low energies were valid for energies above this limit the expected number of events would be 167 while the observed number is only 69.

2.5 Propagation and source identification of UHECRs

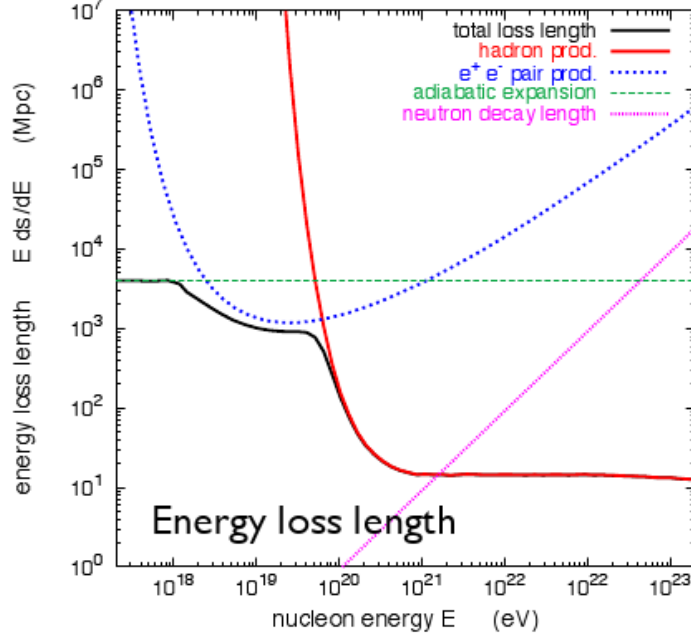


Figure 2.6: Energy loss length for the propagation of extragalactic cosmic rays as a function of nucleon energy. The hadron production corresponds to the GZK effect. Figure taken from [16].

Besides possible interactions with background radiations, cosmic rays are also affected by magnetic fields. Both galactic and extragalactic fields bend the trajectories of cosmic rays but up to now quantitative predictions of the deflection angles are not possible due to the limited knowledge of the strength of magnetic fields, the charge of UHECRs and the location of the sources. Figure 2.8 shows a clear view of the effect of a magnetic field of 10^{-9} G in the trajectories of protons emanating from a point source with different energies. As can be observed propagation of cosmic rays can pass from diffusive to rectilinear propagation, even in the presence of a weak magnetic field, when the primary energy passes from 1 to 100 EeV.

At relative low energies, looking for possible sources of UHECR is a very difficult task since the distribution of arrival directions is isotropic due to the effect of the galactic magnetic field. A lot of efforts has been concentrated to find evidences of anisotropies at the highest energies: claims of correlation between the arrival directions of UHECRs and subclasses of BL Lacs [47]; a flux excess in the region around the Galactic Center in two different energy ranges as reported by AGASA [48] and SUGAR [49] collaborations; evidences of clustering signal at small angular scales for energies above 40 EeV [50].



2. ULTRA HIGH ENERGY COSMIC RAYS

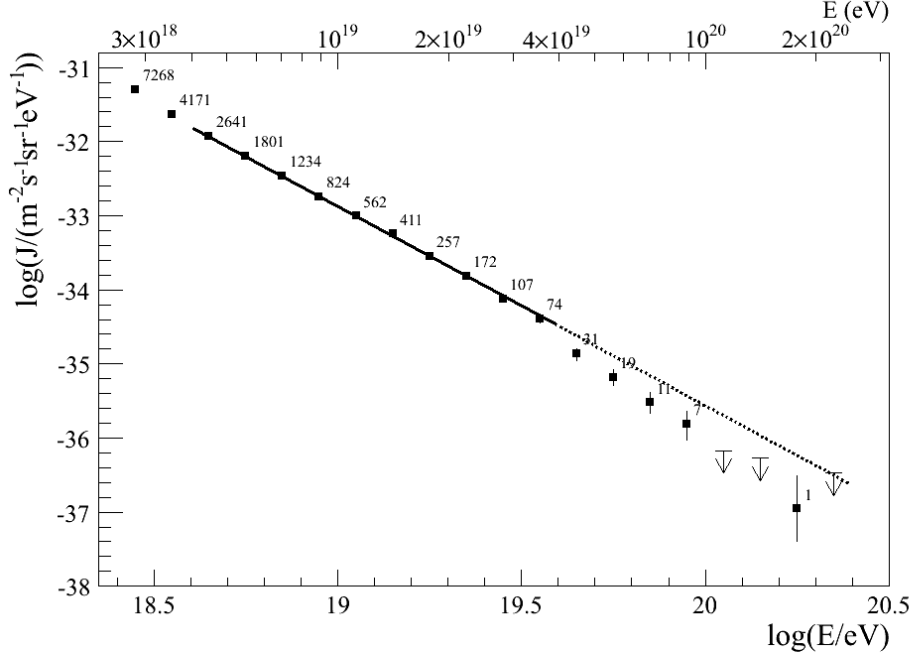


Figure 2.7: Energy spectrum obtained from data taken at the Pierre Auger Observatory. The first part of the spectrum (up to 40 EeV) is fitted to a power-law function which is extrapolated to the highest energies to evidence the flux suppression in this region.

Nevertheless, these claims have not been confirmed by the Pierre Auger Observatory even when nowadays the experiment accumulates the largest statistics [51; 52; 53; 54]. Recently, the Pierre Auger Collaboration has reported a correlation between the arrival direction of cosmic rays with energies above 57 EeV and the position of active galactic nuclei (AGNs) lying within ~ 75 Mpc [55]. Figure 2.9 shows the arrival directions of the 27 highest energy events as circles of radius 3.2° . For 22 out of these 27 events the circle comprises the position of AGNs (red asterisks). This result is compatible with a scenario where UHECRs are accelerated in nearby extragalactic sources, AGNs or also other astrophysical objects with a similar spatial distributions.

2.6 Phenomenology of Extensive Air Showers

As already mentioned, at energies below 10^{15} eV, both charge and mass can be measured directly using space detectors. However above these energies the flux of Cosmic Rays (CRs) is so low that direct detection is not statistically viable. In this case the primary

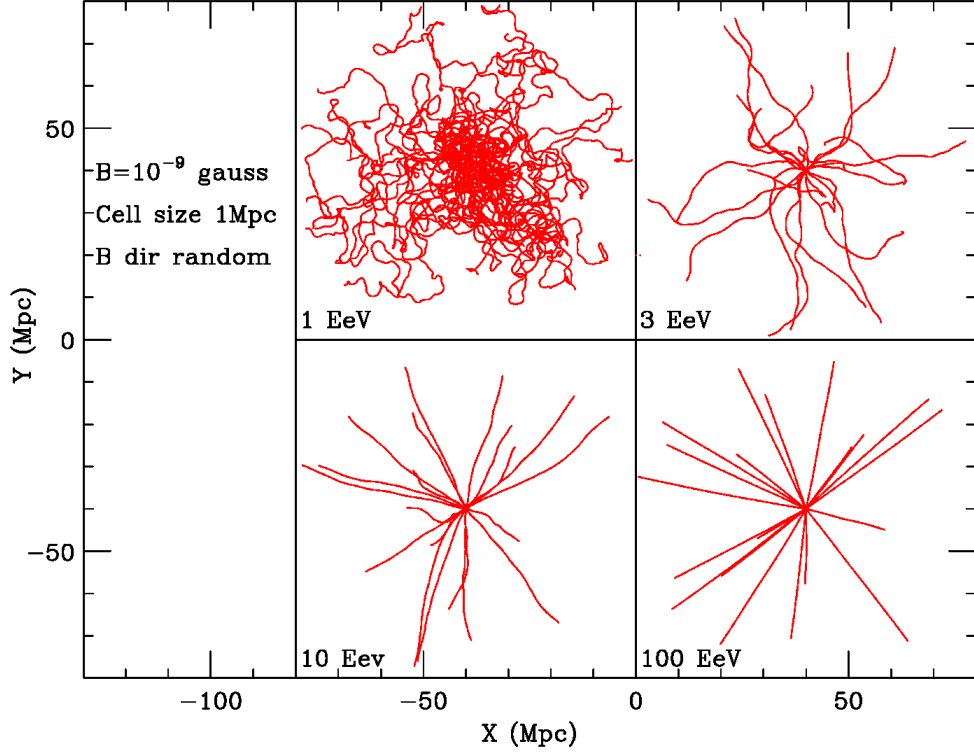


Figure 2.8: Projected view of 20 trajectories for protons of several energies emanating from a point source. The magnetic field is described by cells of size 1 Mpc with mean value 1 nG oriented randomly in each cell. Figure taken from [46].

particle has enough energy to initiate an Extensive Air Shower (EAS) in the atmosphere. The properties of primary CRs of the highest energies have to be deduced from the properties of these showers.

The vertical depth of the atmosphere exceeds 10 times the hadronic interaction length for a proton and is about 30 radiation lengths. The atmosphere is thus a thick calorimeter in which the energy of the leading nucleon decreases to at least 10^{-4} its value, and only a relative small part of it (the one carried by muons, neutrinos and few nucleons) reaches sea level.

If the incoming projectile is a nucleon or a nucleus the primary interaction is of hadronic nature. As a first approximation, if collective effects are neglected, a nucleus of mass number A and energy E is equivalent to the superposition of A independent nucleons, each with energy E/A . The primary interaction produces a large number of secondary particles mainly pions, which give rise to further hadronic interactions



2. ULTRA HIGH ENERGY COSMIC RAYS

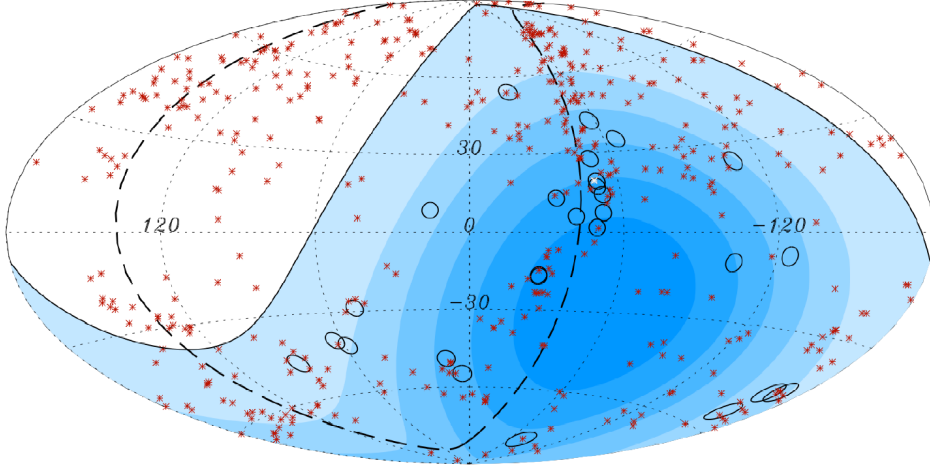


Figure 2.9: Projection of the celestial sphere in galactic coordinates with circles of 3.2° centred at the arrival directions of the 27 highest-energy events detected by the Pierre Auger Observatory. The positions of AGNs with redshift $z \leq 0.017$ are indicated by red asterisks. The solid line represents the border of the field of view (for zenith angles smaller than 60°) for the southern site of the Observatory. The dashed line is the super-galactic plane. Darker blue color indicates larger relative exposure.

forming the so-called hadronic component. The depth at which it takes place and the number of secondary particles, known as multiplicity, depends on the nature and the energy of the primary cosmic ray. The number of hadrons increases with subsequent generations of particle interactions. In each generation about $1/3$ of the energy is transferred to an electromagnetic cascade by the rapid decay of neutral pions into γ rays. Ultimately, this cascade of electromagnetic particles¹ dissipates roughly 90% of the primary particle's energy through ionization of air molecules.

The hadronic component (except a few nucleons) ends up with the decay of charged pions into muons giving rise to the denominated muon component, at intermediate altitudes², when their Lorentz factor γ is such that the decay length $\beta\gamma ct$ becomes comparable to the interaction length, of the order of 50 g cm^{-2} . Muons also inherit the transverse momentum of their parents, of few hundred MeV, and their divergence, the angle that form with the shower axis, is relatively small and strongly anticorrelated to their energy.

¹Throughout this text the term electromagnetic particles refers to γ , e^- and e^+ .

²This height, of around 6 km, is subjected to large fluctuations.

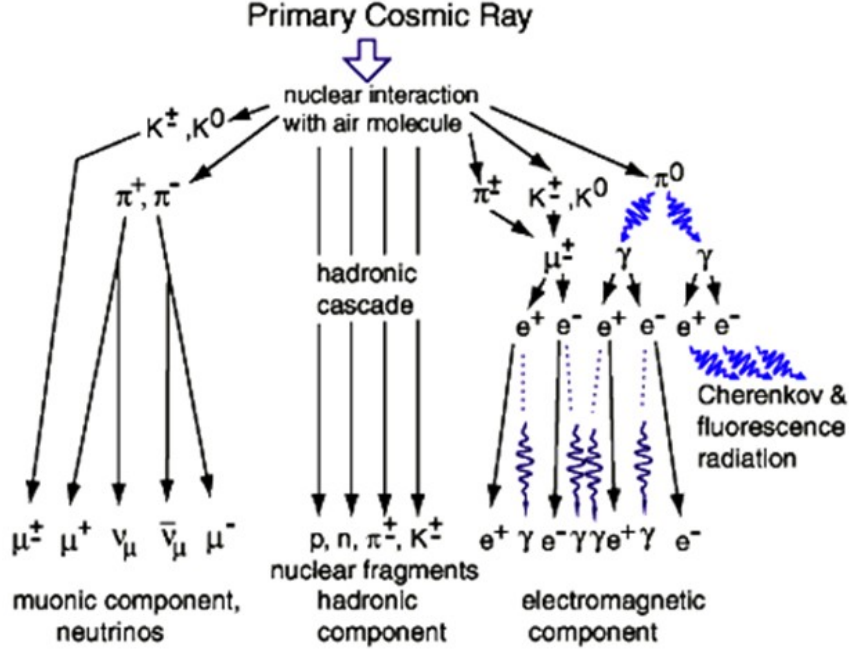


Figure 2.10: Schematic diagram of the principal EAS processes.

The electromagnetic and weak interactions are rather well understood. However, uncertainties in hadronic interactions at ultra high energies constitute one of the main sources of systematic error in the reconstruction of air showers properties using simulations, as it will be shown in 2.6.1. A complete review on the phenomenology of cosmic ray showers can be found in [56]. Figure 2.10 shows a schematics of the principal processes involved in the development of an EAS as it was explained before.

2.6.1 Hadronic processes

As already mentioned, the properties of the primary particles are reconstructed on the basis of measured characteristics of EAS. The quality of such reconstruction depends strongly on the present understanding of complicated EAS physics, specially in the hadron cascade processes. The detailed shower development is too complex to be fully described by a simple analytical model. Therefore Monte Carlo (MC) simulation plays an important role in UHECR physics. The different codes for the simulation uses our present knowledge on interactions, decays and particle transport in matter. The major uncertainties in EAS simulation arise from hadronic interaction models.



2.6.1.1 Collider experiments and UHECRs

Compared to particle physics, UHECR interactions are orders of magnitude beyond what can be achieved in current and near future terrestrial collider experiments. The Large Hadron Collider (LHC) [57] at CERN will provide pp collisions at the unprecedented center-of-mass energy of $\sqrt{s} = 14$ TeV as well as heavy ion collisions at a center-of-mass energy of about 1000 TeV. But it is the energy range the only difference between collider and cosmic ray models?. Is it possible to apply accelerator codes for CR data analysis?

Events triggered and examined at accelerator experiments are those that produce particles with high momentum transfer. Particles with small transverse momenta mostly escape undetected and are responsible for most of the radiation background at LHC experiments. Moreover, CRs collisions are predominantly nucleon-nucleus or nucleus-nucleus collisions, for which acceleration data are available only at much lower energies. Both ATLAS [58] and CMS [59] experiments at LHC are planing to implement additional detectors to cover the forward regions with tracking and/or calorimetry. Additionally a new experiment to study very forward particle emission in the LHC collider, LHCf, has been approved. The results of LHCf will be able to provide the production spectrum of secondary particles in the very forward region, allowing to constrain the Monte Carlo codes [60].

All contemporary hadronic models developed to describe collider and EAS data are based on similar physics ideas. However, collider codes are not generally suitable for immediate application in EAS field, being unable to treat some necessary interactions, predictive calculations of the total inelastic cross section or to be extrapolated into the ultra-high energy region. A comparison of the two most widely hadronic interaction models used in UHECR field, QGSJET [61] and SIBYLL [62] with particle physics at LHC can be found in [63].

2.6.1.2 Hadronic interaction models

Electromagnetic and weak interactions are well understood, but this is not the case for hadron production in collisions of nucleons, pions and kaons with light nuclei, where the lack of experimental data lead to limitations in many UHECR applications. In hadron interactions it is mandatory to distinguish between elastic and inelastic processes, and

inelastic ones are split into diffractive and non-diffractive ones, the later ones usually called minimum bias events.

Nowadays perturbative quantum chromo-dynamics (QCD) is the only strict theory which provides the basis for both qualitative and quantitative description of hadronic production in high energy processes. However, precise perturbative QCD calculations are only possible for processes with large momentum transfer Q^2 (high p_t) or “hard” processes. On the other hand “soft” collisions i.e. low Q^2 (low p_t), produce particles with small transverse momenta, into the extreme forward direction carrying away the largest fraction of the energy. These reactions play a dominant role in the non-diffractive component, previously mentioned, and are very important for air showers, since they carry the energy deep down into atmosphere and thus drive the air shower development.

Hadronic interactions models rely in perturbative QCD formalism in the description of “hard” processes while treating the “soft” ones in a simplified phenomenological approach. The new/contemporary models simulate hadron-hadron collisions as a multiple scattering processes on the basis of the Gribov-Regge theory (GRT) [64]. In GRT an elementary process is treated phenomenologically, as an exchange of a quasi-particle, the Pomeron. Such elementary interactions happen in parallel at the same moment and are believed to be partonic cascades. A good introduction to GRT models is given in [65].

2.6.2 Electromagnetic component

In contrast to hadronic processes, the electromagnetic interactions of shower particles can be calculated very accurately from quantum electrodynamics. The first comprehensive treatment of electromagnetic showers was elaborated by Rossi and Greisen [66]. This treatment has been recently reviewed by Gaisser in a more pedagogical form and a summary is presented in [56].

Electromagnetic particles are the most abundant ones and carry the largest fraction of the total energy. The generation of the electromagnetic component is driven by bremsstrahlung interactions and pair production¹. Eventually the energy per particle drops below a certain energy when Compton scattering starts to dominate over pair production for γ rays and ionization losses over bremsstrahlung for $e^\pm$. Below this

¹Compton scattering, inverse Compton scattering and photo-electric effect are not important until the particles are at energies below of few tens of MeV.



2. ULTRA HIGH ENERGY COSMIC RAYS

critical energy ϵ_c ionization losses start to dominate over particle production and the shower dissipates its energy into the atmosphere. The electrons¹ energy loss rate due to bremsstrahlung is nearly proportional to their energy, whereas the ionization loss rate varies logarithmically with energy. The changeover from radiation processes to ionization losses depopulates the shower. For this reason the electromagnetic component develops fast and reaches a pronounced maximum, in the number of particles, and then start to decrease. The position of this maximum, measured in terms of slant depth, is represented by X_{max} .

As a rough estimate when a 10^{20} eV proton vertical shower reaches the ground, the number of secondary particles with energy above 90 keV in an annular region from 8 m to 8 km from the shower core is of about 10^{11} . 99% of these particles are electromagnetic ones with a typical ratio of γ to $e^\pm$ of 9 to 1. Their mean energy is around 10 MeV and carries 85% of the total energy of the shower at ground level. This scenario completely changes for a primary with a zenith angle $\theta > 70^\circ$. In this case the electromagnetic component becomes attenuated exponentially with atmospheric depth, being almost completely absorbed at ground level. Notice that for a vertical shower the total atmospheric depth is $\sim 1000 \text{ g cm}^{-2}$ about 36 times deeper for completely horizontal shower.

2.6.3 Muon component

The muon content of an EAS is an important parameter strongly related with the nature of the primary. In general, the fraction of the energy remaining in the hadronic cascade decreases exponentially with the number of generations and the energy transferred into muons depends essentially on the number of steps needed to reach the energy level where charged pions decay. A heavy nucleus has less energy per nucleon than a light one of equal primary energy, therefore reaches this level faster and as a consequence gives more muons. The exact ration depends on the model of hadronic interaction.

As compared to EM particles, muons hardly interact and loose small fraction of its energy by ionization. The muonic component differs from the electromagnetic mainly for two reasons. Firstly muons are generated by the decay of cooled charged pions, and thus are sensitive to the baryonic content of the primary particle, and secondly the muon has a much smaller cross section than the electron, and so the muonic component

¹The term electrons refers to both electrons an positrons.

2.6 Phenomenology of Extensive Air Showers

develops differently than the electromagnetic component does. It is important to note that muons do not cascade, i.e. there is not new generation of muons, therefore the number of muons at ground level is much smaller than the number of electrons. There are about 5×10^8 muons above 10 MeV at ground level for a vertical 10^{20} eV proton induced shower. Also the smaller multiple scattering suffered by muons leads to shorter arrival times at the ground than electromagnetic particles.

The ratio of electrons to muons depends on the distance from the core. The $e^\pm$ to $\mu^\pm$ ratio for a 10^{20} eV vertical proton shower varies from 17 at 200 m from the core to 1 at 2000 m. This scenario changes in the case of inclined showers, for zenith angles greater than 60° , this ratio remains almost constant at a given distance to the core. This ratio decreases with the zenith angle, for example at $\theta = 75^\circ$ where the ratio is 400 times smaller than for a vertical shower. Another important difference between vertical and horizontal showers is that the mean muon energy is different. In the case of horizontal shower only high energy muons reach ground level because most of the low energy muons disappear due to energy loss and decay (finite muon lifetime). The average muon energy is 1 GeV for a vertical shower while for horizontal showers is about 2 orders of magnitude greater.

2.6.4 Heitler model

Before the era of the high-power computing, Heitler presented a very simple model of electromagnetic cascade development [67]. At that time more authors, notably Rossi, developed more sophisticated analytical tools, including more physical effects. Such approaches have been well described by Gaisser [68]. Hadronic showers can be modelled using similar approaches.

2.6.4.1 Electromagnetic showers

Heitler's model of electromagnetic showers predicts accurately the most important features of the cascade and the relation of the X_{max} with the energy without the need of considering all the complexities of the showers. Figure 2.11 can be used to show the simplified shower.

In the model, electrons/positrons and photons undergo repeated two-body splittings via bremsstrahlung and pair production respectively. Every particle undergoes



2. ULTRA HIGH ENERGY COSMIC RAYS

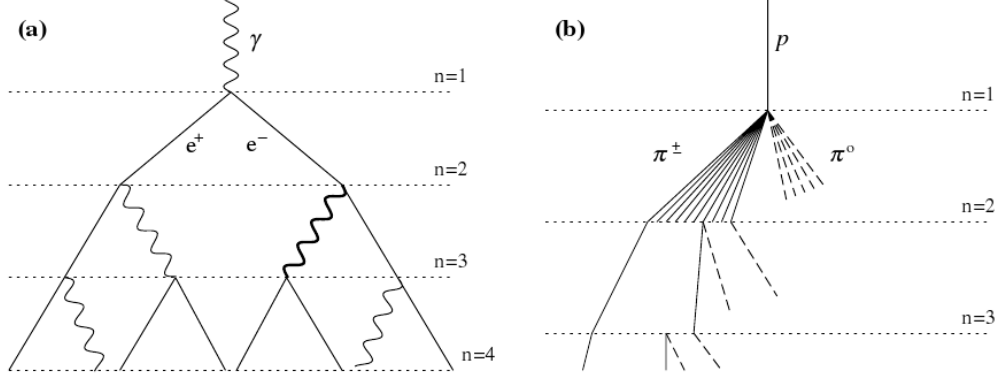


Figure 2.11: Schematic view of an electromagnetic shower (a) and hadronic shower (b). Figure taken from [69].

a splitting after travelling a fixed distance d ¹ and each particle takes exactly half of the energy of the parent particle. As such, after the shower has traversed n splitting lengths the number of particles in an electromagnetic shower is 2^n and their energy is given by $E_0/2^n$ where E_0 is the energy of the particle which initiates the shower.

The number of particles in the shower continues growing until the individual energies of electrons and photons drop below the critical energy ϵ_c ($\epsilon_c = 85$ MeV). At this point the shower reaches its maximum and the particles in the shower begin to fall with the remaining particles dissipating energy by ionisation losses into the atmosphere.

This simplified model does not reproduce accurately all the details of the electromagnetic showers. But two very important features are well accounted for: the final total number of electrons, positrons and photons N_{max} and the depth of maximum shower development X_{max} .

Consider a shower initiated by a photon with energy E_0 . The cascade reaches the maximum size N_{max} when all the particles have an energy equals to ϵ_c , that is $E_0 = \epsilon_c N_{max}$. The number of particles at shower maximum will be $N_{max} = E_0/\epsilon_c$, proportional to the primary energy E_0 . The depth of the shower maximum is obtained after a number of n_c splitting lengths required for the energy per particle to be reduced to ϵ_c . Since $N_{max} = 2^{n_c}$ then $n_c = \ln(E_0/\epsilon_c)/\ln 2$ giving:

¹The average value of d is related to the radiation length of the electron, λ_r by: $d = \lambda_r \ln 2$ [g cm⁻²]. Note that d is in fact the distance over which an electron loses, on average, half its energy by radiation.

2.6 Phenomenology of Extensive Air Showers

$$X_{max} = n_c \lambda_r \ln 2 = \lambda_r \ln(E_0/\epsilon_c) = 2.3 \lambda_r \log(E_0/\epsilon_c) [\text{g cm}^{-2}] \quad (2.5)$$

As can be observed, the depth of the shower maximum is logarithmically proportional to E_0 . The elongation rate D_{10} , defined as the rate of increase of X_{max} with E_0 , predicted by the simple model is:

$$D_{10} = \frac{dX_{max}}{d \log E_0} = 2.3 \lambda_r [\text{g cm}^{-2}/dec] \quad (2.6)$$

This value of D_{10} is in good agreement with that predicted with more complex models. The radiation length of electrons in air is 37 g cm^{-2} , thus the Heitler model predicts an increase in X_{max} of 85 g cm^{-2} per decade in energy for pure electromagnetic showers.

2.6.4.2 Hadronic showers

Air showers initiated by hadrons figure 2.11 (b) are modelled using an approach similar to the Heitler model [69]. Consider a cosmic-ray proton entering the atmosphere with an energy E_0 and an interaction length in air λ_I . After traversing a path length $X_0 = \lambda_I \ln 2$ the proton interacts with an atomic nucleus in the atmosphere, creating a number of pions, known as the multiplicity with an average value which increases with E_0 . The interaction of the primary is an inelastic process where the fraction of the energy directed into the shower is defined as the inelasticity, κ .

Of the number of pions produced, a fraction N_{ch} corresponds to charged pions and a fraction of $\frac{1}{2}N_{ch}$ to neutral pions. The neutral pions decay immediately into two photons before re-interacting¹ with an atmospheric nucleus, initiating electromagnetic showers.

Charged pions continue through the atmosphere interacting until the $\pi^\pm$ energy fall below the critical energy for pions ϵ_c^π which can be approximated as a constant value of $\epsilon_c^\pi = 20 \text{ GeV}$. In this model charged pions are assumed to decay and not interact below ϵ_c^π yielding muons:

$$\pi^\pm \rightarrow \mu^\pm + \nu/\bar{\nu} \quad (2.7)$$

¹Except at Lorentz factors above 10^{11} , that is at energy above 10 EeV.



2. ULTRA HIGH ENERGY COSMIC RAYS

In an hadronic shower X_{max} is the atmospheric depth where electrons and photons of the shower reach their maximum number. The first interaction injects $\frac{1}{3}E_0$ into the electromagnetic shower through the π^0 decay. This is followed by additional subshowers from each interaction point. It is possible to estimate X_{max} using only the first generation of the electromagnetic shower. It is clear that this approximation underestimates X_{max} because it neglects the effects of the subsequent showers, but explains well the elongation rate.

The first interaction produces $\frac{1}{2}N_{ch}\pi^0 \rightarrow N_{ch}\gamma$. Each photon initiates an electromagnetic shower of energy $E_0/(3N_{ch})$. The depth of maximum can be obtained using 2.6 with primary energy of $E_0/(3N_{ch})$ starting at X_0 :

$$X_{max}^p = X_0 + \lambda_r \ln[E_0/(3N_{ch}\epsilon_c)] [g \text{ cm}^{-2}] \quad (2.8)$$

The value of N_{ch} in the first interaction can be parametrised based on pp data as in [69] yielding:

$$X_{max}^p = (470 + 58 \log(E_0/1PeV)) [g \text{ cm}^{-2}] \quad (2.9)$$

It is possible to obtain the number of muons using $N_\mu = N_\pi = (N_{ch})^{n_c}$, where n_c is the number of steps needed to reach the pion critical energy [69]:

$$\ln N_\mu = \beta \ln(E_0(\epsilon_c^\pi)) \quad (2.10)$$

where:

$$\beta = \frac{\ln(N_{ch})}{\ln(\frac{3}{2}N_{ch})} = 0.85 \quad (2.11)$$

The muon size of the shower is then:

$$N_\mu = \left(\frac{E_0}{\epsilon_c^\pi}\right)^\beta \sim 10^4 \left(\frac{E_0}{1PeV}\right)^{0.85} \quad (2.12)$$

For a massive nuclei is possible to use the superposition model. A nucleus with atomic number A and total energy E_0 is equals to A individual nucleons, each with energy E_0/A and interacting independently. The shower properties of a nucleus initiated shower can be expressed in terms of the corresponding quantities of a proton shower with the same total energy E_0

$$N_\mu^A = N_\mu^p A^{0.15} \quad (2.13)$$

$$X_{max}^A = X_{max}^p - \lambda_r \ln A \quad (2.14)$$

2.7 Composition of cosmic rays

As occurs for the energy spectrum, the composition of cosmic rays at relative low energies (less than about 10^{14} eV) can be directly measured with spectroscopes placed high in the atmosphere. Up to these energies the relative abundancies of cosmic rays is rather similar to those observed in the intergalactic medium with some notable relative excesses of light elements which can be attributed to the spallation of heavy nuclei [?]. Beyond this energy cosmic ray studies are based on the observation of EAS. There are several shower properties which correlate with the mass of primary particle. In this section the parameters sensitive to composition, specially those which can be measured with the Pierre Auger Observatory, will be enumerated.

An accurate measurement of the energy spectrum of cosmic rays using surface detectors relies on a good knowledge of mass composition since interaction models used in shower reconstruction predicts different energies for proton and iron primaries with differences around 15% for 10^{20} eV, with protons given the highest value. For the fluorescence technique this effect is less important since it provides a calorimetric energy measurement nearly independent on hadronic models. Nevertheless a small correction of around 10% must be applied to take into account the energy not released in the atmosphere. This correction is again mass dependent with systematic differences around 5% when proton and iron primaries are compared. A precise knowledge of cosmic ray composition is also required to carry out confident anisotropy studies due to the dependence of the effect of magnetic field with particle charge.

Depending on the type of detector, surface array or fluorescence detector, different parameters can be investigated as observables of primary composition.

2.7.1 The shower maximum depth X_{max}

As previously shown in section 2.6.2, the Heitler model for cosmic ray showers predicts that the maximum number of particles is reached at a depth, mass and energy dependent, given by equation 2.14.



2. ULTRA HIGH ENERGY COSMIC RAYS

The most widely used technique in the indirect measurement of primary mass composition is based in the elongation rate theorem, firstly proposed by Linsley [70]. The elongation rate D_{10} , already defined (see section 2.6.4) as the rate of change of the mean value of X_{max} with E_0 , depends on the mean logarithmic primary mass $\langle \ln A \rangle$ [71]:

$$D_{10} = \frac{d}{d \log E_0} \left[2.3 \lambda_r \log \frac{E_0}{A \epsilon_c} \right] + \frac{d}{d \log E_0} [X_0 - 2.3 \lambda_r \log(3 N_{ch})] \quad (2.15)$$

Figure 2.12 presents a summary of the measurements of $\langle X_{max} \rangle$ together with predictions from different models. The Pierre Auger Collaboration has presented a recent measurement of the depth of maximum development in the ultra-high energy region [73] (see figure 2.13) obtained for a high quality sample of hybrid events. As can be deduced from previous plots, the inferred composition is strongly dependent on the selected hadronic models used for composition. While some models suggest a proton dominated composition above 10^{19} eV others indicate a mixed composition. Systematic uncertainties are estimated less than about 15 g/cm² for low energies and less than 11 g/cm² above 10^{18} eV [74].

2.7.2 Fluctuations in X_{max}

The width of X_{max} distributions in a narrow range of energies is another parameter sensitive to composition since heavy nuclei are expected to produce shower-to-shower fluctuations smaller than for protons [75]. In order to use this procedure for mass composition studies a good knowledge of atmospheric profiles is required since, as some studies suggest, fluctuations due to atmosphere changes are quite significant and they could be wrongly attributed to intrinsic X_{max} shower fluctuations [76].

As with the elongation rate, changes in the values of the width of X_{max} as a function of energy can indicate changes in the average mass of the flux of UHECRs at the given energies. Results of the measured fluctuations in hybrid events recorded using the Pierre Auger Observatory, are shown in figure 2.14. As can be observed in this figure an increase of mass composition appears at energies above $10^{18.5}$ eV being pure iron composition at $10^{19.4}$ eV. This is not observed in the elongation rate measurements (see figure 2.13) where a mixed composition is present at even the highest energies. This fact indicates a inconsistent between data and the current models.

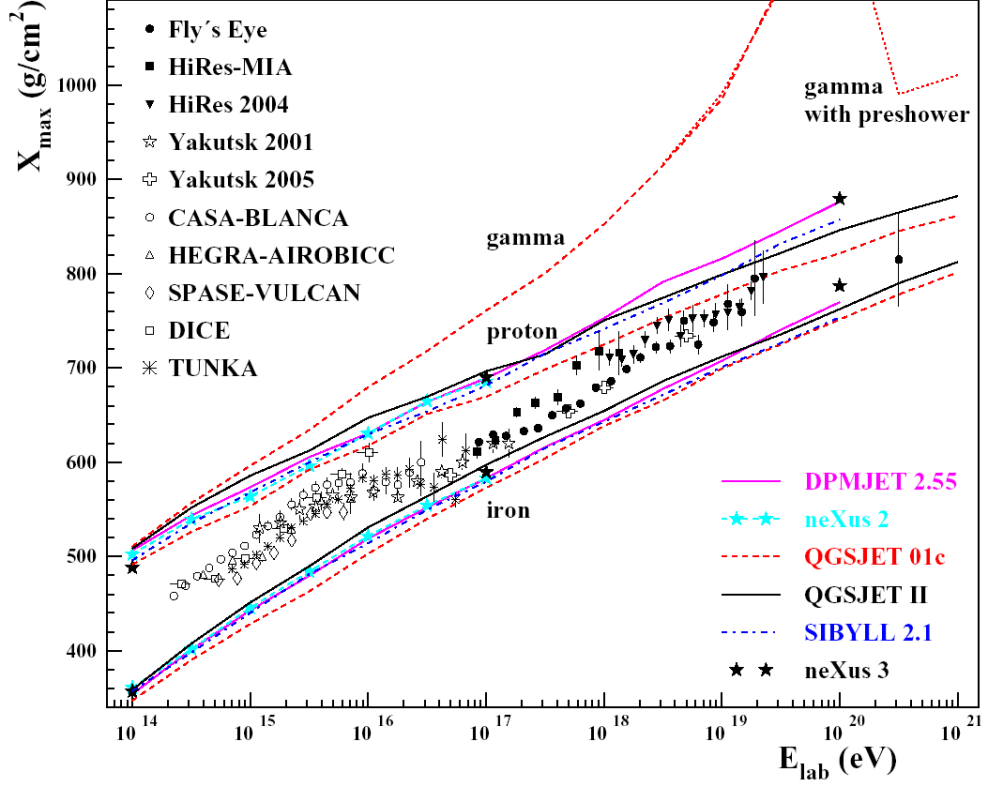


Figure 2.12: Variation of X_{max} with energy measured by different experiments as well as predictions from different models. Figure taken from [72].

The X_{max} parameter is a very powerful observable for mass composition studies but it can only be directly measured using fluorescence telescopes, only operatives during moonless and clear nights. Due to this statistics limitation and also to obtain independent measurements to cross-check the results, observables measured with surface detectors are also used for mass composition studies.

2.7.3 Muon content

It is generally assumed that, at a given primary energy, showers induced by iron nuclei generate a greater fraction of muons at ground level than proton primaries (see Heitler model section). Even when this difference is expected to be enough to separate primaries, the large fluctuations and the differences in the predictions of particular models at energies far away from those confirmed in accelerator experiments make difficult the analysis.



2. ULTRA HIGH ENERGY COSMIC RAYS

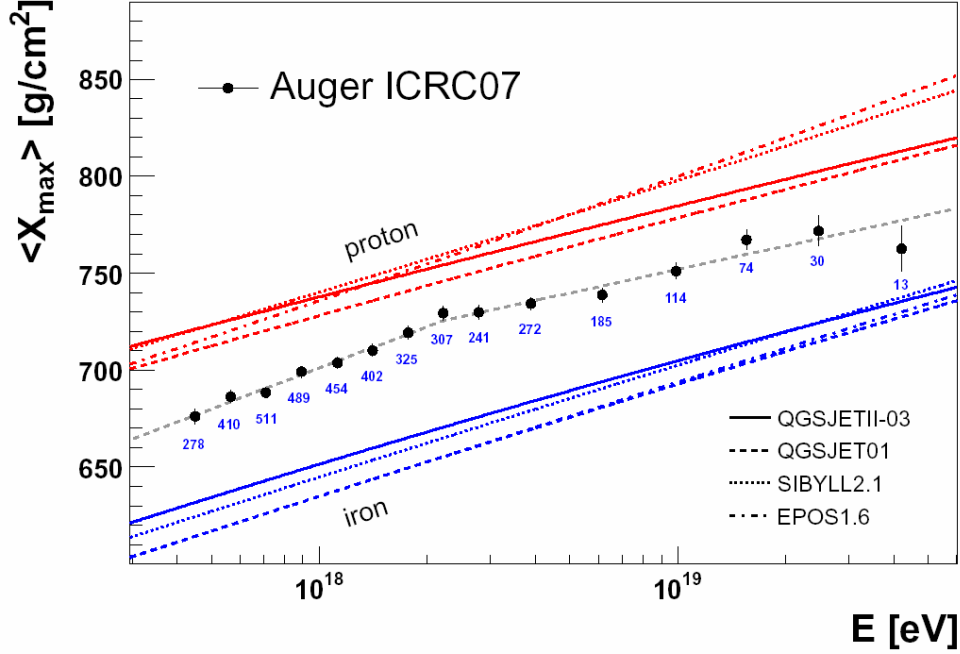


Figure 2.13: Variation of $\langle X_{max} \rangle$ as a function of the energy measured with the Pierre Auger Observatory. Data are compared to predictions from different hadronic interaction models.

Measurements of muon density between 10^{17} and 10^{18} eV carried out with the muon array CASA-MIA [78] predict a changing composition, lighter for increasing energy. The AGASA Collaboration has reported predictions of mass composition from derived measurements of the muon density at 1000 m from the shower core [79] founding a mass composition decreasing with increasing energy between $10^{17.5}$ and 10^{19} eV and consistent with a light component around 10^{19} eV. A major limitation is the accuracy of the measurement, estimated to be around 40%. The hadronic model used for the comparison is also strongly decisive and in this case QGSJET98 was employed.

For higher energies, the Pierre Auger Observatory has the capability to measure the muon content from the time structure of the signals recorded with water Cherenkov detectors [80]. Since the Cherenkov signal is dependent on the particle path length, muons induce peak signals whereas the signal produced electromagnetic particles appears as a continuum spread over large times, i.e. muons generate in the FADC trace isolated and large peaks above a continuous electromagnetic background. The FADC jump $\Delta V(t_i)$ at a given time bin t_i is defined as:

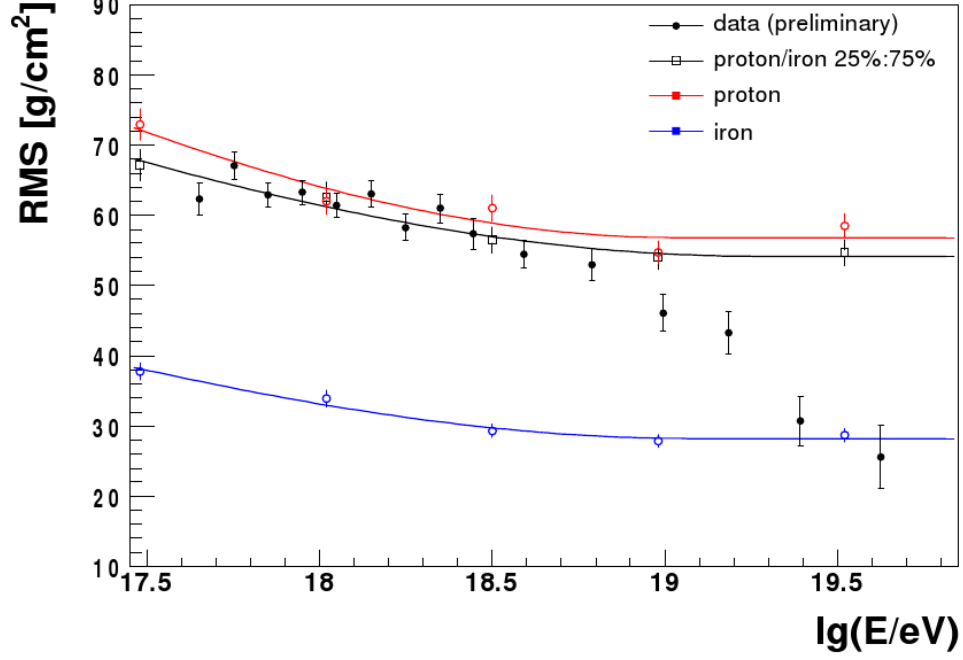


Figure 2.14: Fluctuations of X_{max} as a function of the energy. Data are compared with the predictions the QGSJETII model. Above energies of 10^{18} eV a trend to increasing masses is observed. Figure taken from [77].

$$\Delta V(t_i) = V(t_{i+1}) - V(t_i) \quad (2.16)$$

where $V(t)$ is the FADC signal at the time bin t . The distribution of jumps $\frac{N}{d\Delta V}$ for a sample of real events is shown in figure 2.15 as well as fits to the parameterizations obtained from Monte Carlo simulations for the pure muonic and pure electromagnetic contributions. The number of muons at a given station will be estimated from the area below the curve and dividing this number at a reference distance of 1000 m by the effective detector surface, the muon density at 1000 m $S_\mu(1000)$ is obtained. For interpretations of mass composition, this muon density is compared to predictions for proton and iron induced showers.

2.7.4 Rise time $t_{1/2}$

Due to geometrical effects, velocity differences, multiple scattering deflections, etc., shower particles do not arrive simultaneously at the detector. Muons travel almost in straight



2. ULTRA HIGH ENERGY COSMIC RAYS

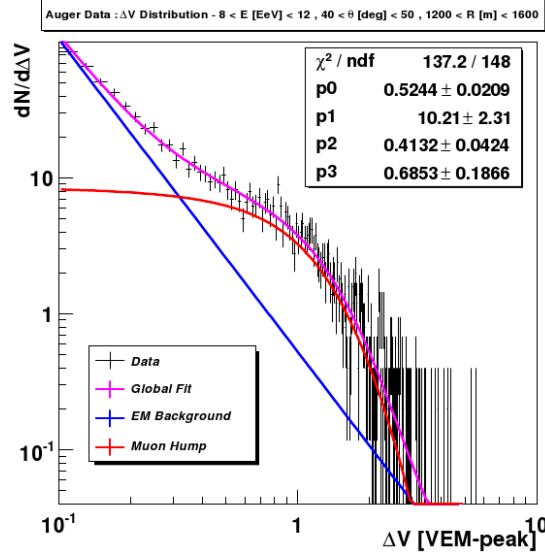


Figure 2.15: Distribution of jumps for events at a given energy, zenith angle and distance to the core range. Lines corresponds to the different fitted functions of muon and electromagnetic signal. Figure taken from [80].

lines whereas the EM component is deflected by multiple scattering. A typical FADC trace generated by a cosmic ray shower is then characterized by a first portion dominated by the muon signal, since the muon component arrives earlier and over a shorter period of time, and an electromagnetic signal with large spread over the whole trace. This shower feature can be measured with the risetime $t_{1/2}$ defined as the time to reach from 10% to 50% of the total integrated signal for each surface detector.

In [81] it was shown that the $t_{1/2}$ fluctuations between showers were larger than expected only from sampling uncertainties. This was the first evidence that $t_{1/2}$ is a mass sensitive parameter. After that in [82] the risetime was calibrated with X_{max} and an elongation rate using the risetime was obtained for energies up to 10^{18} eV. Several studies of $t_{1/2}$ as parameter for mass discrimination have been performed since then. Risetime measurements obtained at Haverah Park for a set of events with an average energy of 10^{19} eV have been reanalyzed and results suggest a large fraction of iron (around 80%) at this energy [83].

Risetime analysis can also be successfully carried out at the Pierre Auger Observatory where the surface stations record the signal in bins of 25 ns as for example in [84] where the risetime at 1000 m from the shower core $t_{1/2}(1000)$ is obtained from the fit:

$$t_{1/2}(r) = 40 + a(\theta) \cdot r + b(\theta) \cdot r^2 \quad (2.17)$$

where $a(\theta)$ and $b(\theta)$ are the fitted parameters. $t_{1/2}(1000)$ as a function of energy is then compared with shower simulations as predicted by different hadronic models (see figure 2.16).

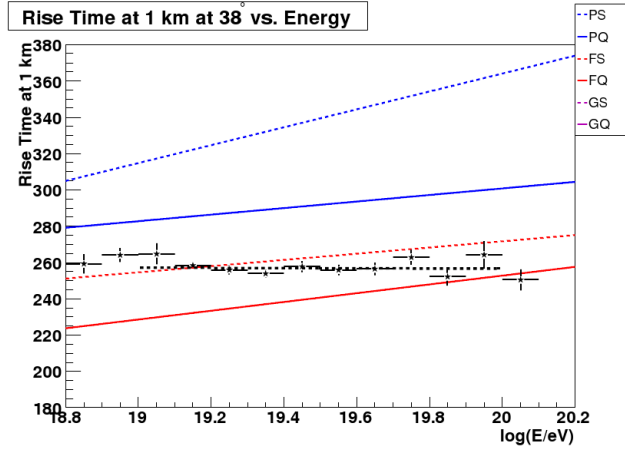


Figure 2.16: Risettime at 1000 m against energy for Monte Carlo predictions (lines) and data (black points) at a fixed zenith angle (38°). Blue = proton, red = iron, solid = QGSJET01, dashed = Sibyll2.1. A dashed black line is the result of a line fitted to the data. Figure taken from [84].

An extensive and detailed analysis on $t_{1/2}$, including fluctuation studies, uncertainties in the measure, detector effects, etc, is being carried out by the Leeds group [85; 86]. Since the risetime is also dependent on the distance of the station to the shower core r and on the zenith angle θ , a new parameter is introduced to eliminate this dependence while the X_{max} dependence remains. This parameter is defined as the average deviation of the risetimes of signals in an event from those evaluated from a fit to the average risetime as a function of core distance for showers at a fixed energy (the benchmark function). The deviation of each signal from the benchmark function is evaluated in terms of the measurement uncertainty $\sigma_{1/2}$:

$$\langle \Delta_i \rangle = \sum_{i=q}^N \frac{t_{1/2} - t_{1/2}^{\text{benchmark}}(r, \theta)}{\sigma_{1/2}(\theta, r, S)} \quad (2.18)$$

where S is the total integrated signal of the station (see figure 2.17(left)). The benchmark function is obtained in the energy range $18.9 < \log E[\text{eV}] < 19.2$ and zenith angles



2. ULTRA HIGH ENERGY COSMIC RAYS

with $\sec \theta < 1.4$ for signals larger than 20 VEM¹ in surface stations with distances to the core up to 1400 m. The fitted function is:

$$t_{1/2}^{\text{benchmark}}(r, \theta) = 10 + \sqrt{a(\theta)^2 + b(\theta) \cdot r^2} - a(\theta) \quad (2.19)$$

The relation of the quantity $\langle \Delta_i \rangle$ with $X_{\max}$ can be established using hybrid events. A linear dependence is found as shown in figure 2.17(right) as well as the fit to data. Using the parameterization of $X_{\max}(\langle \Delta_i \rangle)$ it is possible to measure the maximum depth with surface detector data and its dependence with energy (elongation rate).

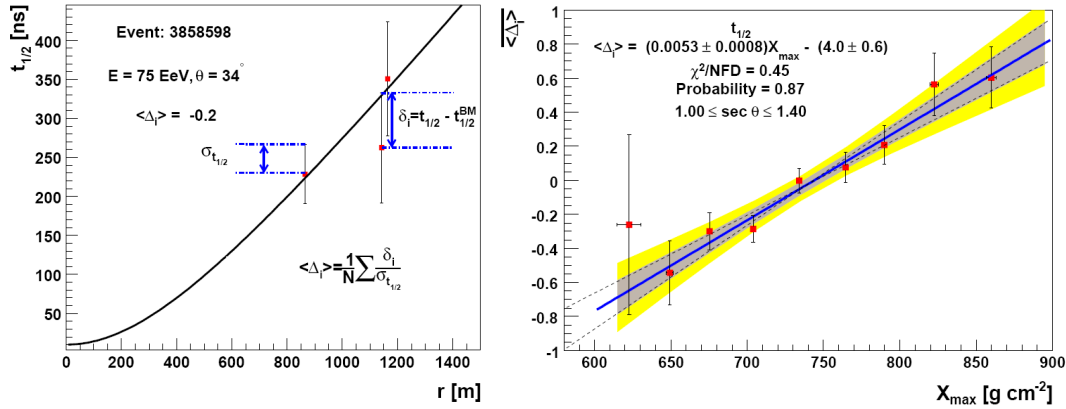


Figure 2.17: Left: Average value of $\langle \Delta_i \rangle$ derived from a single event. The blue line represents the benchmark function and the points the individual measurements of $t_{1/2}$. Right: The average value of Δ_i as a function of $X_{\max}$ for the selected hybrid events. A correlation is seen. Figures taken from [86].

2.7.5 Azimuthal asymmetry in $t_{1/2}$

This technique has been exploited in this work and will be extensively described in subsequent chapters.

2.7.6 Lateral Distribution Function

As will be presented in section 3.4.3.2, the decrease of particle density as a function of distance from the shower axis can be described by a Lateral Distribution Function (LDF). This distribution is determined by the transverse momentum of pions and by the

¹the VEM is the standard unit used in the Pierre Auger Observatory to measure surface signals and will be defined in next chapter.

multiple scattering of the electromagnetic particles. The steepness of this function and also its fluctuations can be used for mass composition measurements since it correlates with the position of the shower maximum X_{max} . Showers developing deeper in the atmosphere are in an earlier stage of development at ground level and therefore EM particles created close to the core arrive this level before they are attenuated.

Detailed measurements of the LDFs were obtained with Haverah Park experiment and have been recently reanalyzed using the QGSJET98 model as reference [87]. The obtained results, based on the assumption of a proton-iron mix, predict a mixture independent of energy in the range $3 \cdot 10^{17}$ to 10^{18} eV with a proton fraction of 34 ± 2 %. John Linsley also obtained precise measurements of the LDF with the Volcano Ranch array. From comparisons with QGSJET98 shower simulations, an iron dominated component is predicted at 1EeV with an estimated iron fraction of $(89 \pm 5(\text{stat}) \pm 12(\text{syst}))\%$ ¹ [88]. Again the systematic uncertainties arising from hadronic models, larger than the statistical ones, impose serious limitations to possible interpretations. If data are evaluated with QGSJET01 the fraction of protons increases around 14% [88].

The array spacing of the surface detector of the Pierre Auger Observatory (1500 m) difficulties the measurement of the steepness of the LDF and its fluctuations on an event by event basis. Nowadays there are plans for the construction of an infill array which should allow to apply this method for mass composition in Auger data.

2.7.7 Relative number of muons and electrons

As previously mentioned, showers induced by iron nuclei generate a greater fraction of muons at ground level than proton primaries with the same initial energy. Furthermore, the number of EM particles at ground for a given energy is dependent on the distance to the shower maximum X_{max} and consequently on the primary mass. As a result, the relative contribution of the EM and muonic component depends on the stage of development of the shower which implies that the muon to electromagnetic ratio at ground level will increase with the primary mass.

At energies around the knee region the KASCADE experiment has determined the composition of cosmic rays by performing a two-dimensional spectrum for the number of muons and electrons at ground level [15]. Results confirm an increase of the mass

¹The systematic uncertainty arising from the lack of knowledge of the energy distribution of events.



2. ULTRA HIGH ENERGY COSMIC RAYS

number with increasing energy but neither of the two simulation models used to compare, QGSJET and Sibyll, are able to reproduce the measured data. This method will be also applied to KASCADE-Grande data in order to extend the results up to 10^{18} eV [89].

2.7.8 Radius of curvature R_c

Since muons travel through the atmosphere almost unaffected by scattering, they are the first particles to arrive to the surface stations. Furthermore, since they are generated at relative early stages they can be modeled as a spherical shower front of particles arising from a point source at the shower axis with a radius of curvature increasing as the shower propagates to the ground and with a narrower time spread than the electromagnetic component. According to this, the arrival of particles at large distances from the axis is expected to be delayed with respect to a hypothetical planar shower front [90]. The arrival times of particles to the surface stations can be fitted in order to estimate the radius of curvature [37].

At a given energy showers generated higher in the atmosphere, corresponding to heavy nuclei with a larger number of muons, will have larger radii of curvature than those induced by lighter nuclei and will produce station triggers more compact in time. This method is also used to search for photon primaries since photon induced showers will have, for a given energy, the smallest radius of curvature [37].

The Pierre Auger Observatory

3.1 Introduction

The Pierre Auger Observatory [91] has been designed to study the properties of UHE-CRs via the EASs they develop in the atmosphere. In order to obtain a full sky coverage, the observatory consists of two sites, one in each hemisphere. The Southern Observatory, already completed, is located near the town of Malargüe, in Mendoza province, Argentina. The Northern Observatory is planned to be constructed in Colorado, U.S.A..

To study with high significance the region above the GZK steepening, where the rate of cosmic rays is extremely low, large collection areas are required. The Southern Observatory consists of an array of 1600 water Cherenkov tanks, known as the Surface Detector (SD), in combination with four fluorescence buildings overlooking the array, the so-called Fluorescence Detector (FD). The SD is arranged on a triangular grid with 1.5 km spacing covering a total area of 3000 km². The Northern Observatory is planned to cover an area even larger of around 20000 km².

This is the so-called hybrid concept, the SD samples the lateral distribution of the shower at ground level while the fluorescence telescopes register the longitudinal development by collecting the faint fluorescence light emitted by the nitrogen molecules excited as the shower crosses the atmosphere. This hybrid operation allows to decrease energy dependency on models while keeping the high statistics and large and stable aperture characteristics of a surface array. A fraction of the detected showers will be



3. THE PIERRE AUGER OBSERVATORY

observed simultaneously by both detectors providing cross-checks and controlling the systematic uncertainties existing in both techniques.

The Surface Detector, always operative, provides high statistics and a well-defined aperture independent of energy, mass composition and hadronic models. The FD only operates during clear moonless nights reducing the duty cycle to approximately 10%. Nevertheless the FD provides many benefits such as a direct measurement of X_{max} , a crucial parameter in composition studies, and a measure of the primary energy nearly model independent.

This chapter starts with a brief introduction to the motivation of the construction of the Pierre Auger Observatory, in particular results provided by previous experiments. In section 3.3 the main features and components of the FD as well as the procedure developed to reconstruct events collected by the FD are briefly illuminated. Section 3.4 is devoted to give a detailed description of the SD. As will be shown in this section, the energy reconstruction of SD events can be performed either using predictions of hadronic models or using a novel technique based on the calibration with the FD. A special section dedicated to the observed asymmetry in the signals recorded by the SD and its possible sources is presented in section 3.6. Finally, the software framework developed by the Pierre Auger Collaboration to analyze cosmic ray events is presented in section 3.5.

3.2 Motivation

The origin and nature of UHECRs as well as the physics behind the acceleration mechanisms and propagation are still not fully understood. Results provided by previous experiments have added even more intrigue to these items. For example, the discrepancies between the spectra measured using different techniques have plagued the cosmic ray field with many theories for the origin of UHECRs and different astrophysical interpretations.

The technique employed to detect cosmic rays is strongly dependent on the energy of the primary. At low energies ($\lesssim 10^{15}$ eV), for which the flux of cosmic rays is high enough, direct detection of the primary CR is feasible. For this purpose particle detectors located on balloon flights, satellites placed in the top of the atmosphere, or even in the Space Shuttle as in the case of AMS [92], are used. For higher energies,

where the flux falls down, indirect techniques are required. Several methods based on the detection of EASs have been developed. These techniques provide the large acceptance needed to measure low fluxes. The features of the EAS can be recorded at ground using arrays of particle detectors, air-Cherenkov light detectors as well as fluorescence telescopes.

Based on the pioneering idea developed by Pierre Auger, Linsley *et al.* constructed in the 1950s [93] the first large scale array of particle counters to sample the spread of shower particles at ground level. This array consisted of 19 scintillators covering an area of 8.1 km². Using this surface array, located in Volcano Ranch, New Mexico, Linsley reported the first measurement of an event with an estimated primary energy over 10²⁰ eV [12]. Several experiments based on surface detectors have followed, gradually increasing the size of the array. Haverah Park (12 km²) [94], Yakutsk (18 km²) [95], SUGAR (70 km²) [96] and AGASA (100 km²) [97], the largest one before the construction of the Pierre Auger Observatory.

Fluorescence detectors collect the near-UV photons released isotropically by N₂ molecules previously excited as shower charged particles dissipate their energy in the atmosphere. This technique, early proposed by Delvaille [98], Suga [99] and Chudakov [100], was firstly implemented in the Fly's Eye detector [101] located in Utah. This detector, which consisted of two fluorescence telescopes separated by 3.4 km, was succeeded by the HiRes detector [102], also in Utah, and composed by two fluorescence stations separated around 12 km.

Up to now, mainly two experiments had provided us with data in the region of extreme energies, AGASA and HiRes, both using different detection techniques. As shown in figure 3.1 the spectra measured by these collaborations show significant discrepancies not only in the absolute flux but also in the prediction of the GZK feature. Figure 3.1, show that the AGASA flux is a factor about 2 larger than the one of HiRes and their measurements predicts a continuous spectrum without evidence of the GZK steepening while HiRes spectrum shows a steepening around 6×10^{19} eV. Both experiments are affected by systematic errors of different nature such as the dependency of the HiRes aperture on the energy and atmospheric conditions or the need to use hadronic models to estimate the primary energy in surface detectors like AGASA. Nevertheless, the discrepancy is not significant since the measurements are dominated by the limited statistics and large systematic uncertainties [103; 104] .



3. THE PIERRE AUGER OBSERVATORY

The Pierre Auger Observatory combines for the first time both detection techniques. This hybrid concept is making possible to control the systematic uncertainties involved in both techniques. The fluorescence telescope reduces significantly the dependency on hadronic interaction models while the surface array allows a large statistics and a stable aperture.

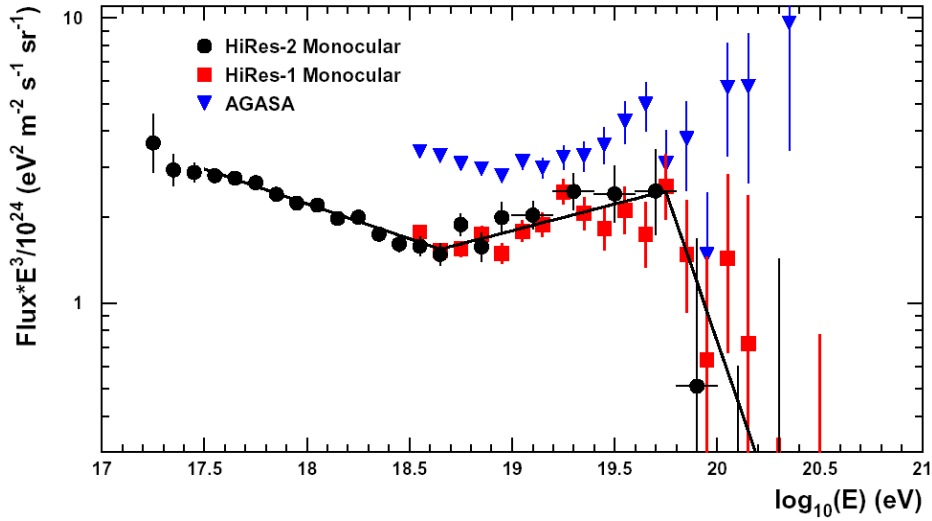


Figure 3.1: The cosmic ray spectrum measured by HiReS and AGASA experiments. The straight lines segments are the results of a fit to a triple power law. Figure taken from [105].

The construction of an observatory with full sky coverage is motivated by anisotropy studies, search of point sources and correlation with astrophysical objects. Since UHE-CRs are weakly deflected in galactic and extragalactic magnetic fields they must point back to their sources. The extremely low rate of the highest energy particles requires large collection areas in order to identify candidate sources. Additionally both locations, when combined, provide a good survey of the galactic center. If anisotropies and the GZK steepening are finally confirmed, different energy spectra could be measured in different regions. A full sky coverage will be essential to characterize the energy spectrum since the distribution of possible nearby sources may not be treatable as a continuum.

Apart from the angular distribution of the arrival directions and the energy spectrum, one of the main objectives in the study of cosmic rays is the determination of the primary composition. Composition studies are crucial to validate models of cosmic ray

propagation, acceleration mechanisms and source candidates. One of the most important parameters in mass composition determination is the X_{max} value, which can be only measured directly using fluorescence detectors. Unfortunately the duty cycle of FD is small and therefore the X_{max} sample is subjected to large statistical fluctuations. Several parameters obtained with the surface array of the Pierre Auger Observatory are being investigated as possible observables for mass composition studies as will be shown in this work. Combining both fluorescence and surface observables more information and sensitivity can be achieved.

3.3 Fluorescence Detector

The Fluorescence Detector FD [106] has been designed to collect the faint fluorescence light emitted by nitrogen molecules excited as the EAS crosses the atmosphere. The FD is composed of four peripheral fluorescence buildings surrounding and overlooking the array of water Cherenkov tanks. The name of the four buildings come from their location in small hills, Los Leones, Los Morados, Loma Amarilla and Coihueco. Each building houses six identical telescopes.

The telescopes are designed following the Schimdt optics technique. Each one consists of a 1.7 m diameter diaphragm aperture where an annular corrector ring made of UV transmitting lenses is located on its outer to increase the effective aperture [107; 108], a spherical mirror of 3.4 m radius of curvature and square shape (3.8m x 3.8m), a spherical PMT camera located on the mirror focal surface and an UV transmitting filter placed at the diaphragm aperture to improve the signal to noise ratio in the fluorescence band between 300 nm and 400 nm [109]. The Schmidt optics eliminates coma aberration and ensures a light spot with a maximum angular size of 0.5, practically independent of the light incident direction.

The PMT camera is an array of 22×20 quasi-regular hexagonal photomultiplier tubes (Photonis XP3062). Each pixel has a size of around 1.5° in such a way that each telescope covers a Field of View (FoV) of 28.6° in elevation and 30° in azimuth. To maximize light collection and to guarantee a smooth transition between adjacent pixels, each PMT is surrounded by an hexagonal set of flat reflecting surfaces. The basic element of this light collector is a *Mercedes* star [110], a simplified version of the classical Winston cone, fixed at the vertex of three adjacent pixels.



3. THE PIERRE AUGER OBSERVATORY

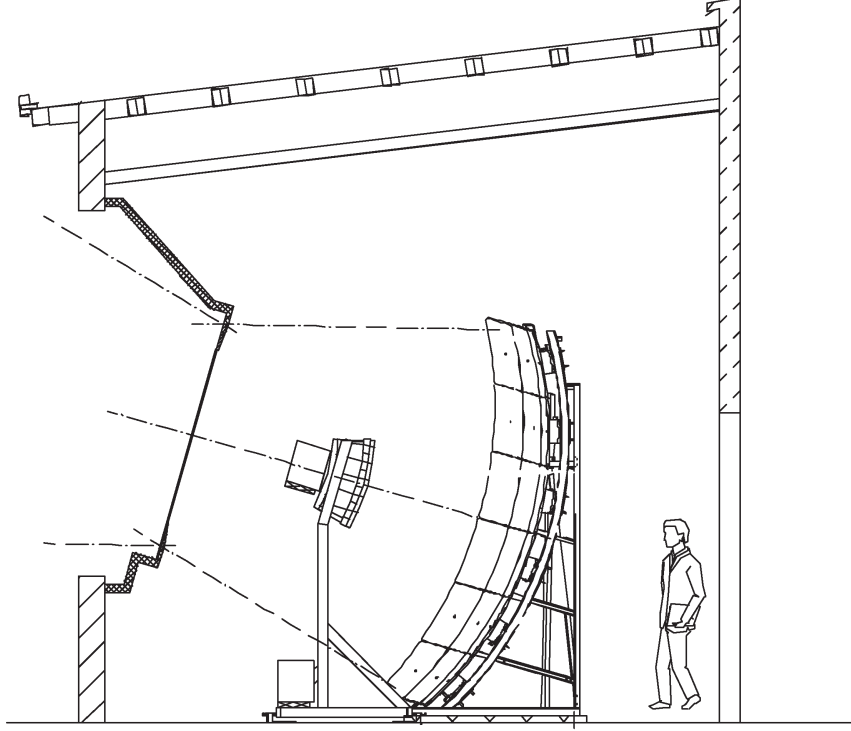


Figure 3.2: Schematic view of a FD telescope. From left to right: shutters, aperture system (including filter and correction ring), PMT camera and mirror.

3.3.1 Data acquisition and trigger conditions

Each PMT tube is equipped with a Head Electronics unit (HE) which drives the analogue signal with low noise and high dynamic range up to one of the 20 front-end boards, each one serving 22 pixels of a camera column, hosted in a crate placed on the floor below the camera.

The signals are digitized continuously by 10 MHz 12 bit ADCs. Four layers of trigger algorithms have been implemented in order to identify event candidates. The digital part of the front-end board is used to implement a First Level Trigger (FLT) which involves decisions at the level of individual pixels. A pixel is marked as triggered if the sum of the digitized signals over 10 consecutive time bins (the bin size is 100 ns) exceeds an adjustable threshold. A Second Level Trigger (SLT) logic is implemented on a separated board which reads the pixel triggers generated for each channel in the

20 FLT boards and searches for geometrical patterns of triggered pixels compatible with the track of a cosmic ray shower. The Mirror PC, a robust, diskless, industry PC associated with each telescope, is used to perform the data readout and the Third Level Trigger (TLT). This software trigger makes a selection, still on telescope level, based on track length and space-time compatibility requirements [111]. After the TLT, data from different telescopes are collected by the Eye PC. Finally a software trigger, the T3 has been implemented on the Eye PC to select shower candidates by performing a fast rough reconstruction which also provides basic shower parameters such as the time when the shower hits the ground or the position of the impact point. This basic information is sent to the CDAS to test for time compatibility with surface stations and to construct a hybrid trigger.

3.3.2 Detector Calibration and Atmospheric Monitoring

An absolute calibration of the fluorescence telescopes is essential to ensure a high quality reconstruction of the primary energy. The measurement of the conversion factor of the FADC counts recorded by the PMTs to number of photons entering the aperture is based on the uniform illumination of the pixels from a diffuse 375 nm light source mounted directly in front of the telescope aperture [112]. Using this end-to-end calibration technique, the signal from a cosmic ray shower recorded by PMTs multiplied by the corresponding calibration constants yields the number of photons entering the diaphragm without needing to know details of the optics or electronics. Besides this absolute calibration, a relative calibration of the PMT is also performed before and after each night of data-taking to study the stability of the fluorescence telescopes [113]. Night to night fluctuations and seasonal trends have been identified and used to make corrections to the absolute calibrations [114].

At present, the systematic uncertainty assigned to the calibration constants, arising from different contributions of the calibration performance, is 9.5% [114]. The absolute calibration has been cross-checked using a portable laser system and results are compatible with the current systematic uncertainties [115].

Besides the optical calibration of the fluorescence telescopes, a good understanding of the atmosphere is essential to obtain a reliable estimate of the primary energy. The Pierre Auger collaboration has developed an extensive array of atmospheric monitoring devices to study the aerosol content, cloud coverage and vertical profiles of temperature



3. THE PIERRE AUGER OBSERVATORY

and pressure. These network includes meteorological radio soundings [116], LIDAR stations [117], cloud cameras, the star monitor FRAM (Photometric Robotic Atmospheric Monitor) [118], and laser-based systems such as the CLF (Central Laser Facility) [119], APF (Aerosol Phase Function) monitors and HAMs (Horizontal Attenuation Monitor) [120].

Data are stored in SQL databases to make them easily accessible during the reconstruction procedure.

3.3.3 Event Reconstruction using the FD

As for SD events, information collected by the FD can be used to reconstruct relevant properties of the primary particle following a two-step process. Firstly, the geometrical reconstruction is performed. A substantial improvement in the geometry is achieved when timing information from surface stations is used in the procedure. In the second step, the primary cosmic ray energy and other relevant parameters are obtained.

3.3.3.1 Geometrical Reconstruction

The geometry of an event registered by the FD is completely defined by the Shower Detector Plane (SDP) and the position of the axis direction within the SDP (see figure 3.3). The SDP is defined as the plane containing the shower axis and the position of the triggered fluorescence building. Since the SDP is seen as a line on the camera, it is estimated by minimizing the angular distances from pointing directions of triggered pixels $\bar{n}_i$ to this plane [121]. The next step in the reconstruction of the shower geometry is to determine the direction of the shower axis within the SDP using timing information. To obtain a precise estimate of the shower axis is very difficult using only information from one fluorescence telescope. The geometry of the shower axis is fully described through the impact parameter R_p , the distance from the shower axis to the telescope, and the inclination of the axis χ_0 (figure 3.3). Both parameters are obtained fitting the arrival times t_i of the fluorescence light to triggered pixels where, depending on shower configuration, a range of (R_p, χ_0) solutions are compatible. To break this degeneracy and find an unique solution timing information provided by surface stations is combined with FD data. The arrival time of the shower to a surface station impose a time constraint and helps to fix the core position. An example is presented in figure 3.4.

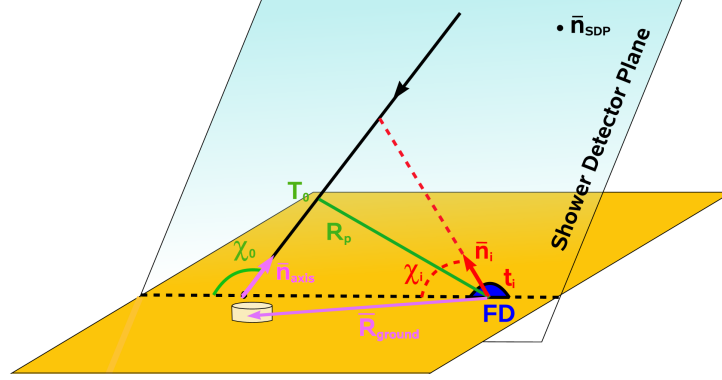


Figure 3.3: Scheme of the geometry of a cosmic ray induced shower. The SDP and R_p , χ_0 and T_0 parameters are shown as well as pixel information $(\bar{n}_i, t_i, \chi_i)$. Figure taken from [115].

The accuracy of the hybrid reconstruction has been studied using Monte Carlo simulations and artificial showers generated by laser shots. The uncertainty on the geometry reconstruction depends on the specific shower geometry but the mean accuracy for hybrid events is about 0.6° [122]. The great improvement achieved in the geometrical reconstruction when timing information from a SD station is used, is illustrated in figure 3.4.

3.3.3.2 Shower Profile and Energy Reconstruction

Light arriving at diaphragm aperture and producing signal in the PMTs has two major contributions: fluorescence light emitted by excited nitrogen molecules and Cherenkov radiation emitted by shower charged particles. During the path of the light to the telescope, photons are affected by both Rayleigh and Mie attenuation due to the scattering with atmospheric molecules and aerosols respectively [123].

The amount of fluorescence light emitted isotropically at a given atmospheric depth is directly proportional to the energy deposited by shower particles at this given level. Furthermore the number of Cherenkov photons, dependent on the number of charged particles, can be considered also proportional to the deposited energy [124].

Light collected from PMTs is firstly converted to the equivalent number of photons at diaphragm as a function of time using the absolute calibration described in section 3.3.2. Using the shower geometry, the fluorescence spectrum of light, the response of the detector at different wavelengths, the contributions of molecular and aerosol



3. THE PIERRE AUGER OBSERVATORY

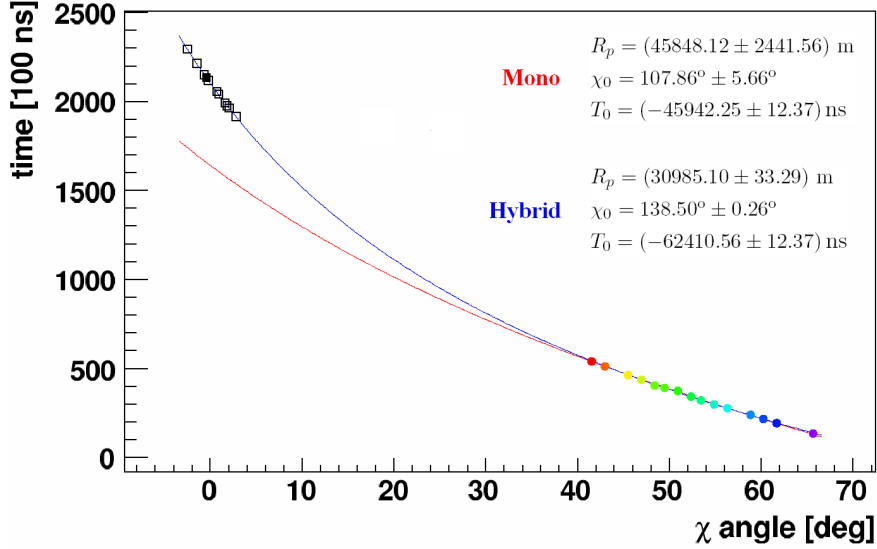


Figure 3.4: Example of a typical event reconstructed using only the FD information (red line) and using also timing information provided by the SD (blue line). Fitted parameters are shown to emphasize the improvement of the hybrid technique. Figure taken from [115].

attenuation and the fluorescence yield¹, this measured number of photons is converted to energy deposited by shower particles as a function of the atmospheric depth. In figure 3.5 the obtained profile of deposited energy of a typical Auger event is shown.

The total energy deposited by the electromagnetic component of the shower can be obtained integrating the energy deposited at each level in the rare case of the full shower profile is observed. Usually, the limited FoV of the telescopes does not allow full observation and therefore the extrapolation to depths outside the FoV is required. The function selected to parameterized the energy deposited as a function of the atmospheric depth X is the well-known Gaisser-Hillas function:

$$f_{GH}(X) = \frac{dE(X)}{dX} = \frac{dE(X)}{dX} \Big|_{X_{max}} \cdot \left(\frac{X - X_0}{X_{max} - X_0} \right)^{(X_{max} - X_0)/\lambda} \cdot \exp \left(\frac{X_{max} - X}{\lambda} \right) \quad (3.1)$$

where X_{max} is the atmospheric depth where the shower reaches its maximum number of particles, $\frac{dE(X)}{dX} \Big|_{X_{max}}$ is the total energy deposited at an atmospheric depth X_{max} , X_0 is the depth corresponding to the first interaction and λ is the interaction length.

¹The fluorescence yield is the number of fluorescence photons per unit of deposited energy [125].

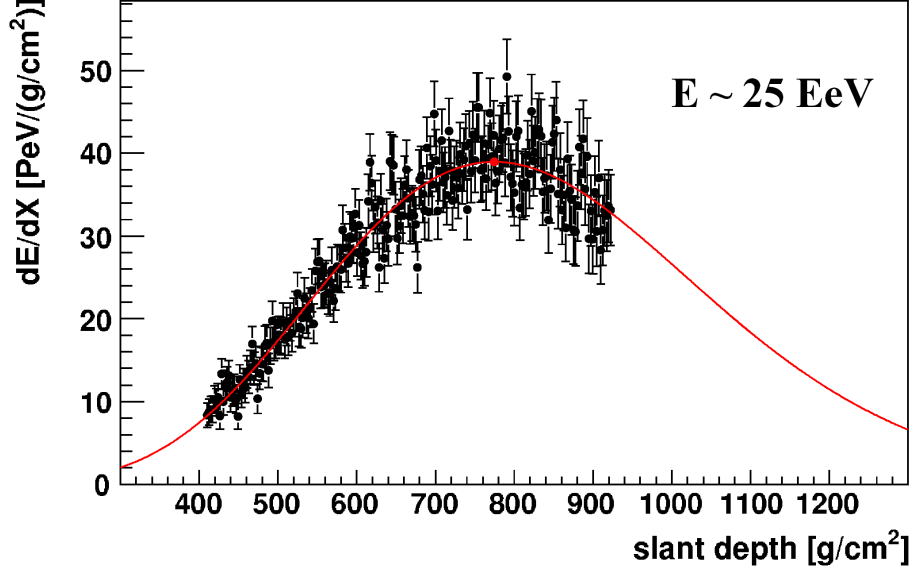


Figure 3.5: Typical longitudinal profile of an event with an energy around 25 EeV observed using the Auger fluorescence telescopes. The red solid line is a fit to a Gaisser-Hillas function.

All these parameters are obtained by minimizing the difference between the expected and observed number of photons.

The total electromagnetic energy E_{em} is then obtained by integrating the Gaisser-Hillas function over the total profile:

$$E_{em} = \int_0^{\infty} f_{GH}(X) dX \quad (3.2)$$

Although most of the primary energy is transferred to the electromagnetic component, a significant fraction is carried by undetected neutrinos and muons that need long path lengths to release their energy. The fraction of this “invisible” energy has been parameterized using shower simulations as a function of the electromagnetic energy E_{em} [126]:

$$\frac{E_{em}}{E_0} = a + b \cdot E_{em}^c [\text{EeV}] \quad (3.3)$$

where E_0 is the estimated primary energy, and a , b and c are constants dependent on the primary mass and interaction model assumed.



3. THE PIERRE AUGER OBSERVATORY

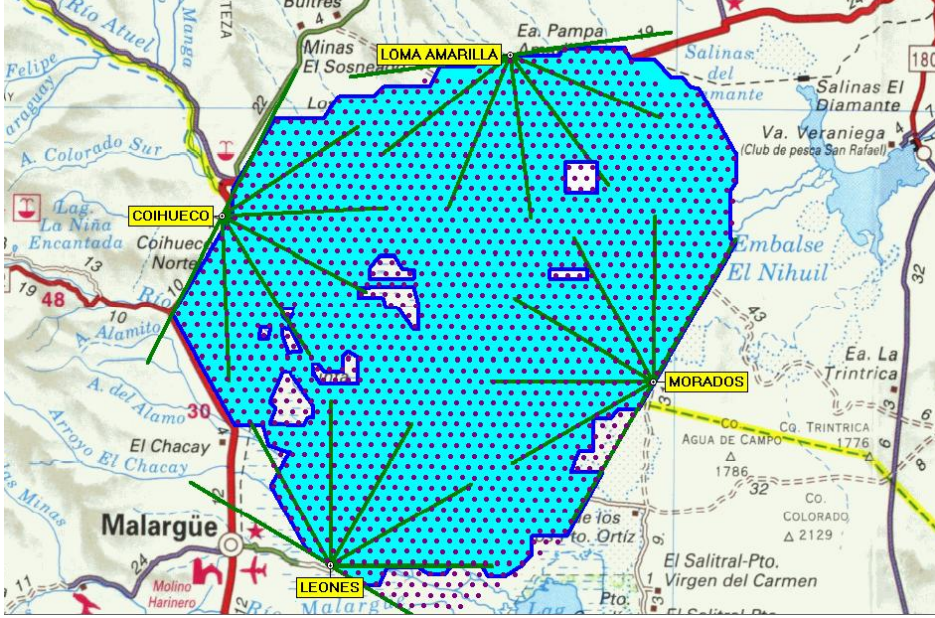


Figure 3.6: Schematics of the surface array. Surface detectors are represented as purple points. The light blue zone marks the status at 13th June of 2008. A few holes remain.

The statistical uncertainty on E_0 has several contributions: those arising from the statistical uncertainties assigned to the Gaisser-Hillas fit, the uncertainty on the shower geometry, atmosphere description and invisible energy [123]. Additionally, a 22% of systematic uncertainty arising also from different contributions, fluorescence yield, calibration constants, etc, affects the absolute energy scale of the Observatory [45].

Besides an energy estimate of the primary energy, the shower profile reconstruction also provides us with a measurement of the X_{max} parameter, very useful for composition studies.

3.4 Surface Detector

The Surface Detector (SD) [127] of the Pierre Auger Observatory consists of an array of 1600 water Cherenkov tanks arranged on a triangular grid with a 1.5 km spacing and covering a total area of 3000 km². A schematics of the array is shown in figure 3.6.

The elemental unit of the SD is a water Cherenkov tank as shown in figure 3.7. Water Cherenkov detectors have been selected against scintillators due to several reasons



Figure 3.7: Picture of a Surface Detector unit.

[91]. Since water-Cherenkov detectors are relatively thicker than scintillators, the first ones are more sensitive to showers arriving at large zenith angles, thereby increasing the sky coverage. Additionally, an array based on this technique can operate for a long period with both high stability and low maintenance.

Cherenkov light is generated by charged particles moving at speed larger than the one of the light in the medium. Assuming a refractive index n , Cherenkov light is generated in a cone with an opening angle $\theta = \arccos(1/n\beta)$. The shower particles at the ground level are mainly photons, electrons and muons with mean energies below 10 MeV for electromagnetic particles and about 1 GeV for muons. Shower charged particles i.e., electrons and muons, generate Cherenkov light in water while photons are converted via Compton scattering and pair production into relativistic electrons which then generate Cherenkov light.

Each Cherenkov detector employed at the Pierre Auger Observatory consists of a rotationally moulded polyethylene tank 3.6 m diameter and 1.55 m height, enclosing a liner filled with 12000 l of high purity water. The liner is a cylindrical plastic bag with a height of 1.2 m which acts as a secondary seal against extraneous light sources. The inner Tyvek[®] surface gives excellent diffuse reflectivity of Cherenkov light. To



3. THE PIERRE AUGER OBSERVATORY

the outside the liner is black and has been specially designed to provide a water and light barrier.

Tanks are equipped with three 9" XP1805 Photonis photomultiplier tubes (PMTs) [128], set into the top surface of the liner and observing the volume of water through three clear plastic windows. Two signals are extracted from each PMT, from the last dynode and from the anode. The dynode signal is amplified by a factor of 32 to match the dynamic range. The anode signal will be used for large signals, typically those seen by tanks relatively close to the core. The readout of the six signals from each tank is digitized at 40 MHz (25 ns bins) using 10 bit Flash Analog to Digital Converters (FADCs) [129]. All the electronics is mounted locally at each station and is contained in a box situated on the top of tank. The local electronics is controlled by a CPU which also control a Programmable Logic Device used to select signals according to the trigger conditions implemented for individual stations. Timing synchronization, crucial for determining the shower direction, is achieved using a commercial Motorola GPS receiver which provides the event time with a precision of around 8 ns.

Power is supplied by two solar panels connected to two 12 V batteries which are situated in a box in the shadow of the tank. A communication antenna connects the tank to the Central Data Acquisition System (CDAS).

Each detector is autonomous and can operate independently of other detectors in the array. Since January 2004 the array has been in stable operation with a steady growing until completion in June 2008.

3.4.1 Calibration of the Surface Detector

Cherenkov radiation generated by shower particles is detected by the three PMTs giving rise to a digitalized signal. The electronic signal generated by a shower particle in a given PMT does not depends only on the light deposited by the particle but also on several parameters such as water quality, PMT response or liner reflectivity. Therefore, equal light deposited by a particle in a water tank or even in the same tank at different times will not generate usually the same signal. The signal measured by each tank must be normalized to a reference calibration unit in order to avoid tank to tank variations or time dependence. The large number of detectors and their separated locations require a simple, robust and automatic calibration procedure.

The selected standard unit is the Vertical Equivalent Muon (VEM or Q_{VEM}), defined as the signal produced by a vertical and central through-going (VCT) muon. The goal of the calibration procedure is to obtain the value of 1 VEM in electronic units. For this purpose, the signals of atmospheric muons, a well understood uniform background across the array, are employed.

Since the SD in its normal configuration has no way to select VCT muons, a reference tank equipped with two movable scintillators has been used to establish the relation between the signal of a VEM and the peak of the histogram obtained from omni directional atmospheric muons crossing the tank (Q_{VEM}^{peak}). Figure 3.8 (left) shows the charge histogram from a SD station, the signal from all 3 PMTs summed, originated by background muons (black solid line) and the charge histogram generated by VCT muons (red dashed line) by coincidence between the two scintillators of the reference tank. The conversion factor between the peak in the charge histogram for background muons and the charge corresponding to VCT muons is a stable parameter independent on temporal tank properties. The peak of the charge histogram corresponding to the sum of the 3 PMT is approximately at 1.09 VEM, i.e. $Q_{VEM}^{peak} = 1.09Q_{VEM}$ and at 1.03 ± 0.02 for each PMT.

To maintain a uniform trigger condition for the whole array, atmospheric muons are also used to established a common reference unit for threshold levels. In this case pulse height histograms (figure 3.8 (right)), those storing the maximum values of the measured FADC are used. Atmospheric muons also produce a peak in this histogram (I_{VEM}^{peak}) which is used as a common reference unit for threshold levels. Additionally, the initial end-to-end gains of the 3 PMTs, i.e I_{VEM}^{peak} , must be roughly equivalent to ensure that the signals recorded from the 3 PMTs are similar in amplitude.

Finally, the calibration procedure must also be able to determine the baselines of all the six FADC inputs and the gain ratio between the dynode and the anode to convert the raw FADC traces into integrated channels.

The calibration is operated online every minute, and sent to CDAS every 6 minutes for monitoring. In addition, high statistics histograms of the charge distribution from muons and average pulse shapes are sent every time a station records an event. Further details about the calibration procedure can be found in the references [130; 131; 132].



3. THE PIERRE AUGER OBSERVATORY

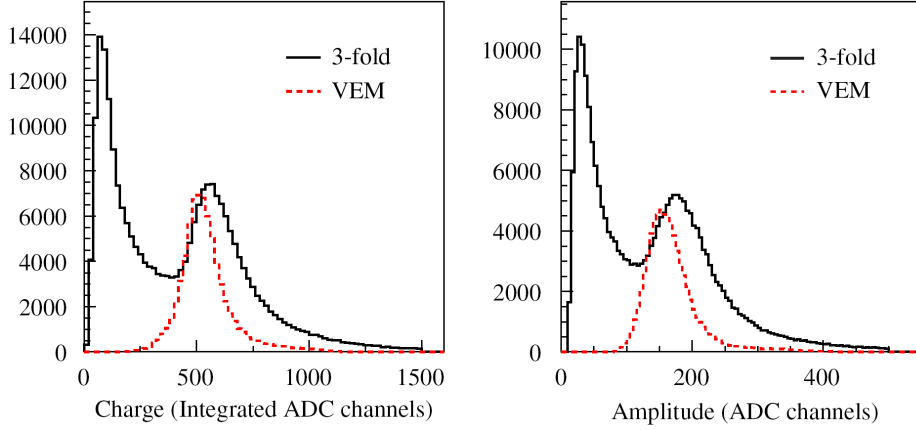


Figure 3.8: Charge and pulse height histograms from an SD station with the signal from all 3 PMTs summed. The dashed histogram is produced by an external muon telescope providing the trigger to select only vertical and central muons. The first peak in the black histogram is caused by low energy particles. The second peak is due to vertical through-going atmospheric muons. Figure taken from [130].

3.4.2 SD Trigger System

Data collected in tanks and other relevant quantities such as baseline values, charge and pulse histograms, dynode/anode ratio, PMT stability, are sent via radio connection. A wireless local area network (LAN) is used to communicate the tanks with four local communication towers located at each fluorescence site. From there, the data are routed through a high capacity microwave link to CDAS where the data storage is organized.

The trigger system [133] has been designed to select with high efficiency ($> 95\%$) events induced by cosmic rays with energies above 10^{18} eV. It consists of five levels of triggers with the first two of them (T1 and T2) implemented at the tank, the so-called low level triggers. The third level trigger T3 is performed by the CDAS based on time and space coincidences of signals from different tank configurations passing T2 level. The highest level triggers, T4 and T5, are implemented offline and are used to select quality physical events.

3.4.2.1 Low level triggers

- T1: The trigger system implements either of two different requirements for the T1 level to identify tanks that could be relevant in the event reconstruction. The

first one is a simple threshold trigger (T1 threshold) satisfied when the signal from all 3 PMTs exceeds a given threshold, $1.75I_{VEM}^{peak}$, at least in one FADC bin (see figure 3.9a). This trigger is quite noisy, with a rate of about 100 Hz, but it is needed to identify fast signals corresponding to the muonic component of highly inclined showers. The second possibility is a Time-over-Threshold (T1 ToT) trigger requiring that the signal from at least 2 out of the 3 PMTs exceeds a threshold of $0.2I_{VEM}^{peak}$ in 13 time bins within a time window of 120 bins (a 3 μ s window). This second condition, with a rate of about 1.6 Hz, is designed to identify small but spread signals like those induced by high energy showers far from the shower core or by low energy showers.

- T2: The second level trigger is also processed by the local tank software. All T1 triggers with the ToT flag are directly promoted to the second level T2 (T2 ToT, see figure 3.9b) whereas the signals from T1 threshold triggers are requested to exceed a higher threshold of $3.2I_{VEM}^{peak}$ in triple coincidence to be accepted as T2 trigger (T2 threshold, see figure 3.9c). This selection criteria reduces the trigger rate to about 20 Hz. Only T2 triggers are considered by the central system to identify groups of stations that are clustered in time and space to form a T3 trigger. T1 stations are included in a T3 event only if they are in time coincidence with it.

3.4.2.2 High level triggers

- T3: This trigger, implemented within the CDAS, requires a certain compactness to tanks passing the T2 level and in time coincidence which could be produce by a single shower. Whenever a station has a T2 trigger, it sends a signal to the CDAS with the registered trigger time. All stations triggered within a time window of 50 μ s centered at the considered station are examined for spatial correlation.

The T3 trigger deals with two different configurations. In one of the T3 modes at least three T2 triggered stations are required around the considered station. At least one of them has to be in one of the six adjacent stations that are closer to the triggered tank used as the time window center (the first ring in figure 3.10). At least two stations have to be within the second ring. The third one has to



3. THE PIERRE AUGER OBSERVATORY

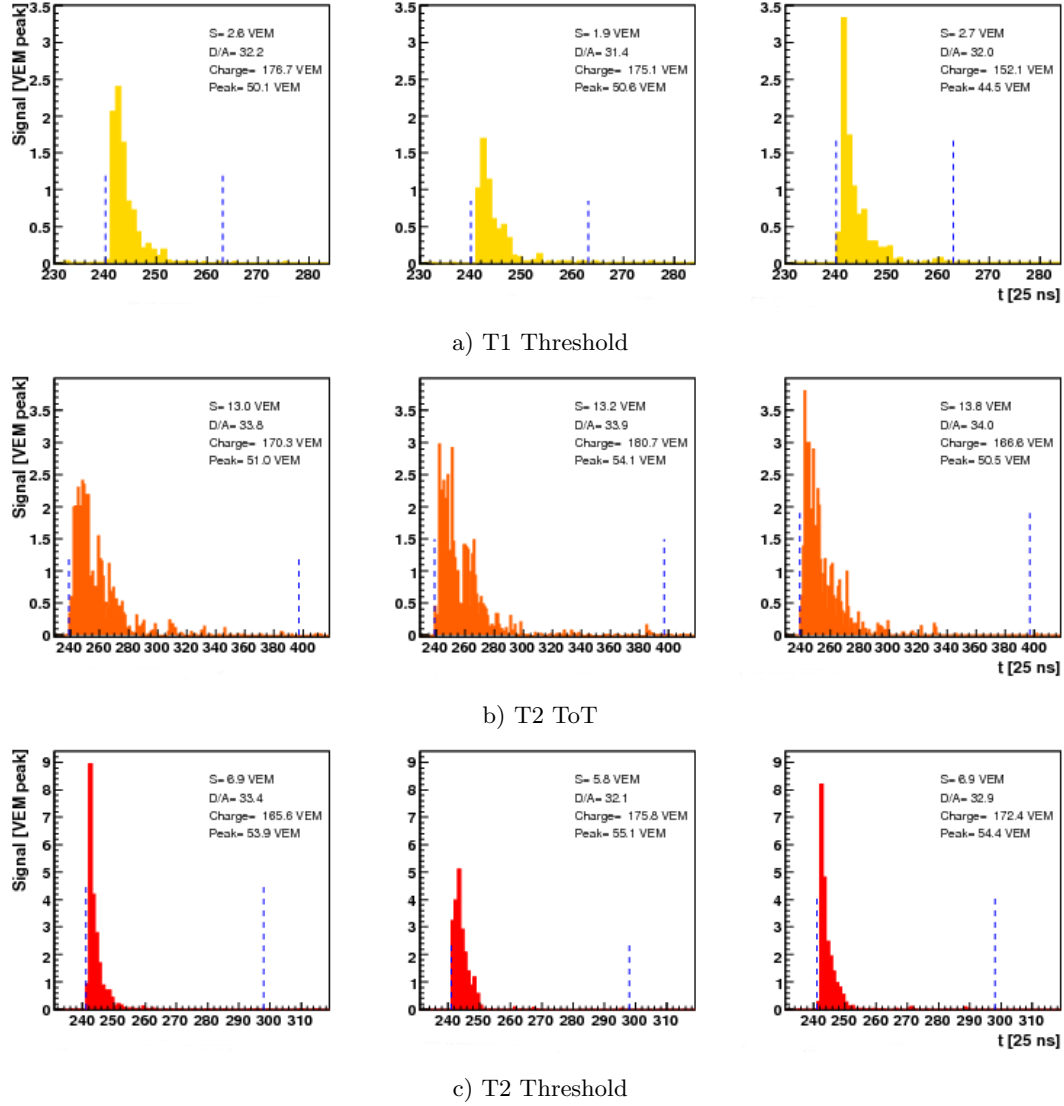


Figure 3.9: Traces of the 3 PMT signals of a tank. a) Satisfying the T1 threshold condition. b) Satisfying the T2 ToT condition. c) Satisfying the T2 threshold condition.

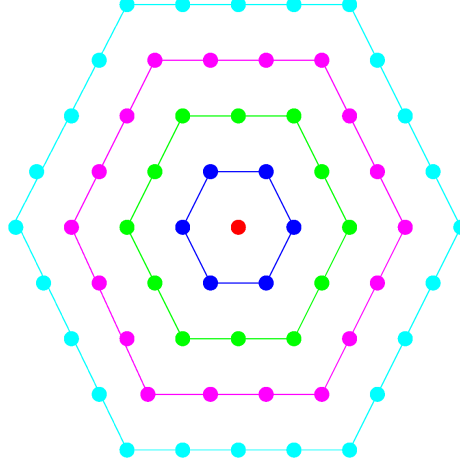


Figure 3.10: The hexagon geometry around the central station used for T3 trigger decisions. Figure taken from [134].

be within the fourth ring. This is known as the 2C1&3C2&4C4 condition where mCn means m triggered stations within the nth ring. This condition is not very efficient for vertical showers (not very inclined showers) since only a few percent correspond to real showers but it is very useful to detect horizontal showers that generate fast signals and have wide-spread topological patterns.

The main T3 trigger condition requires at least three T2 stations in time coincidence that have also passed the ToT condition. They have to fulfill the requirement of a minimum of compactness: the 2ToTC1&3ToTC2 condition, i.e 2 stations within the first ring and 3 within the second one, all passing the ToT condition. This T3 trigger is extremely efficient since 90% of the selected events are real showers.

If the required patterns are found, i.e. the T3 condition is satisfied, the search stops, an event with the T3 flag is formed and stations from the whole array with trigger times in coincidence with the central station are also stored in the event file. Trigger times are considered to be in coincidence if they are within a time window of $(6+5n)\mu s$ from the central triggered station where n indicates the ring number.

- T4, the physical trigger: it has been developed to separate physical events from



3. THE PIERRE AUGER OBSERVATORY

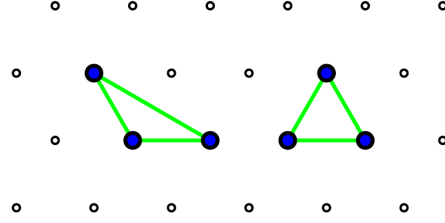


Figure 3.11: The two possible 3ToTC1 configurations. All symmetry transformations of the triangular grid are also valid. Figure taken from [135].

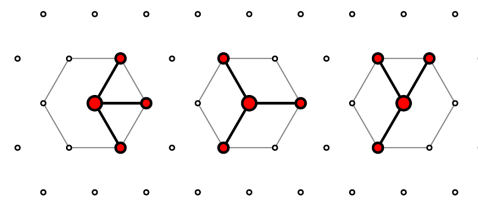


Figure 3.12: The three (minimal) 4C1 configurations. All symmetry transformations of the triangular grid are also valid. Figure taken from [135].

random coincidences in an off-line procedure. Again two different modes have been implemented. The first T4 condition requires at least 3 ToT stations forming a triangle of first neighbours, the 3ToTC1 condition (figure 3.11). This condition is optimized to select vertical showers. In fact, requiring this simple compact configuration more than 99% of the selected candidates are showers with zenith angles below 60° .

The second T4 condition requires a compact configuration of 4 tanks with a T2 trigger each, i.e. at least one fired station has 3 triggered tanks out of its 6 first neighbours, the 4C1 condition (figure 3.12). This condition has been implemented to recover the small fraction of vertical showers lost with the 3ToTC1 condition and also to select events above 60° .

Finally, once the spatial coincidence is verified, compatibility in time between stations is also required to avoid accidental triggered stations. Triggered times must be compatible, within a defined time window, with a plane shower front propagating at the speed of light.

- T5, the quality trigger: This trigger condition has been implemented to filter out events for which the reconstruction may be deficient, mainly due to a missing signal, since the events fall too close to the edge of the array or in a faulty tank. The T5 level requires that the station with the largest signal must be surrounded by 6 working stations at the time of the shower impact. This final trigger ensures that the energy and angular reconstruction can be performed with sufficient accuracy. Furthermore, this quality trigger ensures an easy way to calculate the acceptance of the detector.

3.4.3 Event Reconstruction using the SD

The reconstruction of events detected by the SD is performed in two steps: the geometrical reconstruction, mainly the arrival direction, and the reconstruction of the energy of the primary particle. Timing information from pulses in the Cherenkov tanks are used to reconstruct the arrival direction of the shower and the signal size at a reference distance is used to estimate the energy of the primary cosmic ray ¹. The estimate of the primary energy starts with the obtention of a ground observable parameter, in this case the signal size at 1000 m from the shower core. The conversion between this estimator and the primary energy can be obtained by two different ways: from parameterizations based on shower simulations or from a calorimetric calibration by means of the FD.

Apart from the incoming direction and the energy, another relevant parameter of primary cosmic ray is the mass. As will be discussed later, timing information of the SD is very valuable for the analysis of the composition.

3.4.3.1 Geometrical Reconstruction

An EAS can be considered as a front of particles propagating with the speed of light c along a straight line determined by the arrival direction of the primary particle. In a first step this shower front is approximated by a planar disk perpendicular to the shower axis direction $\vec{a}$. Accidental triggered stations are removed from physical events (i.e those events passing the T4 level) by requiring an arrival time compatible with a planar shower front traveling at the speed of light. A detailed description of the criteria designed to select only stations belonging to the air shower can be found in [136].

¹The signals are also used to obtain the impact point at ground level, i.e. the shower core position.



3. THE PIERRE AUGER OBSERVATORY

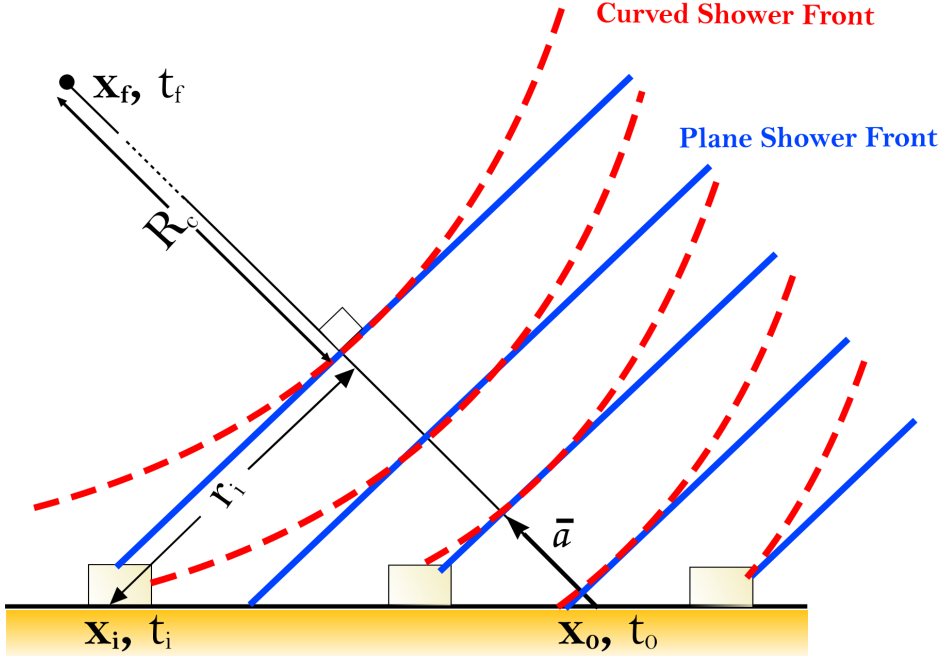


Figure 3.13: Shower geometry including parameters used in the SD reconstruction.

This first approximation to a planar shower front is used to determine the axis direction, given by the normalized vector $\vec{a}$, using an algorithm based on timing information. In this approximation the expected $t(\vec{x}_i)$ at which the shower plane is passing through some chosen station located at $\vec{x}_i$ can be inferred as (see figure 3.13):

$$c(t_0 - t(\vec{x}_i)) = (\vec{x}_i - \vec{b}) \cdot \vec{a} \quad (3.4)$$

where $\vec{b}$, which is a first estimate of the shower core position $\vec{x}_0$, is set as the signal-weighted barycentre of triggered stations and t_0 is the arrival time of the estimated core position.

The function to be minimize is the sum of squares of the time differences between the measured signal start time t_i and the predicted one $t(\vec{x}_i)$ given by equation 3.4:

$$\chi^2 = \sum_i \left[\frac{t_i - t(\vec{x}_i)}{\sigma_{t_i}} \right]^2 \quad (3.5)$$

where σ_{t_i} is the station variance in the start time t_i [137]. This equation has three free parameters, two components of $\vec{a}$, the third component is given by the normalization

constraint, and t_0 . An approximate linear solution can be obtained if the z-component of the stations is neglected [135].

The more realistic approximation is obtained assuming a spheric (curve) shower front with radius of curvature R_c (figure 3.13). In this case the expected arrival time $t(\vec{x}_i)$ will be given by:

$$c(t_0 - t(\vec{x}_i)) = (\vec{x}_i - \vec{x}_0) \cdot \vec{a} + \frac{\rho_i^2}{2R_c} \quad (3.6)$$

where $\rho_i = |\vec{a} \times (\vec{x}_i - \vec{b})|$ is the perpendicular distance to the axis. A first approximation to R_c is obtained assuming that the time propagation of the shower front is described as an expanding sphere centered at the position of the first interaction, without any assumption about the ground impact point [135]. Since the position of the shower core is required, the curvature fit is only performed after the LDF fit described in the next section which provides the core position.

3.4.3.2 Energy Reconstruction via the LDF

In the Volcano Ranch experiment the energy of an event was obtained from the total number of particles, this last one estimated by integrating the signal over all distances using an empirically determined Lateral Distribution Function (LDF). However, the large fluctuations in the shower development and inaccuracies in the assumption of the LDF resulted in large uncertainties in the determination of the energy. To avoid this problem, Hillas [138] proposed to use the signal at some reference distance from the core for which the particle density is subjected to smaller fluctuations than those of the total number of particles.

On the other hand, in [139] it is demonstrated that at this optimal core distance r_{opt} , which depends mainly on the spacing of the array, the fluctuations in the expected signal due to a lack of knowledge of the LDF are minimized. For example, at the Haverah Park instrument an optimal core distance of 600 m was used. For the array spacing used in the Pierre Auger Observatory (1500 m) the optimum distance has been found to be a 1000 m. This is shown in figure 3.14 where a single simulated event is reconstructed 50 times using the LDF given by equation 3.7 with different values of the slope parameter β . As can be appreciated, the different parameterizations used to describe the expected signal crossing at a distance close to 1000 m. Finally, it should



3. THE PIERRE AUGER OBSERVATORY

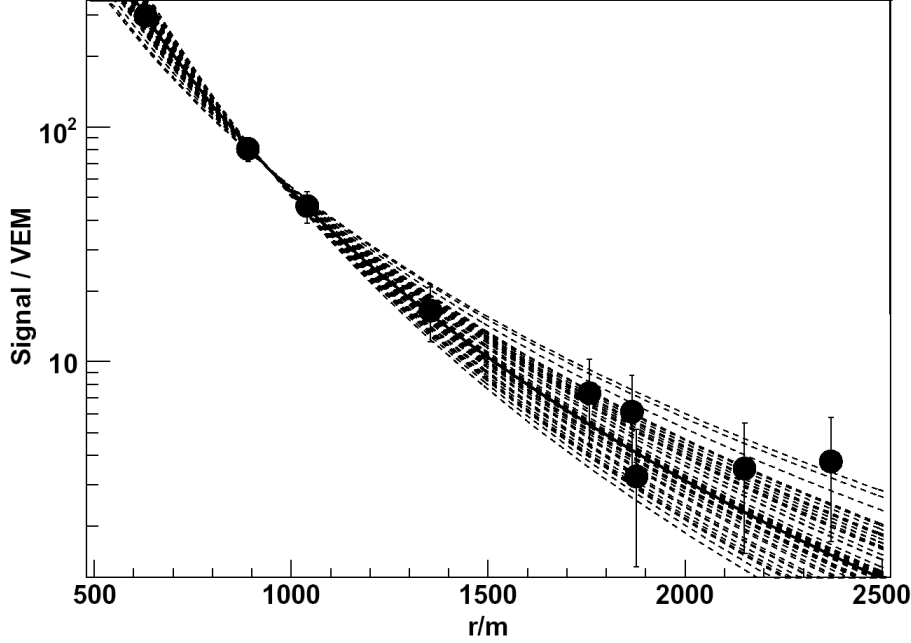


Figure 3.14: Station signals as a function of distance (dots) for a simulated event and the reconstructed LDFs using different values of β (dashed lines). Figure taken from [139].

be noticed (pointed out) that the optimal distance r_{opt} fluctuates from event to event increasing up to 1500 m when there are saturated stations.

At the Pierre Auger Observatory several parameterizations for the LDF have been investigated [140]. The selected LDF used to described the expected signal $S(r)$ measured in VEM units for a station at a core distance r is a modified NKG function [141; 142]:

$$S(r) = S(1000) \left(\frac{r}{1000} \right)^\beta \left(\frac{r + 700}{1000 + 700} \right)^\beta \quad (3.7)$$

where r is measured in m. $S(1000)$, defined as the signal in VEM units at 1000 m from the shower core is chosen as the normalization factor of the LDF. β is the slope of the LDF and it is given by [135]:

$$\beta = \begin{cases} a + b(\sec \theta - 1), & \sec \theta < 1.55 \\ a + b(\sec \theta - 1) + f(\sec \theta - 1.55)^2, & \sec \theta > 1.55 \end{cases} \quad (3.8)$$

The values of the constant parameters a , b and f can be found in the reference.

The parameters which best fit the measured signal S_i at a given station to the expected signal $S(\rho_i)$ are obtained by minimizing the χ^2 function:

$$\chi^2 = \sum_i \left[\frac{S_i - S(\rho_i)}{\sigma_{S_i}} \right]^2 + \sum_{i, \text{ zero signal}} S(\rho_i) \quad (3.9)$$

The signal accuracy σ_{S_i} has been parameterized as a function of the axis zenith angle θ as $\sigma_{S_i} = (0.32 + 0.42 \sec \theta) \sqrt{S_i}$ [143]. The first term of equation 3.9 runs over stations with signal while the second term has been added to take into account stations with zero signal but with a non-negligible probability to measure a signal lower than the threshold. Several procedures have been investigated to include in the fit also stations with a saturated trace due to the proximity to the shower core [144; 145; 146]. From this minimization process, the shower size $S(1000)$ and a precise estimate of the core position are obtained.

Finally, to estimate the energy of the primary cosmic ray, a conversion from $S(1000)$ to energy is required. For a reconstruction using only SD data, a conversion is obtained using predictions from Monte Carlo simulations strongly dependent on hadronic models and assumptions about primary mass.

3.4.3.3 Energy Reconstruction via Calibration with the FD

One of the main advantages of the hybrid design is the capability to assign a calorimetric energy, measured using the fluorescence technique, to the $S(1000)$ parameter measured with the surface array. In this way the large dependencies on hadronic models presented in the SD-only reconstruction of the primary energy are removed.

The energy calibration procedure [45] used to relate the fluorescence energy with the $S(1000)$ parameter is described next. In a first step, the dependency of $S(1000)$ with zenith angle has to be known to obtain an absolute energy estimator independent on the arrival direction. At a given energy, the signal at ground $S(1000)$ will be smaller as the zenith angle increases due to the attenuation of particles. The dependency, parameterized as an attenuation curve, is studied using the Constant Intensity Cut (CIC) Method [148] which relies in the assumption that the cosmic rays flux is nearly isotropic.

Using this attenuation curve a zenith-angle independent variable, the S_{38° parameter, to be related to the energy is obtained. The S_{38° parameter is defined as the



3. THE PIERRE AUGER OBSERVATORY

$S(1000)$ signal that would be measured if the air shower arrives from a zenith angle of 38° . For a given shower arriving at a zenith angle θ and with a measured $S(1000)$ signal, the corresponding S_{38° parameter is obtained as:

$$S_{38^\circ} = \frac{S(1000)}{CIC(\theta)} \quad (3.10)$$

where $CIC(\theta)$ is the attenuation curve parameterized as:

$$CIC(\theta) = 1 + a \cdot (\cos^2 \theta - \cos^2 38^\circ) + b \cdot (\cos^2 \theta - \cos^2 38^\circ)^2 \quad (3.11)$$

where a and b are fitted parameters.

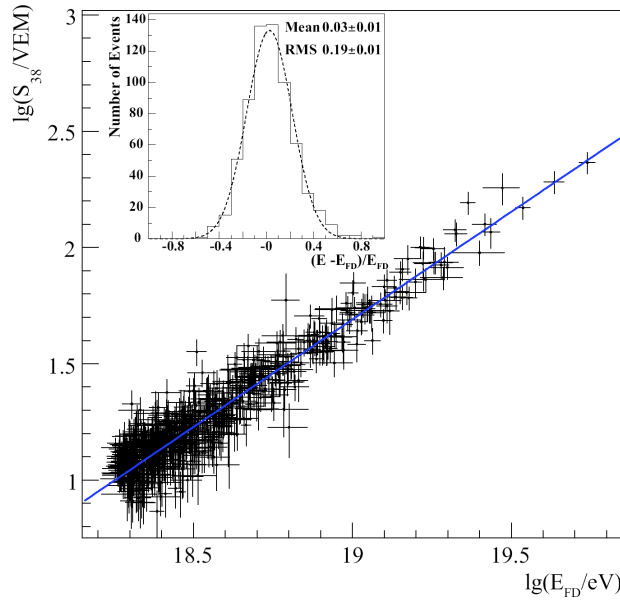


Figure 3.15: Calibration curve showing the correlation between $\log E_{FD}$ and $\log S_{38^\circ}$. Figure taken from [45].

The relation between the SD energy estimator S_{38° and the calorimetric energy is studied using a high quality sample of hybrid events, those registered by both detectors (see figure 3.15). The statistical uncertainty in S_{38° arises from three different contributions: the uncertainty in the parametrization of the attenuation curve, the angular accuracy of the event and the uncertainty in the measured $S(1000)$. The total uncertainty in S_{38° is about 16%. The statistical uncertainty on E_{FD} has also several

contributions: the accuracy of the hybrid geometry, the uncertainty arising from the Gaisser-Hillas profile fit, the uncertainty due to the invisible energy correction and the uncertainty in the aerosol content measurement. Summing all these contributions, the total uncertainty in E_{FD} is about 9%. The relation between both parameters has been parameterized by the calibration curve:

$$\log E_{FD} = A + B \cdot \log S_{38^\circ} \quad (3.12)$$

where A and B are fitted parameters. Using this relation, the energy of a SD-only event will be calculated from its $S(1000)$ parameter without the large uncertainties arising from model dependencies¹.

The fluorescence energy E_{FD} and in consequence the derived energy for a SD event is affected by a total systematic uncertainty of 22%. It arises from different contributions such as the one coming from the fluorescence yield uncertainty and its dependency with atmospheric properties, those related with the optical calibration of the telescope, reconstruction procedure or invisible energy [45].

3.5 Offline Software Framework

The Offline Software Framework [149] has been developed within the Pierre Auger Collaboration to provide the necessary tools for the analysis of data gathered by the observatory, including simulation and reconstruction tasks, taking into account the large and dispersed number of physicists involved in the project and the enormous number of objectives to be fulfilled. The framework, implemented in C++, also allows members replace existing algorithms with his own in a simple manner.

The Offline Framework consists of three principal parts:

- **Modules:** The different tasks to be developed within the Pierre Auger Collaboration can be usually factorized into several sequences of self contained processing steps. Each one of this tasks are carried out by processing algorithms, the so-called modules, which can be assembled and sequenced through instructions

¹The FD energy is obtained by integrating the energy deposited by shower particles in the atmosphere with a small correction to take into account the energy carried by muons and neutrinos passing into the ground. This correction, known as the missing energy, is model-dependent since it is parameterized using shower simulations (see section 3.3.3).



3. THE PIERRE AUGER OBSERVATORY

provided in an XML file. This modular design allows users build up a wide variety of applications by combining modules as well as an easy way to exchange some modules by user's own ones. Next, a typical example of the structure of a sequencing file employed for reconstruction of SD events is presented:

```
<!-- A sequence for an SD only reconstruction -->
<sequenceFile>
  <enableTiming/>
  <moduleControl>
    <loop numTimes="unbounded" save="yes">
      <module>EventFileReaderOG</module>
      <module>EventCheckerOG</module>
      <module>SdCalibratorOG</module>
      <module>SdEventSelectorOG</module>
      <module>SdPlaneFitOG</module>
      <module>LDFFinderOG</module>
      <module>SdEventPosteriorSelectorOG</module>
      <module>SRecUserModule</module>
      <module>SdRecPlotterOG</module>
    </loop>
  </moduleControl>
</sequenceFile>
```

- An **event data** model which accumulates all simulation and reconstruction information (raw, calibrated, reconstructed and Monte Carlo data). Furthermore, through this event data model, modules can communicate and relay data to one another.
- **Detector description:** which provides a unified interface with non-event information about the configuration and performance of the observatory as well as atmospheric conditions as a function of time.

Most of the analysis developed in this thesis has been carried out within the Offline Software Framework. A number of modules have been developed to complement those provided by default and to perform the desired calculations.

3.6 Asymmetry in Extensive Air Showers

The footprint of an EAS in the plane of the ground has elliptical symmetry centered on the shower axis. In principle a simple transformation projecting the observed parameters into a plane perpendicular to the shower axis should restore the expected circular

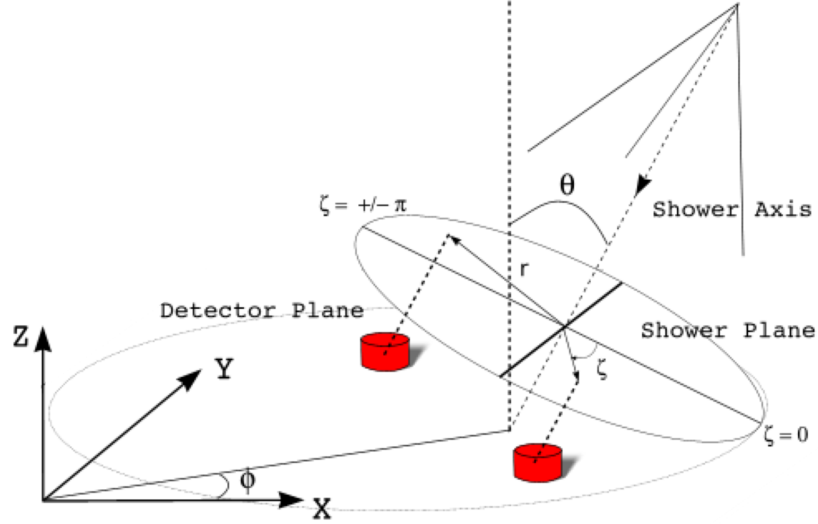


Figure 3.16: Schematic view of the shower geometry. The incoming direction of the primary particle defines two regions, “early” ($-\frac{\pi}{2} < \zeta < \frac{\pi}{2}$) and “late” region ($\frac{\pi}{2} < \zeta < -\frac{\pi}{2}$).

symmetry [147] and hence only the radial distance of the particles need to be considered. The assumption that the signal of the surface detectors tanks (proportional to the density of particles) only depends on their perpendicular distance to the shower axis is crucial in the ground array technique, for example in the primary energy determination using the LDF (see section 3.4.3). However this assumption is a good approximation at small zenith angles and small core distance.

For non-vertical showers this symmetry is broken because a portion of the cascade develops further as the shower disk passes through the ground. Then an asymmetry appears since the magnitudes measured by the tanks, viewing the shower at a different azimuthal angle in the shower plane, will be different due to the different stage of development. Therefore, the magnitudes measured by the stations of the surface detector are dependent on the azimuth angle ζ in the shower plane (see figure 3.16). As will be discussed later, in a surface array a geometrical contribution to the asymmetry takes place due to the fact that the effective surface area (top + side) of the tank depends on ζ .

In this section the origin of this asymmetry in signal intensity (i.e. particle density) will be discussed. Signal asymmetries were first observed at the Haverah Park Array where an in-filled portion of the array allowed accurate core location for a subset of



3. THE PIERRE AUGER OBSERVATORY

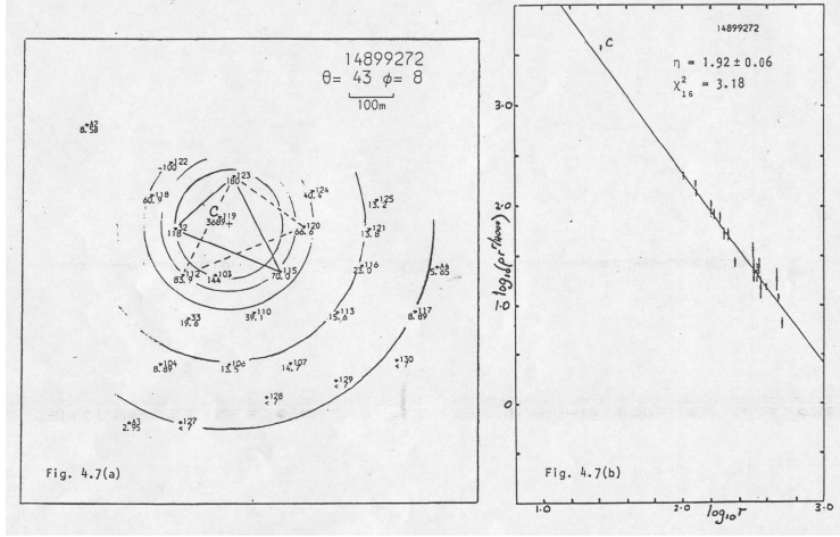


Figure 3.17: Event 14899272 recorded by the Haverah Park array with reconstructed energy $E \sim 3 \times 10^{18}$ eV and $\theta = 43^\circ$. Left: Data in the shower plane and axis position (cross). Right: Fit result to the equation 3.13.

events [150]. Figure 3.17 (a) shows the data in the shower plane for an event with $\theta = 43^\circ$ and $E \sim 3 \times 10^{18}$ eV. Also in this figure, the position of the axis (marked as a cross) and the corresponding detectors at the same distance to the axis are shown. In this particular event the detectors which lie at the same distance to the core did not record the same density of particles. The radial dependence of the signal was fitted to a lateral distribution in the form of a modified power law which was found to describe the data well up to 800 m from the shower core:

$$S(r) = kr^{-(\eta + \frac{r}{4000})} \quad (3.13)$$

The result of this fit is shown in figure 3.17 (b). The obtained slope parameter was $\eta = 1.92 \pm 0.06$. Remarkably, the fit was not able to account properly the data ($\chi^2 = 3.18$).

This behavior was also observed in several inclined showers at Haverah Park [150]. It was pointed out that in some cases the circular symmetry of the density of the particles is broken. As a result, events with $\chi^2 > 3$ were not used in the analysis presented in [150].

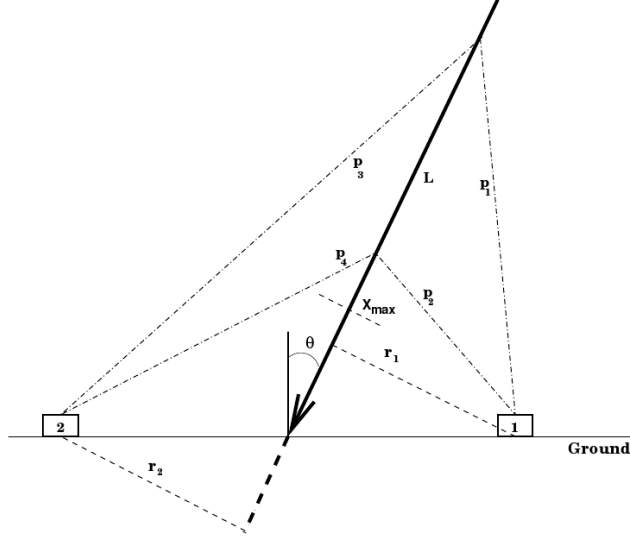


Figure 3.18: Schematic view of a shower with zenith angle θ . An early tank (1) will see a greater difference in paths lengths (P_1P_2) than a late one (2), (P_3P_4).

3.6.1 Sources of azimuthal asymmetry

The study of the asymmetry and its possible sources is a complex problem. There are several processes which contribute to the observed asymmetry to a greater or lesser extent. As the zenith angle increases, a tank lying in the late region (see figure 3.16) will see a portion of the shower front that have passed through a larger atmospheric depth than the one observed by an early tank. The intervening atmosphere traversed by the shower will cause the attenuation of the electromagnetic particles. Therefore the early tanks will measure a larger signal, even if they are at the same core distance in the shower plane.

Another contribution to the asymmetry arises from the fact that tanks lying far from the shower core have different effective areas, viewed in the shower plane, depending on their azimuthal angle ζ .

The above two sources of asymmetry are the most important ones and they will be discussed in detail. However there are other effects that contribute in a smaller extent to the asymmetry. For example early tanks will see a greater difference in paths lengths of particles than late tanks, see Figure 3.18. Also, a late tank is at smaller angular distance from the shower axis than an early one.

Finally the geomagnetic field can produce an asymmetry in the shower footprint if



3. THE PIERRE AUGER OBSERVATORY

the charged particles have traveled enough distance. This effect is only important in very inclined showers which are muon dominated. The effect of the geomagnetic field plays a very important role in the reconstruction of horizontal showers ($\simeq 70^\circ$) [151].

3.6.1.1 Geometrical asymmetry

In the analysis presented in [152; 153] it was shown that there is an asymmetry which arise because tanks that lie far from the shower core have different effective areas projected in the shower plane. This effect, which depends on the tank core distance, comes from the fact that in average the particles in the shower diverge from the axis, due to the multiple scattering, and then the flux received in the early tanks is more vertical than that received in the late ones (see figure 3.19).

Following [152] one can easily understand that there should be a early/late asymmetry in the ground plane. When projecting the ground densities onto the shower plane, as the direction of the individual particles is not known, the shower direction must be used as can be seen in figure 3.19. The particles which hit the ground in the early region will be seen more vertical than the ones in the late region, this is the so-called geometrical asymmetry.

As demonstrated in [152] it is possible to obtain an asymmetry factor comparing the density of particles in a vertical shower with an inclined one. The approach followed in [152] was to use a simplified model of the shower particle flux. For this purpose the direction of the particles is expressed in terms of the distribution of momenta unit vectors in the shower frame. This distribution, using cylindrical coordinates, can be represented by $N(r, \zeta, z)[r; \phi; z; p_r; p_\zeta; p_z]$ with $N(r, \zeta, z)$ being a global normalization factor and (p_r, p_ζ, p_z) the normalized projections on the 3 axis of the particle momentum $\vec{P}(r, \zeta, z)$. Assuming in a first approximation that the shower is stationary (i.e. that the development of the shower is negligible) and has cylindrical symmetry, the distribution only depends on the distance r to the axis, that is $\vec{P}(r)$ and $N(r)$. It also assumed that the divergence in the ζ axis is negligible compared to the radial divergence, $p_\zeta \ll p_r$.

The ground density of particles can be calculated as the flux through the ground, which simply corresponds to the distribution of $-p_z$. Under the above approximations the density in a vertical shower can be expressed as:

$$\rho_v(r) = \langle -p_z(r) \rangle N(r) \quad (3.14)$$

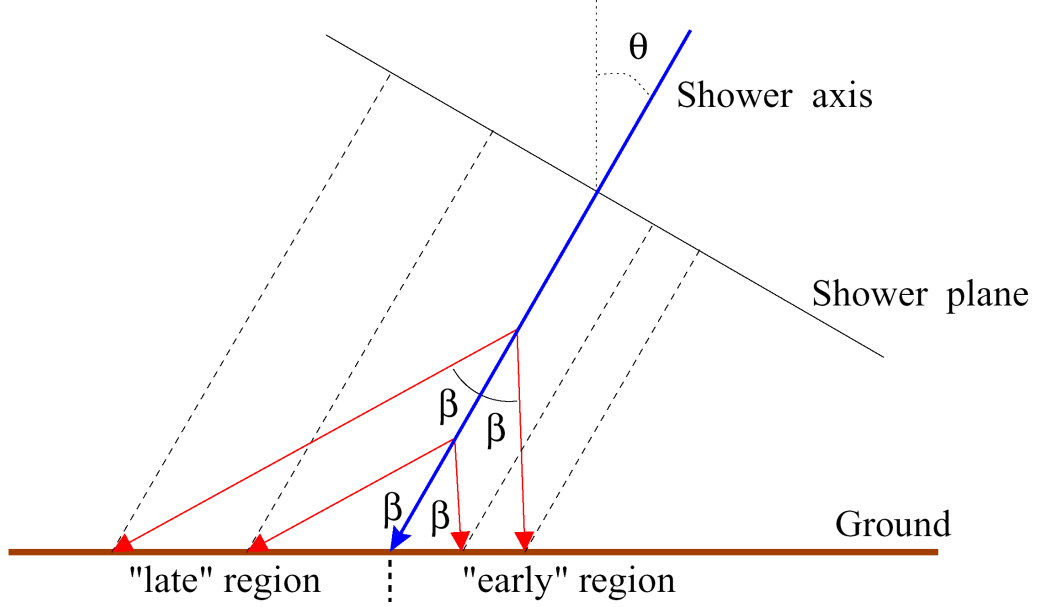


Figure 3.19: Particles going away from the shower axis with symmetric angles β . When projecting the ground densities on the shower plane, a purely geometric asymmetry appears. Figure taken from [152].

where ρ_v represents the vertical density.

An inclined shower corresponds to a θ rotation around the y axis. The momentum vector after this rotation, expressed in cartesian coordinates is:

$$\vec{P}(r, \zeta) = (p_r(r) \cos \zeta \cos \theta + p_z(r) \sin \theta) \vec{e}_x + p_r(r) \sin \zeta \vec{e}_y \quad (3.15)$$

$$+ (-p_r(r) \cos \zeta \sin \theta + p_z(r) \cos \theta) \vec{e}_z \quad (3.16)$$

which now depends on r and ζ . The ground density is again computed by the distribution of $-p_z$ but projected into the shower plane which corresponds to $-p_z / \cos \theta$, leading to:

$$\rho_\theta(r, \zeta) = (p_r(r) \cos \zeta \tan \theta - p_z(r)) N(r) \quad (3.17)$$

where the sub-index θ stands for an inclined shower. This expression allows to obtain an asymmetry factor comparing 3.14 to 3.17:

$$A(r, \phi) = \frac{\rho_\theta - \rho_v}{\rho_v} = \left\langle \frac{p_r(r)}{-p_z(r)} \right\rangle \tan \theta \cos \zeta = A(r) \cos \zeta \quad (3.18)$$

therefore the purely geometric asymmetry expected is roughly $A(r) \cos \zeta$.



3. THE PIERRE AUGER OBSERVATORY

Assuming that the development of the electromagnetic component of an inclined shower is the same as in a vertical one at the same accumulated atmospheric depth, a shower with zenith angle θ is comparable to a vertical shower rotated by the θ angle and evaluated at the atmospheric depth of $X_g/\cos\theta$, where X_g is the atmospheric depth of the ground. Then, the purely geometrical asymmetry must be reproducible with a vertical shower at the modified depth, that is, the observed asymmetry in an inclined shower should be the same as the one obtained by a vertical shower rotated by the θ angle but each particle projected in the rotated ground along its own direction, not along the shower axis.

To check the model, the authors used simulations obtained with AIRES/SIBYLL 2.2.1. Figure 3.20 shows the comparison of the projected density of the electromagnetic energy, calculated as described before, to the direct observation of an inclined shower on a ground at 900 g cm^{-2} in different slices of r and two different zenith angles. As it can be seen at 30° the asymmetry is very well reproduced, at 40° , the longitudinal attenuation begins to play an important role at large distances.

The geometrical asymmetry has an important effect when the density of the particles is convolved with the detector response [153]. The Auger surface detectors are water Cherenkov tanks, and part of the signal comes from particles entering the side walls of the tank. As it was described before, the flux received in the early tanks is more vertical than that in the late ones, then the signal deposited by the particles entering the top of the tank will be larger in an early tank than in a late one, while it is the opposite for particles entering the side of the tank, i.e. the signal will be larger in a late tank than in a early one for particles entering the side of the tank.

The contribution due to the geometrical effect of the total observed asymmetry in the signal of the Auger tanks of the SD was analyzed by P. Billoir *et al.* in [153]. In this analysis, simulations of the response for two different detector shapes were done. Calling α the divergence angle of the particles (see figure 3.21) and A_{top} and A_{side} the top and lateral areas of the tank, the effective area can be defined as $A_{top} \cos(\theta - \alpha) + A_{side} \sin(\theta - \alpha)$ for early tanks, and $A_{top} \cos(\theta + \alpha) + A_{side} \sin(\theta + \alpha)$ for late ones, assuming that the tanks lie on the longitudinal axis. Then, for a giving intrinsic density in the shower, tanks with a top area much larger than the side wall area will receive more particles in the early region, while tanks with side wall area much larger than the top one, will receive more particles in the late region.

3.6 Asymmetry in Extensive Air Showers

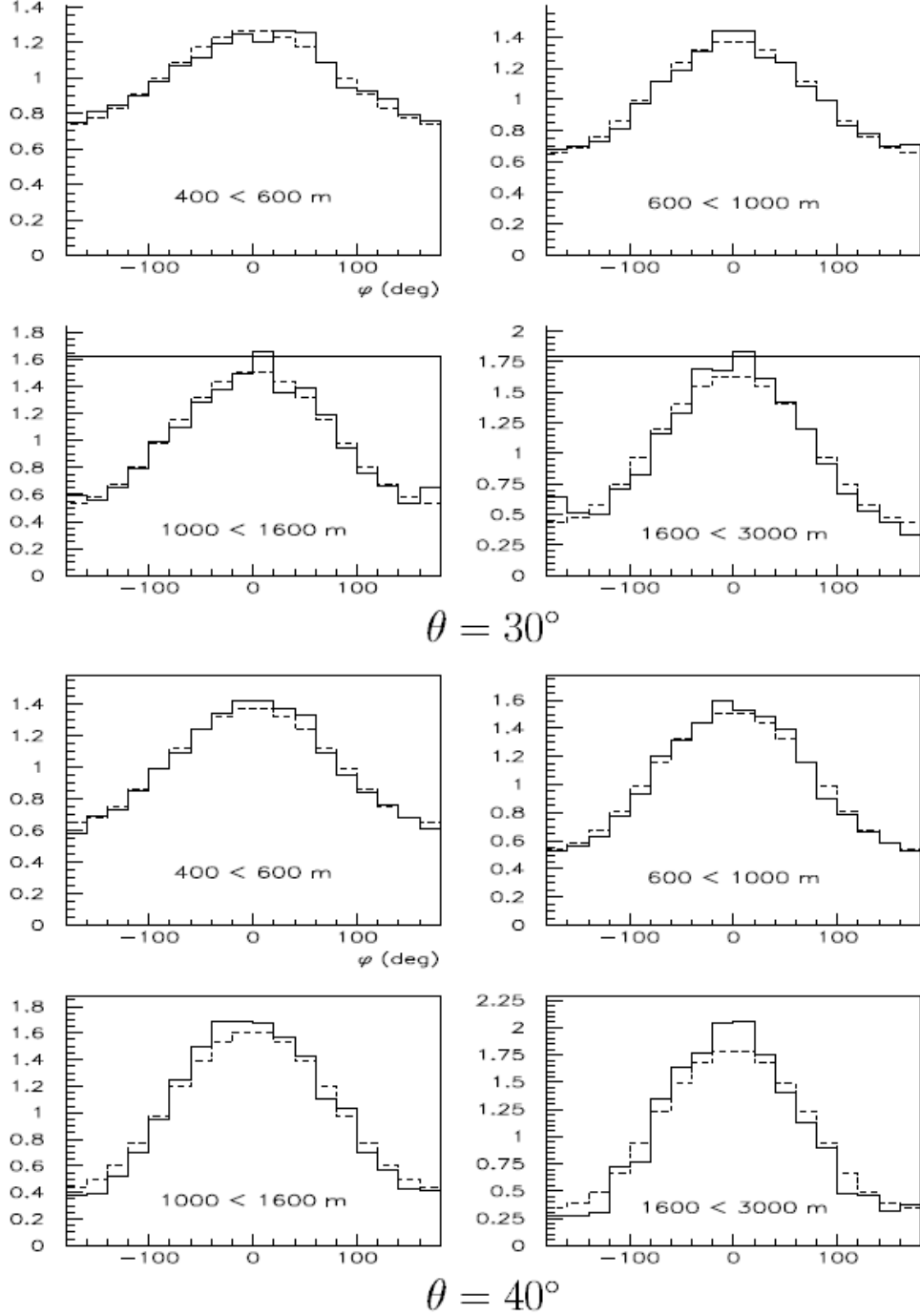


Figure 3.20: Comparison of the ground asymmetry (electromagnetic energy density as a function of ζ angle) observed in simulated inclined shower (solid lines) to the prediction of a vertical shower at the same depth, after projection (dashed line). Figure taken from [152].



3. THE PIERRE AUGER OBSERVATORY

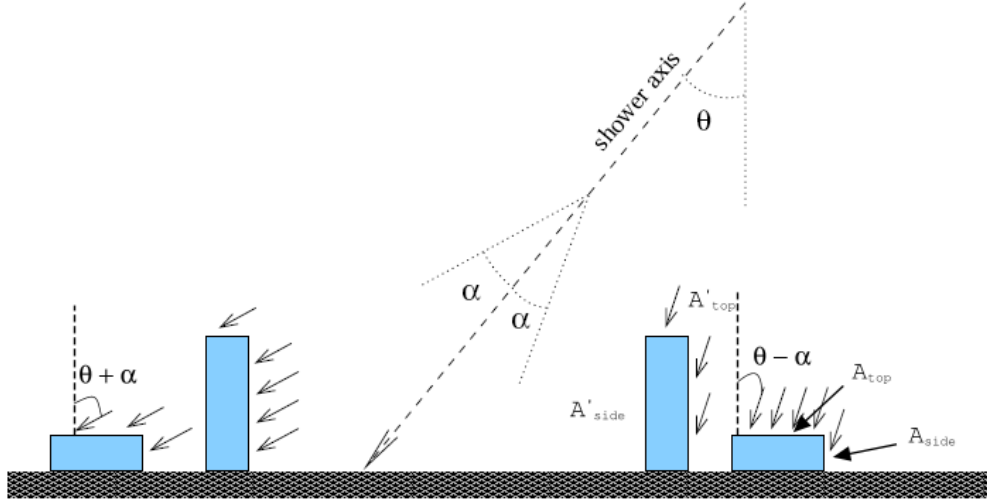


Figure 3.21: The geometrical forward-backward asymmetry for photons and electrons.

It is important to note that the Cherenkov light deposited by muons is roughly proportional to their path in water, and they have generally enough energy to go through the tank, as a consequence the signal produced by muons entering the side of the tank is larger than the one produced if they enter the top of the tank. This compensates the geometrical effect, whatever the incident direction and therefore there is no geometrical asymmetry for muons.

The tank shapes simulated in this analysis were, one with the nominal shape (radius 1.8 m, water height 1.2 m) and another with an inverted shape with the same total volume (radius 0.9, height 4.8 m). The tank response was investigated for four different zenith angles and several core distance intervals. Taking into account that the muon/electromagnetic ratio increases with zenith angle and with the core distance, the expected behavior is:

- for flat-shaped tanks, more signal in the early region for any θ value, with the asymmetry going to zero for vertical showers, because there is no geometrical asymmetry, and for high zenith angles, because the shower is muon dominated.
- for high-shaped tanks, a reversed asymmetry compared with flat-shaped tanks, where the geometrical effect dominates, with the same dependence with the zenith angle.

3.6 Asymmetry in Extensive Air Showers

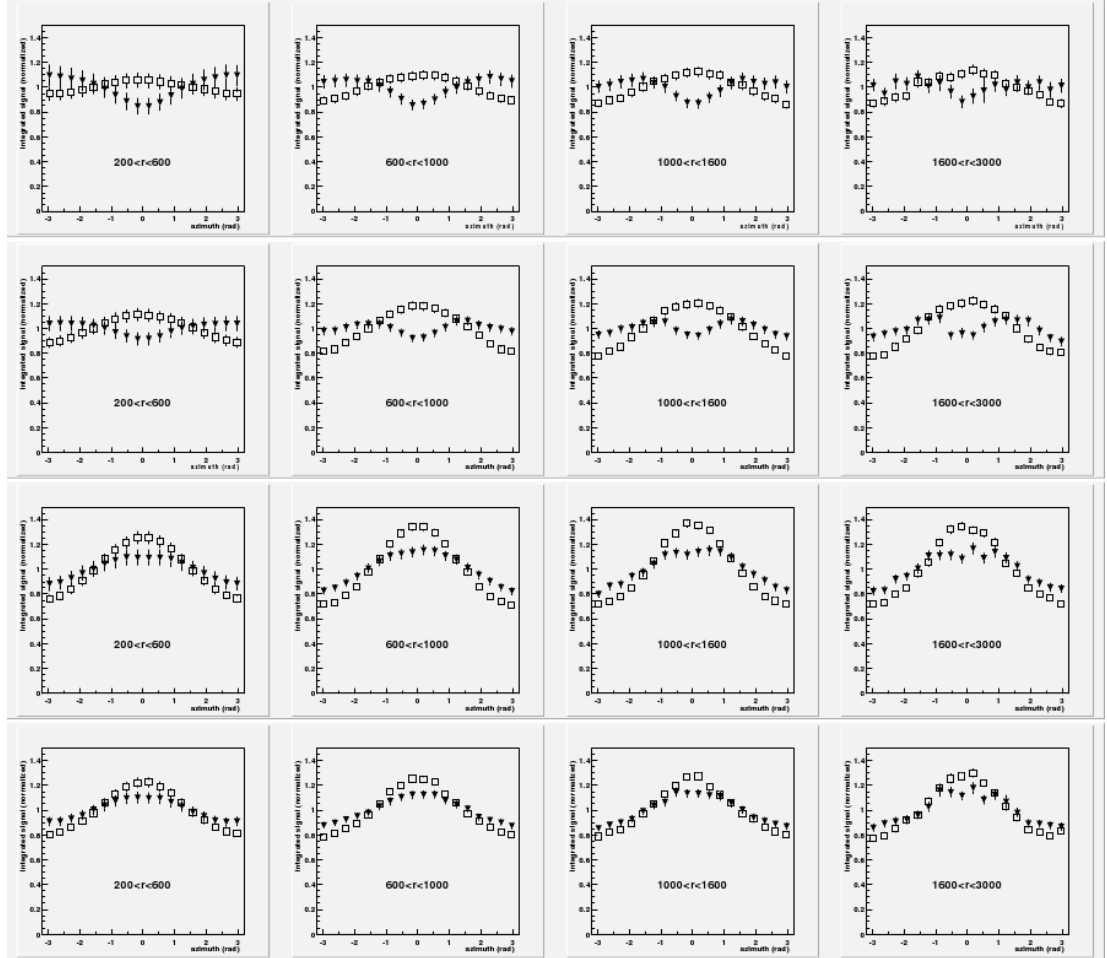


Figure 3.22: The azimuthal dependence of the integrated signal at $\theta = 25, 36, 53, 66$ (from top to bottom) at different distances from the shower axis, for normal-shaped tanks (flat-shaped) (open squares) and high-shaped tanks (filled triangles). The average signal is normalized to the same value. Figure taken from [153].



3. THE PIERRE AUGER OBSERVATORY

Figure 3.22 shows the azimuthal dependence of the total integrated signal for four different zenith angles, $\theta = 25^\circ, 36^\circ, 53^\circ, 66^\circ$ at different distances from the shower axis, for normal-shaped tanks (open squares) and high-shaped tanks (filled triangles), the averaged signal is normalized to the same value. This figure exhibits the expected behavior, the asymmetry in both cases increases with the zenith angle and with the distances to the core. The reversed asymmetry is clearly seen at low zenith angles for high-shaped tanks, showing where the geometrical asymmetry dominates. As the zenith angle increases the asymmetry for high-shaped tanks has the same behavior as the flat-shaped tanks, demonstrating that the geometrical effect no longer dominates, and therefore the origin of this asymmetry is the atmospheric attenuation. Then it can be concluded that the geometrical asymmetry dominates for low zenith angles up to about $\theta = 35^\circ$.

3.6.1.2 Shower evolution and atmospheric attenuation

In addition to the previously mentioned geometric effect there is a contribution to the total observed asymmetry because the shower continues its development in the late region as it passes through the plane of detection [154; 155]. To understand this asymmetry it is necessary to take into account the difference in path traveled by the particles between the early and late regions.

Two stations at the same distance R in the shower plane correspond to different distances r_1 and r_2 in the ground plane (see figure 3.23). Particles which reach the late station (r_2) belongs to a different stage of shower development (which corresponds to a slant depth of $t + \Delta t$, where t is the slant depth of the ground) with respect to the early station (r_1). As shown in figure 3.24 the additional path traveled by the particles Δt as a function the core distance measured in the shower plane R . As can be seen in this figure Δt increases with the zenith angle. For example $\Delta t = 300 \text{ g cm}^{-2}$ for a late station at a 1300 m from the shower axis with incoming zenith angle of 40° .

For inclined showers the slant depth depends on ζ , inducing the asymmetry features in the observed signal as shown in [154]. The azimuthal dependence can be obtained if one considers the dependence on the slant depth transversed by the shower, defined as:

$$t' = \int_{z'}^{\infty} \rho(z') dz' \quad (3.19)$$

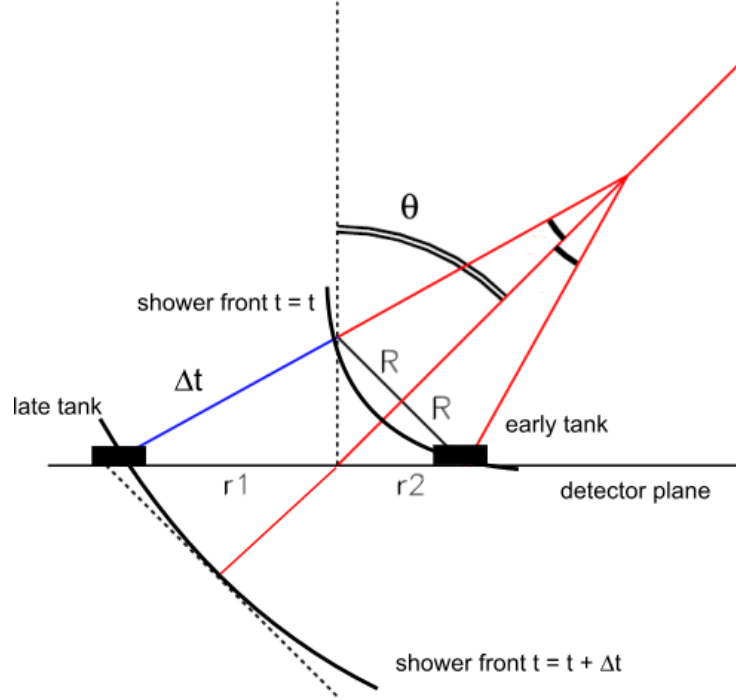


Figure 3.23: Difference in the path Δt traveled by the particles at ground, for two points at the same distance R from the shower axis measured at the shower plane. Their corresponding distances in the detector plane are $r1$ and $r2$.

The actual structure of a shower is quite complex but following [156; 157] an inverted cone is a reasonable approximation. The intersection of an inclined cone with a plane parallel to the ground gives:

$$\frac{z}{\cos \theta} - L = (z' - L) \left(1 - \frac{r'}{z' - L} \tan \theta \cos \zeta \right) \quad (3.20)$$

where L and θ are indicated in Fig. 3.25, and z' , r' and ζ are the coordinates in the shower frame with $x' = r' \cos \zeta$. With this expression, the relation of the vertical atmospheric depth and the slant depth in terms of the azimuthal angle ζ can be obtained. If the cone angle α is assumed nearly independent of z and using the fact that $\tan \alpha = r'/(z' - L)$, then:

$$\frac{dz}{\cos \theta} = dz' (1 - \tan \alpha \tan \theta \cos \zeta) \quad (3.21)$$



3. THE PIERRE AUGER OBSERVATORY

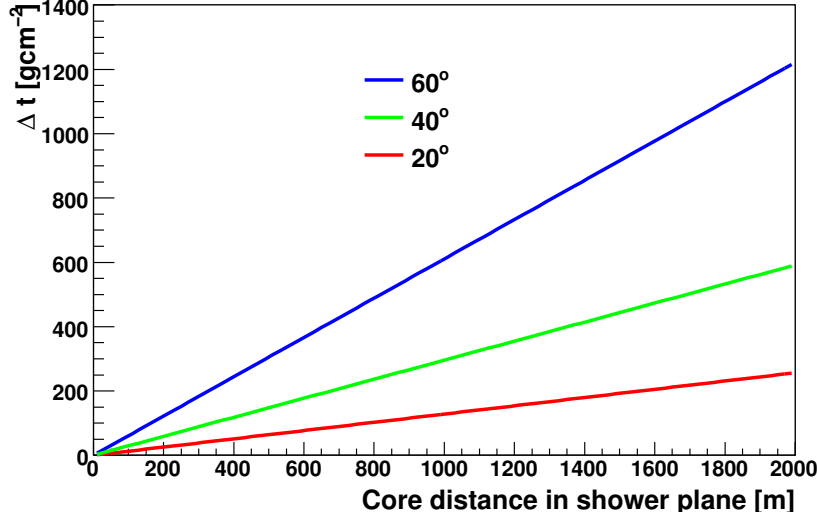


Figure 3.24: Maximum difference of path travelled by particles Δt , as a function of the distance to the shower core in the shower plane, for three different zenith angles.

which is equivalent to:

$$\frac{t}{t'} = \cos \theta (1 - \tan \alpha \tan \theta \cos \zeta) \quad (3.22)$$

Equation 3.22 can be approximated by:

$$t'(\zeta) = t_s + \Delta t_s(\zeta) \simeq t \sec \theta (1 + B \cos \zeta) \quad (3.23)$$

$$B = B_0 \tan \theta \quad (3.24)$$

$$t_s = t \sec \theta \quad (3.25)$$

$$\Delta t_s(\zeta) = t_s B \cos \zeta \quad (3.26)$$

Using this approximation it is possible to obtain the dependence on ζ of the particle density due to the attenuation [155]. In a vertical shower, the density of particles depends on the core distance and the vertical atmospheric depth $\rho(r, t)$, in the case of an inclined shower $\rho(r, t'(\theta, \zeta))$. It is possible to obtain explicitly the ζ dependence using a Taylor expansion around t_s :

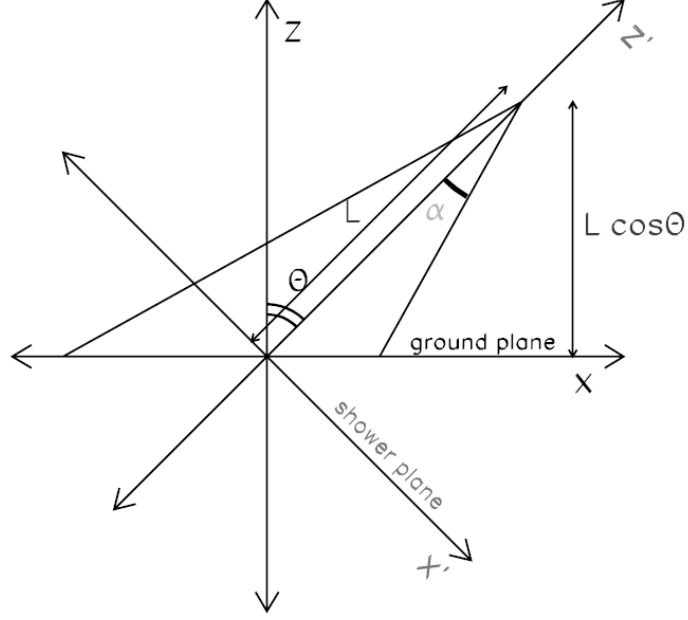


Figure 3.25: Intersection of a cone with the ground plane.

$$\rho(r, \zeta) = \rho(r, t'(\zeta)) = \rho(r, t_s + \Delta t_s(\zeta)) \quad (3.27)$$

$$\rho(r, \zeta) = \rho(r, t_s) + \left. \frac{\partial \rho(r)}{\partial t'} \right|_{t_s} \Delta t_s(\zeta) + \dots \quad (3.28)$$

$$\rho(r, \zeta) = \rho(r, t_s) + \left. \frac{\partial \rho(r)}{\partial t'} \right|_{t_s} t_s B \cos \zeta + \dots \quad (3.29)$$

$$\rho(r, \zeta) = \rho(r, t_s) \left(1 + \left. \frac{\partial \log \rho(r)}{\partial t'} \right|_{t_s} t_s B \cos \zeta + \dots \right) \quad (3.30)$$

Keeping only up to the first order of the expansion the particle density can be approximated by:

$$\rho(r, t_s, \zeta) = \rho(r, t_s) (1 + C(r) \cos \zeta) \quad (3.31)$$

which has to be evaluated at t_s and where $C(r) = \left(\left. \frac{\partial \log \rho}{\partial t'} \right|_{t_s} \right) t_s B$. This expression is qualitatively the same as equation 3.18, where now the factor $C(r)$ depends on the functional form of the dependence of the particle density on the core distance.



3. THE PIERRE AUGER OBSERVATORY

To study the conditions for which the asymmetry due to attenuation dominates over the geometrical one, simulations of the response of spherical tanks were performed in [158]. In this particular case, asymmetry is only due to the longitudinal development of the shower and to the attenuation of the electromagnetic particles. Figure 3.26 shows the results of these simulations for the electromagnetic component (γ, e^+, e^-) (red circles), muons (blue squares) and the total signal (black triangles) for the same zenith angles as in figure 3.22. As it would be expected no asymmetry is found for muons but the crucial point is the asymmetry in the EM component due to the attenuation of particles in the atmosphere is clearly observed. Furthermore, although the geometrical asymmetry dominates at low zenith angles there is also a contribution due to the attenuation of the electromagnetic particles for distances to the axis > 1000 m, even at $\theta = 25^\circ$.

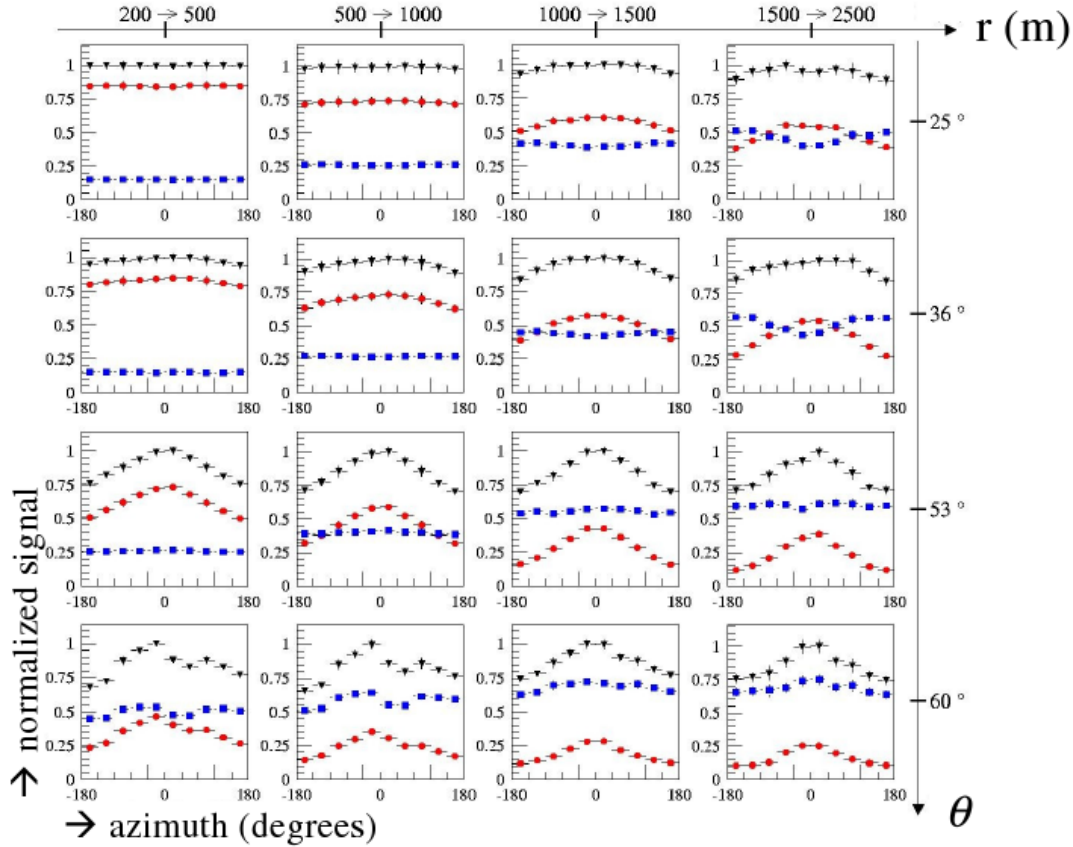


Figure 3.26: The azimuthal dependence of the integrated signal, at different distances from shower axis, for normal-shaped tanks and high-shaped tanks.

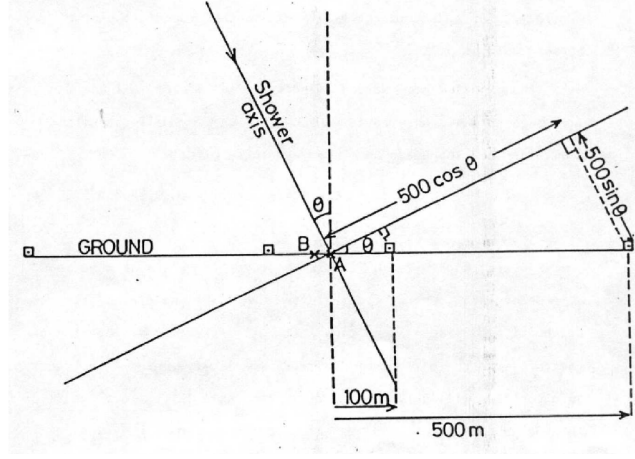


Figure 3.27: Scheme of a shower with zenith angle θ . The true core position is located in A. The signal in the early tank is greater than that in the late one due to the attenuation of shower particles. The circular assumption shifts the core to position B.

3.6.2 Effects of asymmetry in shower reconstruction parameters

The asymmetry effect in the signal intensity is difficult to observe as it was pointed out in [150]. This is due to the fact that the circular symmetry assumption in ground array technique produces a shift of the core position to the early part of the shower, as illustrated in figure 3.27. In this figure the true core position A is reconstructed in position B producing the mentioned shift.

This shift was also observed in the Pierre Auger Observatory [159]. For a set of hybrid events it is possible to compare the reconstructed core position using the fluorescence technique with the surface technique affected by the asymmetry. Figure 3.28 shows the difference of estimated core location from SD and hybrid analysis on the shower plane. The systematic shift caused by the asymmetry effect can be seen in bottom panel of this figure and is about 40 m in average. In principle, this uncertainty would not affect directly the $S(1000)$ estimation. If core location is systematically shifted, the core distance of some stations are underestimated and other stations are overestimated and as a consequence the LDF fit compensates this systematic effect. The influence of the reconstructed core position on $S(1000)$ was studied in [160] using shower simulations and the effect obtained is less than 4% in the $S(1000)$ value.

The asymmetry in the total signal of the water Cherenkov tanks of the SD is parametrized as [161]:



3. THE PIERRE AUGER OBSERVATORY

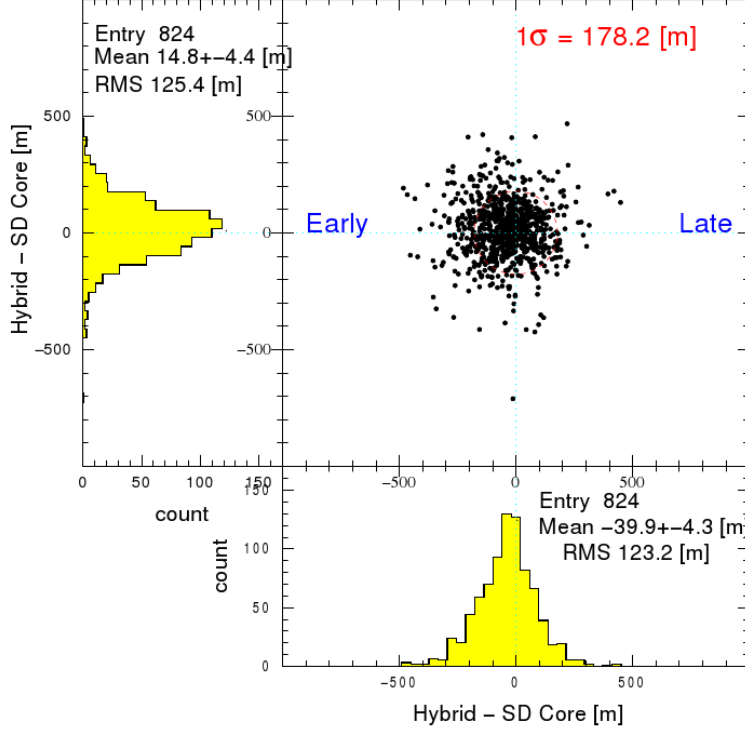


Figure 3.28: Difference of estimated core location from SD and Hybrid analysis on the shower plane. The origin of the scatter plot is the estimated core location from Hybrid analysis and black dots indicate that from SD. Red circle in the scatter plot shows a 1σ of 2-dimensional-Gaussian fit. Effect of the asymmetry can be seen in bottom panel. Figure taken from [159].

$$S(r, \zeta) = \bar{S}(r)(1 + \alpha(r) \cos \zeta) \quad (3.32)$$

where $\bar{S}(r)$ is the signal size at $\zeta = \frac{\pi}{2}$ where there is no asymmetry and $\alpha(r)$ is fitted to the function $\alpha(r) = \alpha_m \tan^{-1}(r/r_0)$ with two free parameters, α_m and r_0 . Results obtained with simulations can be seen in figure 3.29. This parametrization is implemented as an asymmetry correction in the LDF fit included in the CDAS code.

3.6 Asymmetry in Extensive Air Showers

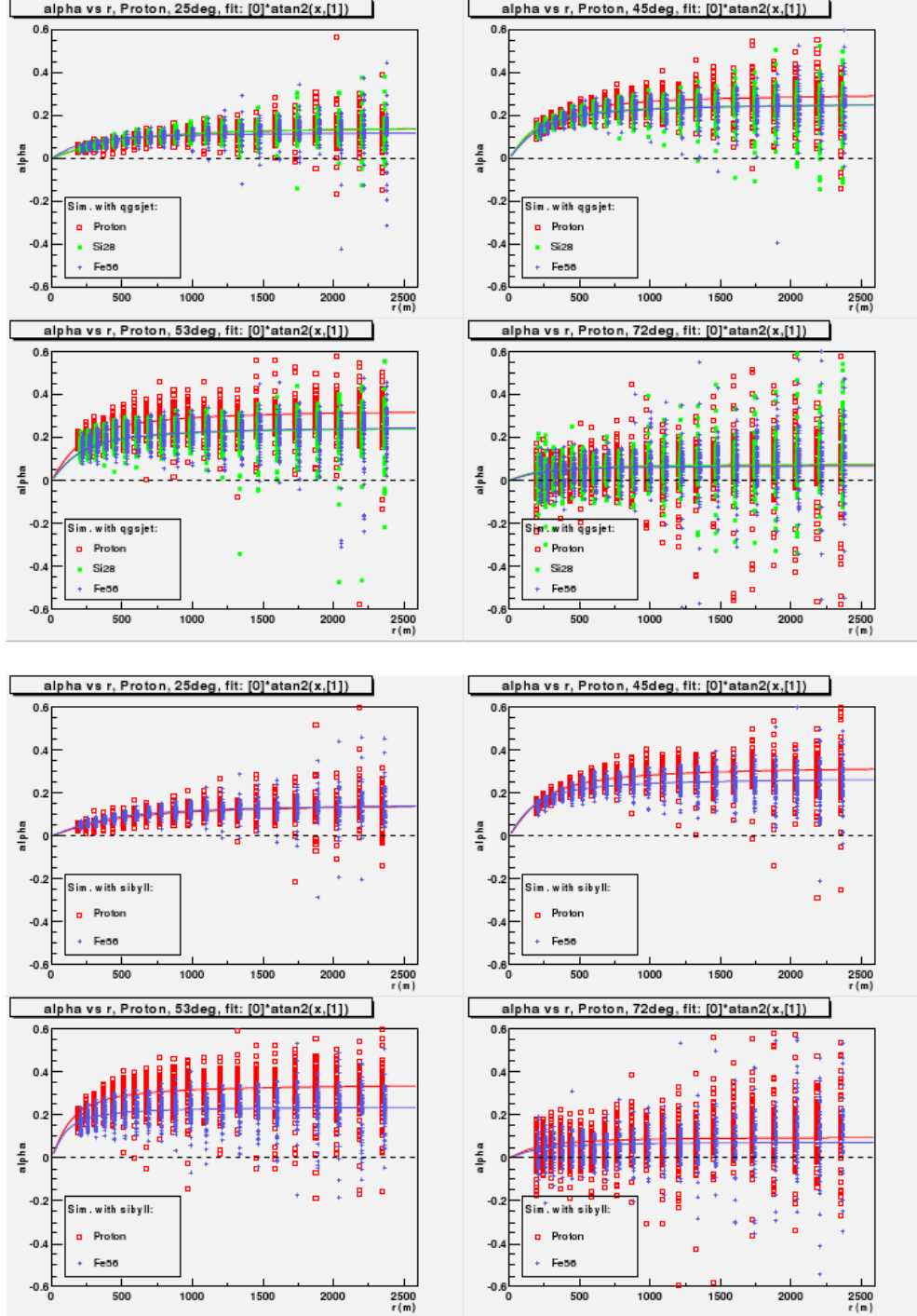


Figure 3.29: Asymmetry in the LDF as a function of the core distance, for different primaries using QGSJET01 (top) and SIBYLL2.1 (bottom) for hadronic interactions. The curves are the results of the fit with $\tan^{-1}$ function. Figure taken from [161].



3. THE PIERRE AUGER OBSERVATORY

Mass dependence of the Asymmetry in the Time Structure of the SD Signals

4.1 Introduction

Apart from the general features of the asymmetry of the total integrated signal, discussed in section 3.6, asymmetry in the time structure of the trace has been also observed. In this chapter this time asymmetry in the surface detector signal and its dependence on the primary mass are studied in detail. A novel technique for the determination of mass composition of UHECRs has been developed in this work (see also [162]).

As illustrative example, the integrated signals of event #184599 are shown in figure 4.1. This event, recorded by the Engineering Array (EA)¹ on Friday 26th April 2002, was the first one where the asymmetry in the time distribution of the signal was observed. Traces of the 11 stations of the EA are plotted in figure 4.2. The effect of the attenuation is clearly seen since late stations present shorter rise-times than early ones. The analysis of the 11-fold event #184599 demonstrated that the traces of the SD stations deliver rich information about shower profile. The signal and rise-time distributions show clearly the attenuation of the electromagnetic component of the shower.

¹The elements of the Pierre Auger Observatory were validated in an Engineering Array which consisted of two fluorescence telescopes and 32 surface detector tanks [163].



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

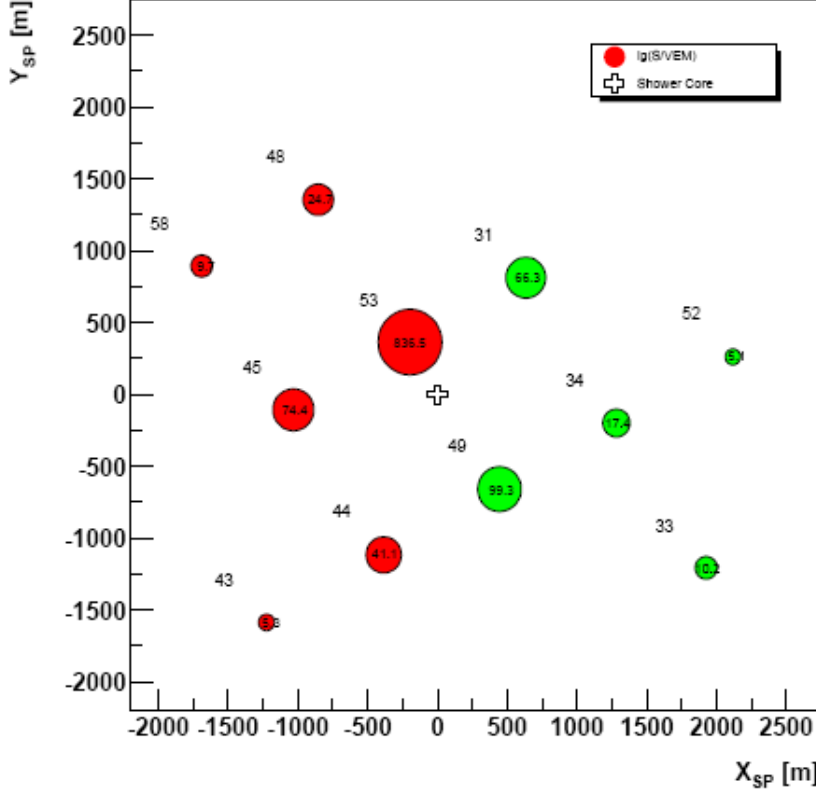


Figure 4.1: Integrated detector signals in the shower frame of event #184599. The size of the circles are proportional to $\log(S/[VEM])$ and the numbers are given as inset accordingly. Green indicates early arrival times while red corresponds to late arriving particles in the detectors. The open cross marks the position of the shower core.

A detailed analysis of the event #184599 can be found in [164].

Azimuthal asymmetry in time structure of non-vertical showers collected from the EA of the Pierre Auger Observatory were found to depend on zenith angle [165]. Rise-time measurements from the EA, as a function of the azimuth angle ζ for two different bins of $\sec \theta$ are shown in figure 4.3.

As shown in [82], the time structure of the shower front, derived from signals produced by particles entering the surface detector tanks, is related with the longitudinal development of the cascade. This can be explained by simple geometrical considerations. The difference in the path length of particles of the shower front is greater in the case of a shower penetrating deeper into the atmosphere, i.e greater X_{max} (figure 4.4 a), than in the case of a shower with smaller X_{max} (figure 4.4 case b). As a result the time

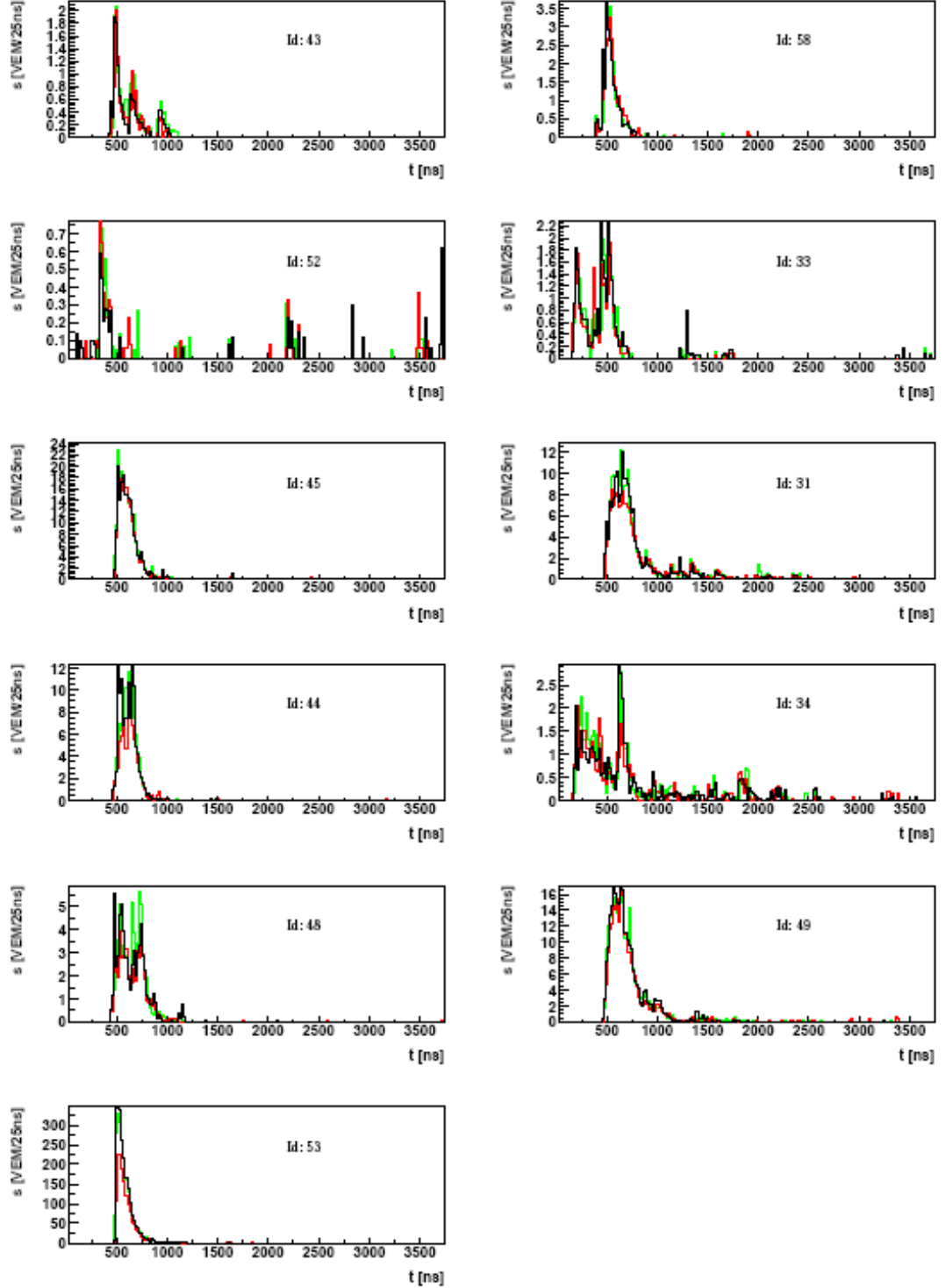


Figure 4.2: Calibrated FADC traces of the SD stations of the event #184599. The colour code represents the different PMTs: PMT1 = red, PMT2 = green, PMT3 = black. Stations 31, 33, 34, 49 and 52 corresponds to early stations. Stations 43, 44, 45, 48, 53 and 53 to late stations.



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

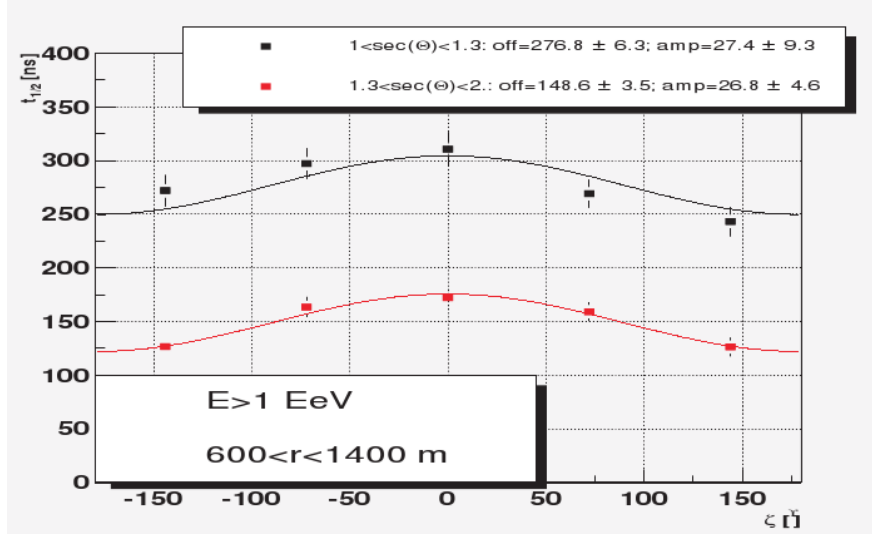


Figure 4.3: Risetime as a function the azimuth angle ζ for two $\sec \theta$ bins from events recorded with the EA of the Pierre Auger Observatory.

of signal deposition increases with the depth of the shower maximum. Also, as it was previously explained, the electromagnetic component of the shower is more attenuated and loses more energy than the muonic component and therefore, the muon component tends to arrive earlier and over a shorter period of time than the electromagnetic one.

In this chapter it is shown that the asymmetry in risetime $t_{1/2}$ (section 4.3) is related to the stage of shower development and therefore it is possible to obtain an asymmetry factor which depends on the incoming angle θ ¹. This so-called asymmetry longitudinal development will allow us to define new observables sensitive to mass composition of the primary cosmic ray. This chapter is organized as follows; in section 4.2 the relationship between the asymmetry in the time distribution and shower evolution will be demonstrated. The risetime, which is the parameter chosen to apply the asymmetry technique will be described in section 4.3. In section 4.4 the Monte Carlo simulation sample used to develop the technique will be presented. All the necessary steps needed to obtain the mass composition parameters from the asymmetry analysis will be detailed in section 4.5. In section 4.6 the optimum set of cuts for the asymmetry analysis will be described. Finally in section 4.7 the method to extract the primary mass composition of a sample will be shown.

¹In fact the dependence study will be on $\sec \theta$ which is proportional to the slant depth.

4.2 Asymmetry in the time structure as an indicator of shower evolution

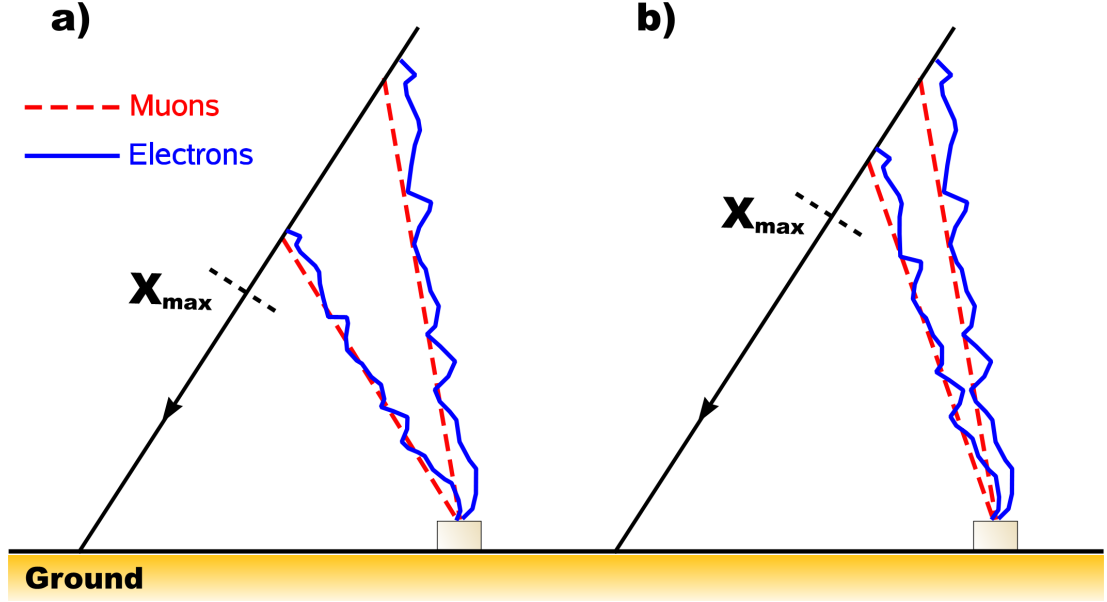


Figure 4.4: The time structure measured in the surface detector is related with X_{max} . The smaller is the X_{max} value the smaller is the difference in path lengths for particles arriving at the detector, case b). Muons travel in straight line while EM particles are subjected to multiple scattering and therefore the muon contribution dominates the first portion of the signal.

4.2 Asymmetry in the time structure as an indicator of shower evolution

Most of the observables sensitive to composition aim at being like a snapshot of the shower development and thus, they are correlated with both X_{max} and the observation depth t . The time distribution of the signals recorded by the surface detector in EAS experiments contains implicitly the information of the shower development. Therefore it is natural to expect a dependence of the mean values of the possible time parameters and of their azimuthal asymmetry with the atmospheric depth traversed by the shower and thus with the zenith angle of the incoming cosmic ray direction.

The sensitivity of timing parameters to primary composition can be explained on the basis of the dominance in the different time regions of the signal of the electromagnetic and muonic component. The first portion of the signal is dominated by the muon component which tends to arrive earlier and over a shorter period of time than the electromagnetic particles, which are spread out on time.

The relation between asymmetry and shower evolution is sketched in figure 4.5,



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

where three different scenarios are presented for a shower with a given X_{max} value, arriving at three different zenith angles. The attenuation (early-late) of the electromagnetic component depends on the difference in the path travelled by particles to reach the detector stations. In case a, i.e. vertical shower, there is no difference in the paths of the EM component so there is no asymmetry. As the zenith angle increases (case b) the difference in the attenuation of the EM component due to different travelled paths increases and, as a result, early-late asymmetry appears¹. At very large zenith angles (case c) the EM component has been strongly absorbed before reaching the detector. Therefore, the asymmetry decreases with θ since the larger is the angle the smaller is the contribution of the EM component. Note that the muonic component is basically asymmetry free.

Then for a given zenith angle the asymmetry provides information of the stage of the development of the shower. According to the above arguments, for a given X_{max} the behavior of the asymmetry with $\sec \theta$ would be reminiscent of the longitudinal development of the shower, and it is expected to observe a maximum correlated with the X_{max} value. This “longitudinal development of the asymmetry” is strongly dependent on the primary type.

4.2.1 Slant depth dependence of the asymmetry in the time structure

As it was shown in section 3.6.1.2 the azimuthal dependence of the slant depth t' at a fixed core distance $r = r_0$, assuming that the azimuth angle correction is small, can be approximated by:

$$t'(r = r_0, \zeta) = t \sec \theta (1 + B \cos \zeta) = t_s + \Delta t_s(\zeta) \quad (4.1)$$

where t is the vertical atmospheric depth at observation level.

A generic time distribution for vertical showers $\tau(r, t)$ depending on atmospheric depth and core distance r , becomes for inclined showers:

$$\tau(r, t) \rightarrow \tau(r, t'(\zeta, \theta)) \quad (4.2)$$

¹As it was shown in section 3.6 at zenith angles smaller than 35° there is an additional effect affecting the asymmetry when using water Cherenkov detector for surface arrays, due to a combination of the geometry of these stations and the arrival direction of individual particles (see section 4.6).

4.2 Asymmetry in the time structure as an indicator of shower evolution

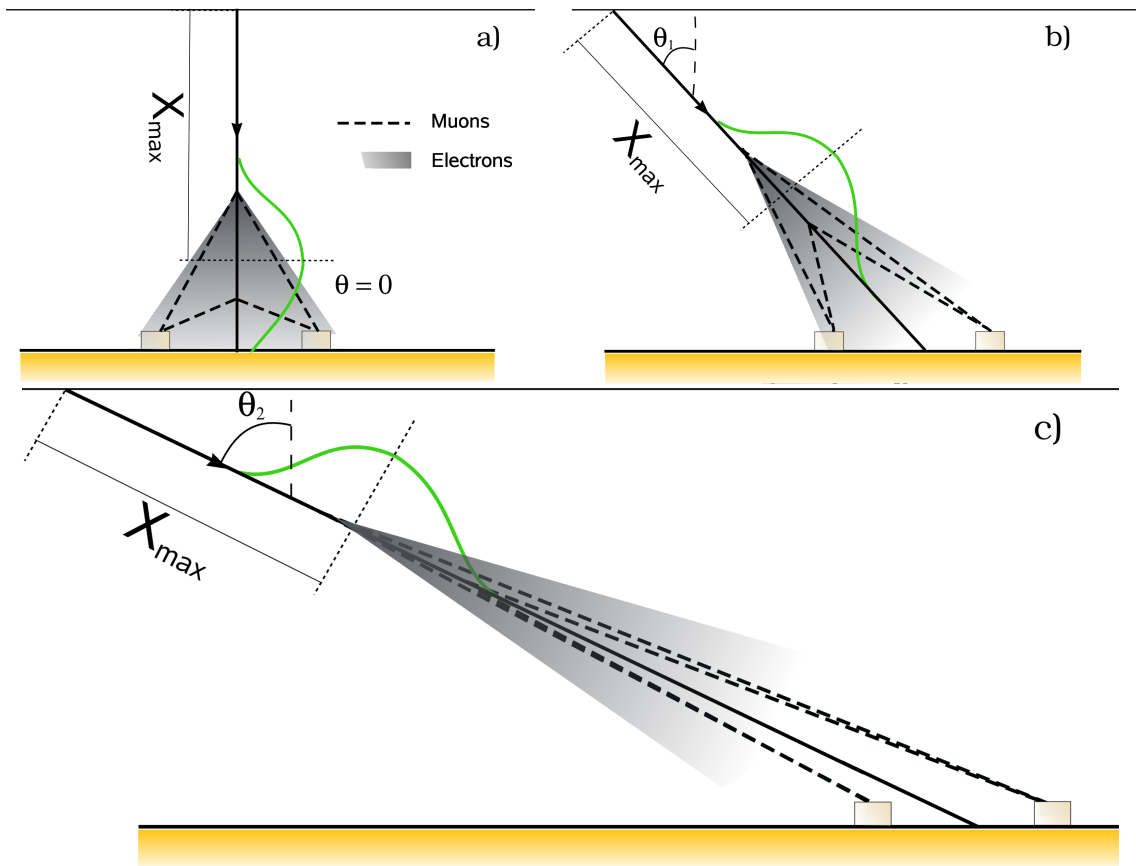


Figure 4.5: Schematic view of shower development when arriving at three different zenith angles



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

Following [154], it is possible to treat this dependence using a Taylor expansion of $\tau(r, t'(\zeta, \theta))$ around $t_s = t \sec \theta$. In a first approximation, since term depending on the azimuth angle is small compared to the total slant depth, only up to the first order is considered which is equivalent to using a linear function in $\cos \zeta$ to describe asymmetries.

$$\tau(r, \zeta) = \tau(r, t'(\zeta)) = \tau(r, t_s + \Delta t_s(\zeta)) \quad (4.3)$$

$$\tau(r, \zeta) = \tau(r, t_s) + \frac{\partial \tau}{\partial t'}|_{t_s} \Delta t_s(\zeta) + \dots \quad (4.4)$$

$$\tau(r, \zeta) = \tau(r, t_s) + \frac{\partial \tau}{\partial t'}|_{t_s} t_s B \cos \zeta + \dots \quad (4.5)$$

$$\tau(r, \zeta) = \tau(r, t_s) \left(1 + \frac{\partial \log \tau}{\partial \log t'}|_{t_s} B \cos \zeta + \dots \right) \quad (4.6)$$

Keeping only the terms up to the first order of the expansion, equation 4.6 can be expressed as:

$$\tau(r, \zeta) = a + b \cos \zeta \quad (4.7)$$

with

$$a = \tau(r, t \sec \theta) \quad (4.8)$$

$$\frac{b}{a} = B \frac{\partial \ln \tau}{\partial \ln t'}|_{t_s} \quad (4.9)$$

The asymmetry factor b/a which depends on t_s , can be used as a measure of the logarithmic variation of τ with slant depth. This parameter is also an indicator of the variation with the slant depth of the ratio between the EM component and the muonic component, and hence it depends on the primary particle type. The variation of the asymmetry factor b/a with $\sec \theta$ will allow to find new observables useful for determining the mass composition. As will be shown below the risetime of the signal follows equation 4.7 and the ratio b/a can be determined experimentally.

4.3 The risetime

The time structure of the shower front is evaluated from the signals generated when the particles traverse the surface detectors of the Auger Observatory. The spread in arrival

times of these particles is measured from a fixed segment of the integrated signal over time. Traditionally, the chosen segment corresponds to the time between the 10% to 50% of the total integrated signal, which is called the risetime $t_{1/2}$. These percentages are the same as the ones used in previous experiments (e.g. Haverah Park [94]) where the signals were recorded using analogue technology. Although there is no physical reason for this particular time interval it has been found very useful to describe the time structure of the shower front¹.

The risetime has been used for composition analysis within the Pierre Auger Collaboration for establishing new limits in the UHE photon [37] and neutrino fluxes (author?) [38]. More recently, as shown in 2.7.4, $t_{1/2}$ parameter is being used to infer the X_{max} value for events measured with the SD. The details of the procedure can be found in [85] and [86].

The use of $t_{1/2}$ in the asymmetry analysis has several advantages. Firstly the fraction of the signal selected in the risetime contains information of both muonic and EM components, which is not the case for other portions of the signal. For example $t_{10\%} - t_{30\%}$ is more dominated by muons and present less asymmetry effect [166] than $t_{1/2}$. On the other hand the falltime, $t_{50\%} - t_{90\%}$ is dominated by the EM component and it is affected by strong fluctuations. Secondly $t_{1/2}$ is a relative measurement, i.e. the value of the risetime depends on the signal deposited by the muons and EM particles in the corresponding tank, for this reason $t_{1/2}$ provides a measure of the ratio between the muon to EM contents in the tank.

4.4 Monte Carlo simulated data

The study of the sensitivity of the method proposed here to separate primary species is based on samples of simulated showers generated with AIRES 2.8.3 [167]. This Monte Carlo code performs a realistic shower simulation which includes electromagnetic algorithms and links to different hadronic models. For air showers with the highest energies the number of secondaries becomes so large ($> 10^{11}$) that simulation is prohibitive in both computing time and disk space. A. M. Hillas [168] introduced a non uniform statistical sampling mechanism which allows to reconstruct the whole EAS from a small

¹Indeed several studies had been carried out for different portions of the signal (see for example [166]).



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

representative fraction of secondaries that are full tracked. This technique is called “statistical thinning”. AIRES includes an extended thinning algorithm, which is explained in detail in [167]. For the high energy interactions two of the most widely used models, QGSJETII(03) [169] and SIBYLL 2.1 [62; 170] were used.

The generated data sample contains 2×10^4 proton and iron¹ showers splitted in energy and angle as follows:

- $\log(E/\text{eV}) = 18.5, 19.0, 19.25, 19.5, 19.75$ and 20 .
- θ (deg) = $32, 36, 41, 45, 49, 53, 57, 60$ and 62 .

The AIRES output consists of a set of ASCII tables with information of the tracked particles and a ground particle file with all the information (energy, direction, impact point, etc) of the particles which reach the observation plane. With the file containing the information of particles at ground level, the simulated response of the Pierre Auger Observatory Surface Detector can be obtained. The simulation of the detector response was carried out with the official Offline framework of the Pierre Auger Observatory [149]. Due to the thinning procedure it is necessary to do some smoothing of the density of particles at ground around the position of each tank. This “unthinning” process is described in (author?) [147]. Recently a study [171]. As explained in 3.5 the software Offline Framework is based in a sequence of modules which perform specific tasks. The sequence for the simulation of a SD Event is shown next.

```
<sequenceFile>
  <enableTiming/>
  <moduleControl>
    <loop numTimes="1" pushEventToStack="yes">
      <module> EventFileReaderOG </module>
      <loop numTimes="1" pushEventToStack="yes">
        <module> EventGeneratorOG </module>
      <!--
        Loop to regenerate particles in batches. The max number
        of particles per batch is set in the shower regenerator
        config file.
      -->
```

¹Details of the simulations can be found in <https://www.auger.unam.mx/AugerWiki/AiresSim>.

```

    <loop numTimes="unbounded" pushEventToStack="no">
      <module> CachedShowerRegeneratorOG </module>
      <module> G4TankSimulatorOG          </module>
    </loop>
    <module> SdSimulationCalibrationFillerOG </module>
    <module> SdPMTSimulatorOG              </module>
    <module> SdFilterFADCSimulatorMTU       </module>
    <module> SdBaselineSimulatorOG         </module>
    <module> TankTriggerSimulatorOG        </module>
    <module> TankGPSSimulatorOG            </module>
    <module> CentralTriggerSimulatorXb     </module>
    <module> CentralTriggerEventBuilderOG  </module>
    <module> EventBuilderOG                </module>
    <module> EventFileExporterOG          </module>
  </loop>
</loop>
</moduleControl>
</sequenceFile>

```

In this sequence all the necessary tasks needed to simulate an Auger event are shown, from the unthinning (CachedShowerRegenerator), the tank simulation (G4TankSimulatorOG), the necessary calibration constants, the trigger simulation for that event, etc.

The Offline Framework was also used for the reconstruction of the simulated event. The reconstructed values of the primary energy, arrival direction and core location of the shower have been used in this analysis, to treat the simulated data as if it were real data.

A typical example of the reconstructed footprint, in the shower plane, of one of the simulated events is shown in figure 4.6. In particular, this example corresponds to a 10^{19} eV proton shower with zenith angle $\theta = 45^\circ$. The triggered tanks are represented by circles with size is proportional to the signal intensity. The line indicates the incoming direction of the primary. In figure 4.7 it is shown the simulated signal produced by the muonic and electromagnetic components for an early station (right) and a late one (left) of the previous event. It is clearly seen the smaller contribution of the electromagnetic component in the late station than in the early one.



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

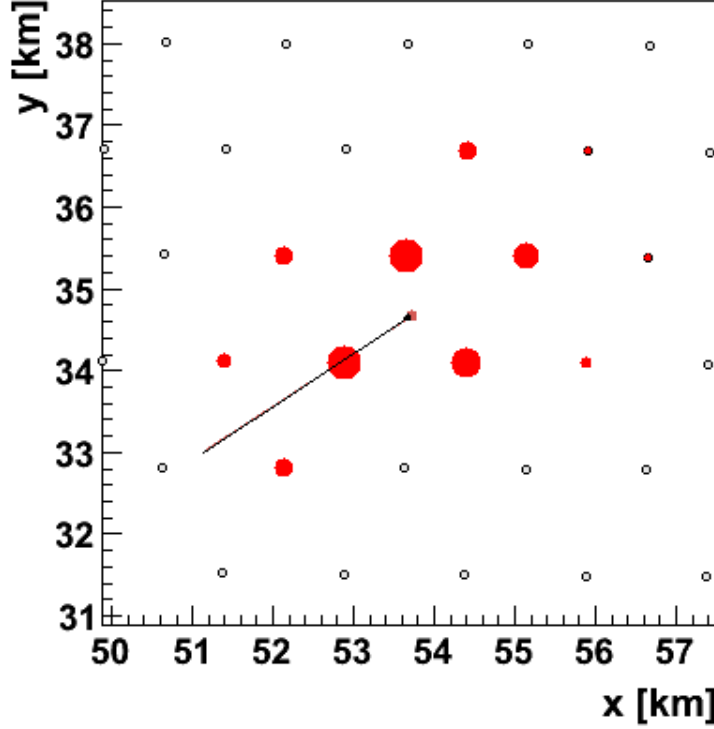


Figure 4.6: Footprint of a 10^{19} eV proton shower with $\theta = 45^\circ$. The triggered tanks are represented with circles with area proportional to the signal intensity.

In general for any analysis involving data taken with the SD of the Pierre Auger Observatory several cuts are applied to ensure the quality of data. Firstly the reconstructed events are required to satisfy the so-called T5 trigger level (see section 3.4), which assures that the station with the largest signal is surrounded by six working stations. So as to be in the regime of detector full efficiency a minimum energy of 3.16 EeV and a maximum zenith angle of 62° are required. In addition, several quality cuts have been applied to discard badly reconstructed events.

In addition to the cuts applied to the event selection, quality cuts on station data are also necessary. These cuts are related to the core distance and the total signal of the station. Essentially the stations are required to be far enough from the core to avoid saturation problems. Currently, saturation of the SD stations occurs for signals larger than 1000 VEM. Stations at very large core distance are also discarded to prevent highly fluctuating signals and assure the station is inside the region for which risetime

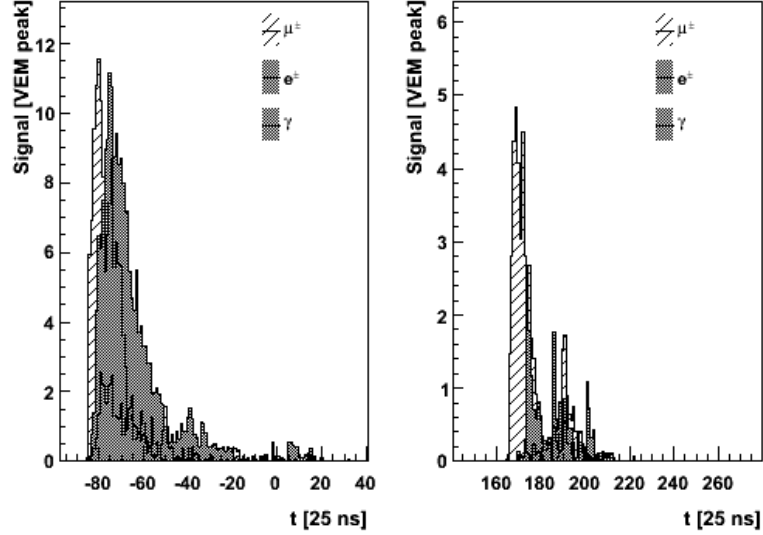


Figure 4.7: Simulated signal produced by the muon and electromagnetic components in an early station (right) and a late one (left).

grows linearly with r . A lower cut in the integrated signal is applied to reject noisy stations.

In section 4.6 the analysis of the efficiency and the discrimination power as a function of the core distance and total signal of the stations will be shown. In this analysis the following cuts on the SD stations have been found to be optimum for mass separation.

- Core distance r within the interval $0.5 - 2.0$ km.
- Integrated signal size $S > 10$ VEM.

4.5 Asymmetry parameters for mass analysis

In this section the longitudinal development of the asymmetry will be defined and several related parameters will be presented as well as their sensitivity to the primary mass analyzed.

Due to the intrinsic fluctuations of extensive air showers and the limited sampling of the shower front recorded by the SD, it is not possible to obtain the mass composition on a shower-by-shower basis. That is, instead of measuring the asymmetry in individual showers, the approach used in this work consists of using the mean value



of the asymmetry factor, as defined above 4.9, for all showers in a certain interval of energy and zenith angle.

4.5.1 The longitudinal asymmetry development

As anticipated in 4.2 the longitudinal evolution of the asymmetry is defined as the evolution of the asymmetry factor b/a with $\sec \theta$. This evolution is reminiscent of the well known longitudinal shower development usually measured by fluorescence detectors. The procedure used to get the average longitudinal development of the asymmetry, can be summarised in the following steps:

1. Select events in bins of reconstructed energy and θ values. For these, the risetime of those stations passing the corresponding cuts is determined. For each bin of E , $\sec \theta$ and ζ , the mean value and the standard deviation of the corresponding $t_{1/2}/r$ distribution, is calculated. It is worth mentioning that risetime grows linearly with the core distance (see section 4.3) and thus the $t_{1/2}/r$ value is more sensitive to asymmetry and a large range of r can be used for the analysis.
2. For each (E, θ) bin, a fit of $\langle t_{1/2}/r \rangle$ to a linear cosinus function of ζ (eq 4.7) provides the asymmetry factor b/a . Figure 4.8 shows an example of the dependence of $\langle t_{1/2}/r \rangle$ on the azimuthal angle of Monte Carlo samples for both primaries, proton (left) and iron (right) at $E = 10^{19}$ eV and four zenith angles, 32° , 45° , 53° and 60° (from top to bottom). The lines represent the result of the fits to eq. 4.7. From these plots it can be seen that the average value of $\langle t_{1/2} \rangle$ over ζ decreases with the zenith angle as expected. Besides, the asymmetry increases with the zenith angle up to a maximum value and then decreases for larger angles.
3. For each primary type and energy interval the dependence of the asymmetry factor on $\sec \theta$ is obtained, that is the longitudinal development of the asymmetry (see figure 4.9).

The values of b/a versus $\sec \theta$ are fitted to a Gaussian function, which plays the same role as the Gaisser-Hillas function in the longitudinal shower development. The asymmetry development can be described by means of three parameters as indicated in Figure 4.9. XAsymMax, the position of the maximum asymmetry, i.e the $\sec \theta$ value

4.5 Asymmetry parameters for mass analysis

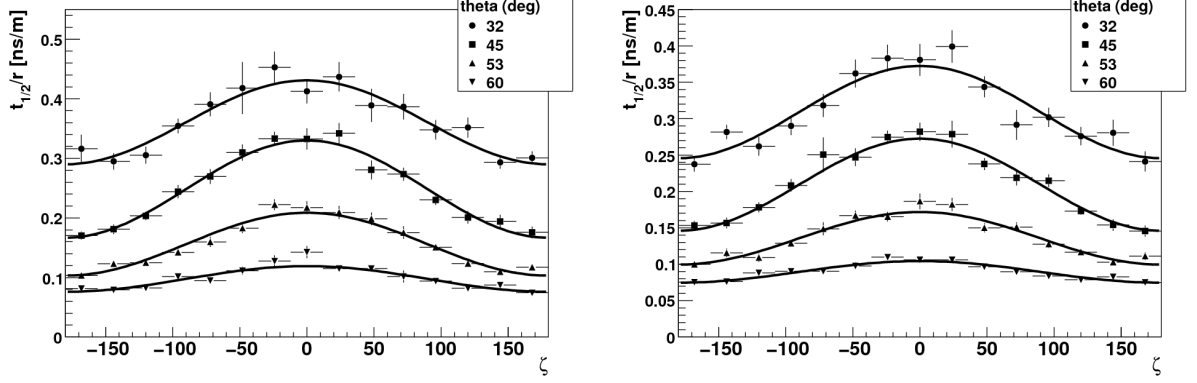


Figure 4.8: Risetime vs. ζ for primary energy 10 EeV and different angles (32° , 45° , 53° and 60° , from top to bottom) for protons (left) and iron nuclei (right).

for which b/a maximizes; AsymHeight, the height at maximum, i.e. the maximum b/a value for the corresponding energy, and XAsymWidth the half width at half maximum of the Gaussian function, related with the speed of the evolution of the b/a parameter.

In figure 4.10 the asymmetry longitudinal development at primary energies $\log(E/\text{eV}) = 18.5, 19.0, 19.25, 19.5, 19.75$ and 20 . (from top to bottom and from left to right) for proton (red dashed line) and iron (blue solid line) primaries is shown. It can be seen that the asymmetry evolution is different for proton and iron primaries. Protons have larger values of XAsymMax and AsymHeight while the XAsymWidth seems to be the same for both primaries.

The different behavior of the longitudinal development of the asymmetry can be understood in terms of the different longitudinal development and the muon content of showers initiated by each primary type. For the same primary energy proton initiated showers penetrate deeper into the atmosphere than iron ones. That is at the same zenith angle and primary energy proton showers are “younger” at ground as compared with iron ones, and therefore a larger fraction of electrons survive reaching the observation level. For this reason proton showers needs to go through a larger amount of atmosphere to reach the point where the attenuation, between early-late regions reaches the maximum. This fact translates into larger values of XAsymMax for proton showers.

On the other hand, iron showers have more muons than proton showers at the same energy and the contribution of electrons to the total signal is smaller at the detector



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

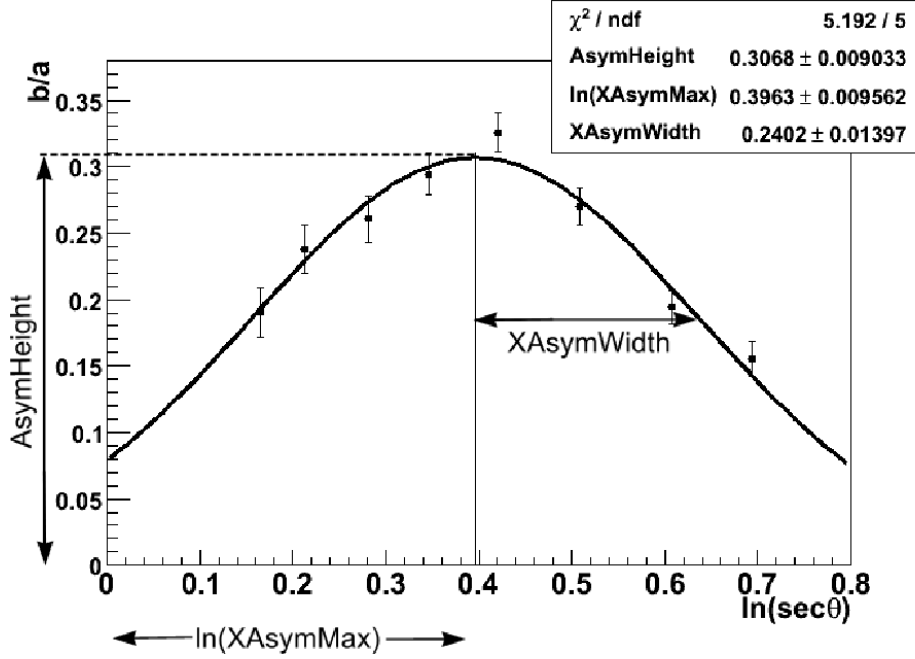


Figure 4.9: Asymmetry longitudinal development for 10^{19} eV protons. The three asymmetry parameters describing the longitudinal development are shown.

level. Therefore the muon to electron signal ratio increases and the asymmetry factor, b/a , decreases as well as AsymHeight which also decreases for iron showers.

The XAsymWidth parameter takes similar values for both primaries due to the fact that the speed of the asymmetry development depends on the evolution of the EM component in the atmosphere which is the same for both proton and iron showers.

4.5.2 Energy dependency and hadronic models

Once the parameters of the asymmetry longitudinal development are obtained, their dependence with the primary energy can be studied. In figures 4.11, 4.12 and 4.13 the values of the parameters from the above mentioned fit have been represented against the primary energy. Results for both hadronic models are included in the plots. The error bars of the points come from the fitting uncertainties.

As shown in figure 4.11 XAsymMax grows linearly with $\log E$. This is the expected behavior taking into account that the X_{max} value increases with $\log E$ (see section 2.7), and therefore as the energy increases the position of the shower maximum is closer to the ground and thus more EM particles reach the detector level. As a consequence the

4.5 Asymmetry parameters for mass analysis

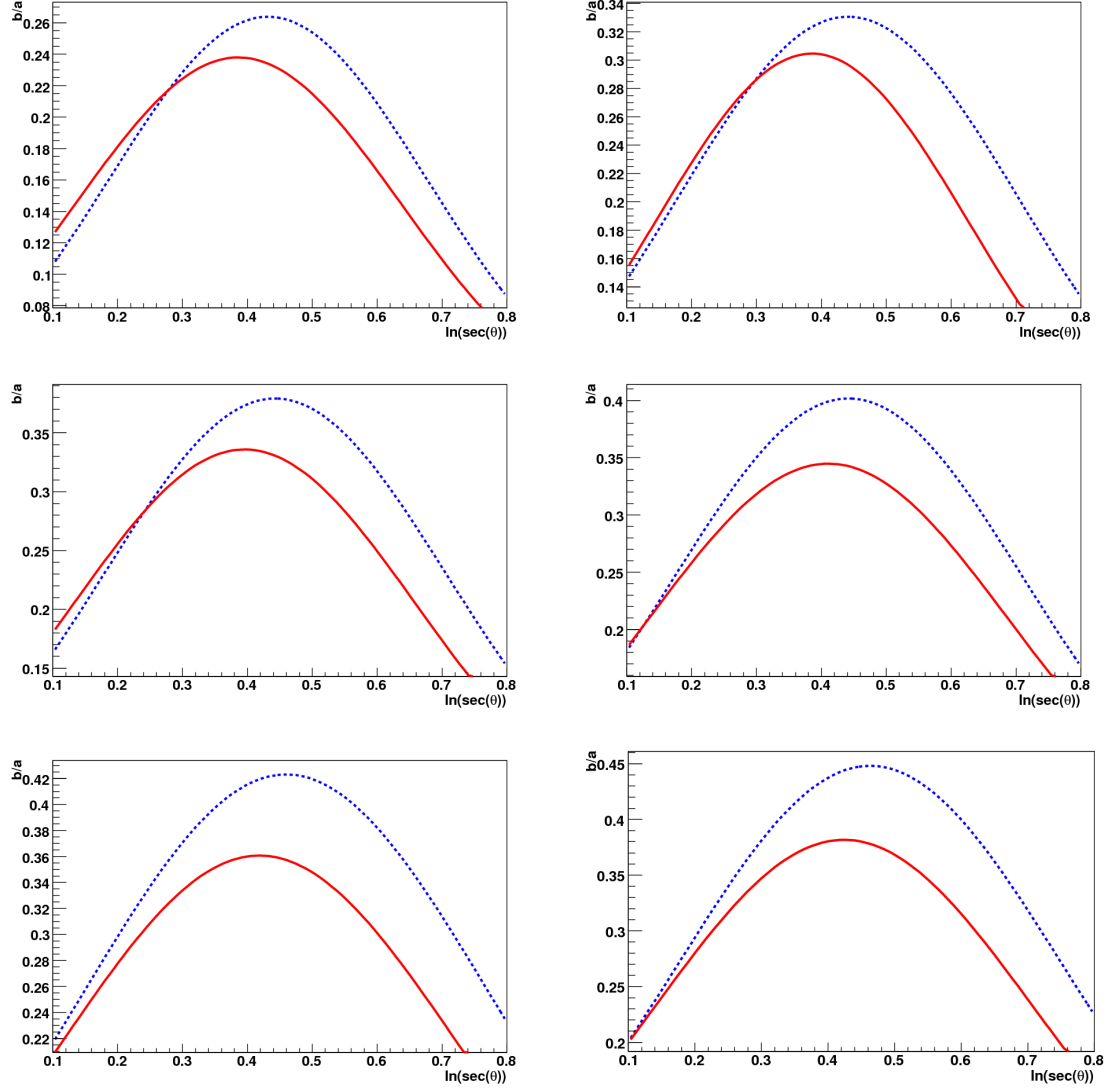


Figure 4.10: Asymmetry longitudinal development at primary energies 3.16, 10, 17.8, 31.6, 56.3 and 100 EeV (from top to bottom and from left to right). Monte Carlo protons (red dashed line) and iron nuclei (blue solid line) are shown.



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

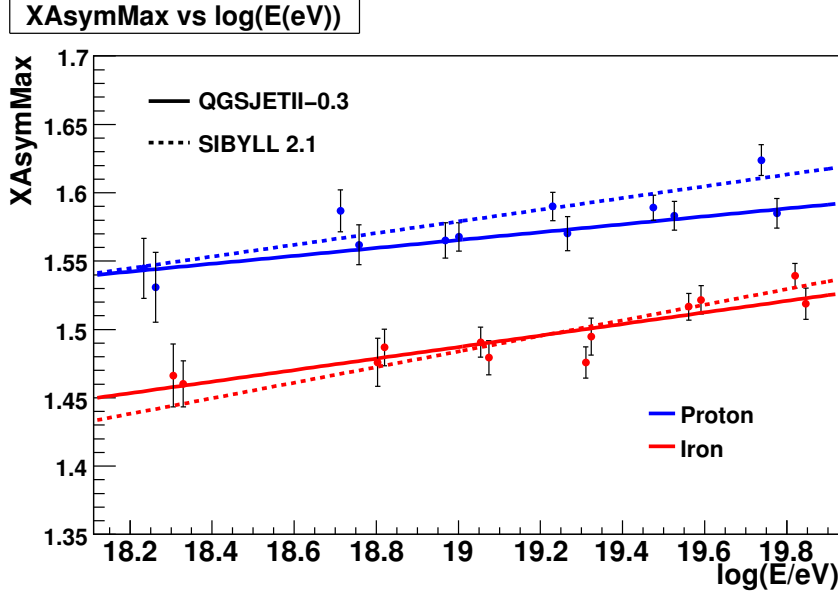


Figure 4.11: XAsymMax vs. $\log(E/eV)$ for both hadronic models; SIBYLL and QGSJETII. The points are the result of the MC analysis for both proton and iron. Lines represent the fit to a linear function.

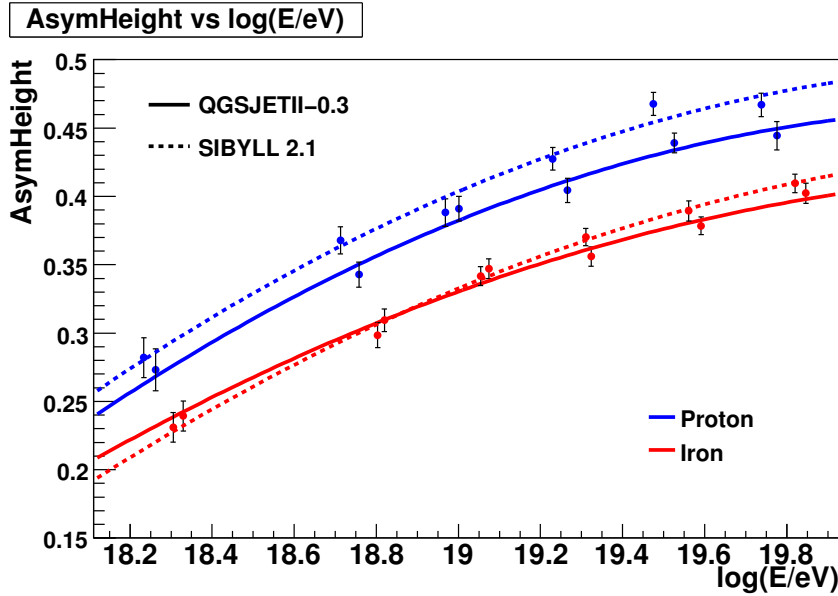


Figure 4.12: AsymHeight vs. $\log(E/eV)$ for both hadronic models; SIBYLL and QGSJETII. The points are the result of the MC analysis for both proton and iron. Lines represent the fit to a quadratic function.

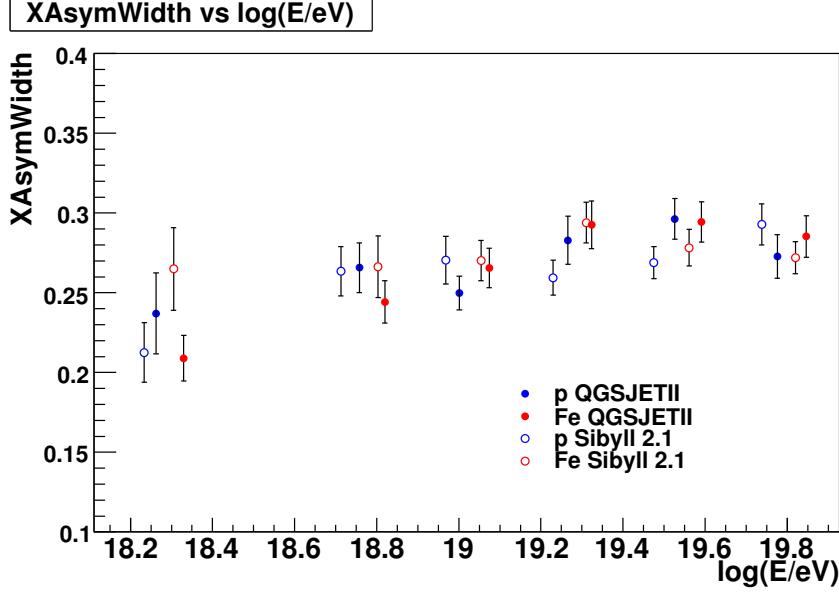


Figure 4.13: XAsymWidth vs. $\log(E/\text{eV})$ for both hadronic models; SIBYLL and QGSJETII. The points are the result of the MC analysis for both proton and iron.

zenith angle for which the relative attenuation early/late maximizes (i.e. XAsymMax) increases. The similarities between the energy dependence of X_{max} and XAsymMax are clearly seen. The corresponding linear fits (continuous lines) of both primary types are clearly separated, thus allowing discrimination of heavy and light primaries.

On the other hand the AsymHeight parameter, figure 4.12, shows a dependence with $\log E$ which nearly follows a parabolic function. The number of both electrons and muons increases with the energy but in a different way (see section 2.6.4). The number of electrons increases more rapidly ($N_e \propto E$) than the number of muons ($N_\mu \propto E^{0.9/0.85}$). Therefore the asymmetry factor increases with the energy and the AsymHeight also increases.

Finally, XAsymWidth is nearly independent on the energy with a value very similar for both primaries and therefore this parameter does not allow separation between primaries.

As can be seen in the previous figures, predictions for iron do not show a remarkable dependence on the hadronic interaction model. On the contrary, there is a clear difference for proton primaries, particularly at high energies. This is also a feature of the X_{max} vs E plots (elongation rate) where a similar behavior at high energies is



observed for both models [172].

4.5.3 Relationship of the asymmetry parameters with X_{max}

As it was shown in the previous section the behaviour of XAsymMax reminds that of the X_{max} parameter. Thus, it is desirable to study the correlation between the asymmetry mass-sensitive parameters measured with a surface detector and the shower maximum depth. To this end, the steps described above were repeated but instead of grouping events by primary energy, they were grouped in bins of X_{max} . The X_{max} values used in these plots, are those of the simulated extensive air shower.

In figure 4.14 the correlation with X_{max} is shown for the three parameters using both hadronic interaction models QGSJETII(03) and SIBYLL 2.1. As expected from figures 4.11 and 4.12, XAsymMax and AsymHeight present a strong correlation with X_{max} which is nearly independent on the hadronic model. This is an encouraging result confirming that the observed azimuthal asymmetry is a reliable mass estimator. Certainly the correlation of the SD parameters with the position of the shower maximum might be also useful to provide a measurement of $\langle X_{max} \rangle$ using only the data of surface detector. See section 5.4 for a more detailed study including correlation in real data.

4.6 Cut optimization from sensitivity and efficiency

At energies above 3.16 EeV the detector is fully efficient for any type of primary, and therefore in principle it is possible to choose the station selection criteria so as to maximise the separation between proton and iron. For example close to the core there is no asymmetry, between early-late stations, therefore including close station will not add any useful information for primary separation. The same happens if stations with low signal and far for the core are included in the analysis, in this case the fluctuations and effects of the detector response make difficult to extract useful information.

4.6.1 Sensitivity

The optimum combination of criteria for station selection is obtained on the basis of the best sensitivity to primary discrimination. For this purpose it is possible to define a sensitivity parameter as the difference between the values of the three asymmetry

4.6 Cut optimization from sensitivity and efficiency

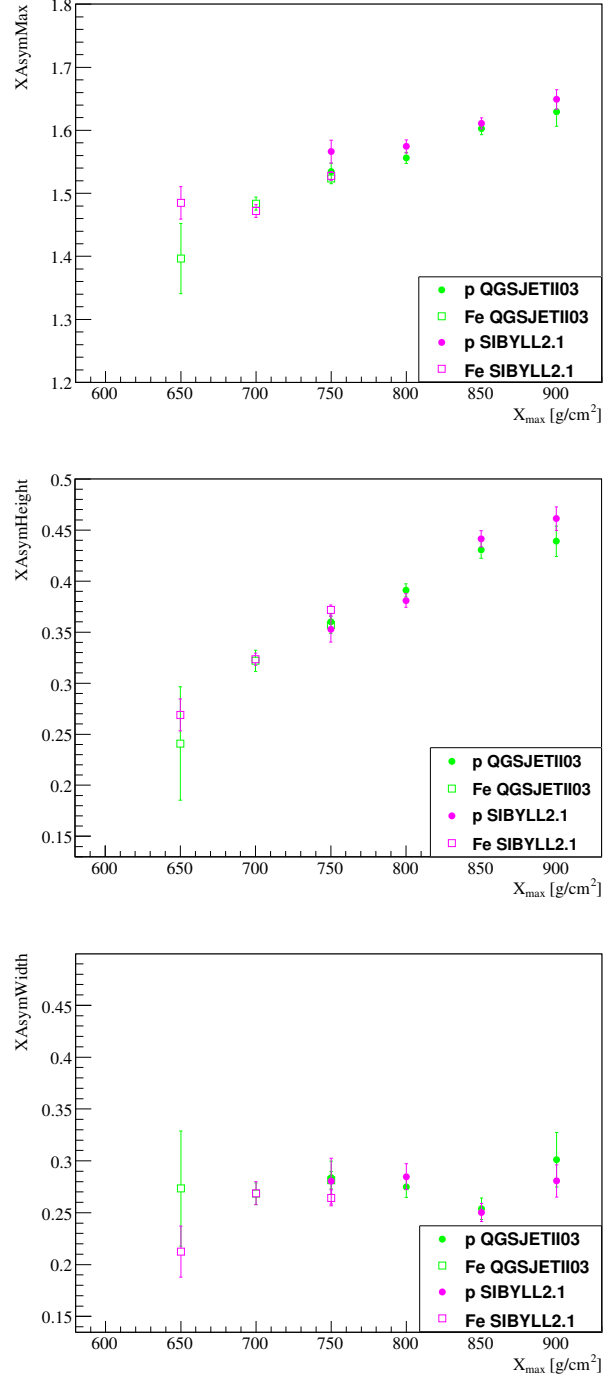


Figure 4.14: Correlation between $\langle X_{max} \rangle$ and the three parameters of the asymmetry longitudinal development: XAsymMax (upper plot), AsymHeight (middle plot) and XAsymWidth (lower plot) for two different primaries, iron (squares) and proton (solid dots), and for two hadronic models, QGSJETII-03 (green) and SIBYLL 2.1 (pink).



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

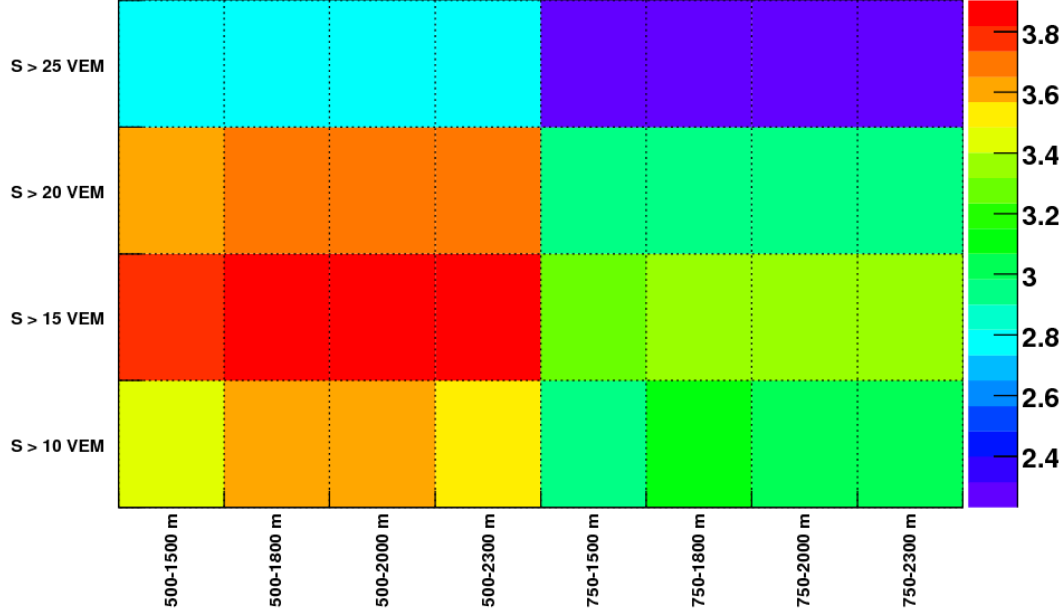


Figure 4.15: Sensitivity parameter (equation 4.10) for different sets of cuts for 10^{19} eV showers.

parameters for proton and iron measured in units of their corresponding errors. As will be discussed in the next chapter the $XAsymMax$ parameter has turned out to be the more efficient and reliable one and therefore the sensitivity parameter is defined as:

$$S = \frac{XAsymMax_p - XAsymMax_{Fe}}{\sqrt{(\Delta XAsymMax_p)^2 + (\Delta XAsymMax_{Fe})^2}} \quad (4.10)$$

The subindex p and Fe represents proton and iron respectively and $\Delta XAsymMax$ their errors obtained from the fit to a gaussian function. The S parameter is calculated for different combinations of cuts in core distance and signal size of the station, and the optimum combination is selected for the one which maximizes S . As an example, results of equation 4.10 for different sets of cuts are presented in figure 4.15 for 10^{19} eV. In general it is possible to achieve proton-iron separation ($S > 1$) for any set of reasonable cuts.

4.6.1.1 Efficiency

The cuts in core distance and signal of the stations has to be optimized to the value of maximum sensitivity that is the maximum separation between proton and iron, but under the constrain of keeping the efficiency at a reasonable level and ensuring no primary mass bias due to the selection is produced. In other words the number of stations selected after the cuts for proton and iron showers are the same.

To this end an efficiency parameter has been defined as the number of stations that survive the cuts divided by the complete sample of the stations without applying any cut:

$$\epsilon = \frac{\# \text{Stations after cut}}{\# \text{Stations with no cuts}} \quad (4.11)$$

Simultaneously to the sensitivity the efficiency of each set of cuts for proton and iron showers has been calculated. In figure 4.16 the efficiency (equation 4.11) for different cuts as a function of the zenith angle and for both primaries is shown. As shown in this figure the efficiency decreases with increasing zenith angle due to the lower signal in stations for high zenith angles. At higher energies this effect is no longer present since even at large angles the signal of the stations is high enough to fulfill the required cut. In figure 4.17 it is shown the efficiency mixing all the zenith angles for the same set of cuts (as in figure 4.15) for proton (left) and iron (right) showers at 10^{19} eV. In these plots no significant bias is observed on the selection toward either of the primary species. A possible increase in the efficiency could be obtained when stations at higher distances from the core are allowed, but this is not correlated with an increase in the sensitivity. Taking into account the above analysis on sensitivity and efficiency the optimum combination of cuts in the signal size and core distance interval is:

- Stations are required to be in the core distance interval 0.5 - 2 km.
- A lower cut in the integrated signal of 10 VEM is applied.

4.7 Extracting the mean primary mass

As already shown, the asymmetry longitudinal development, described by the three parameters defined above, is sensitive to mass composition. The dependency of these



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

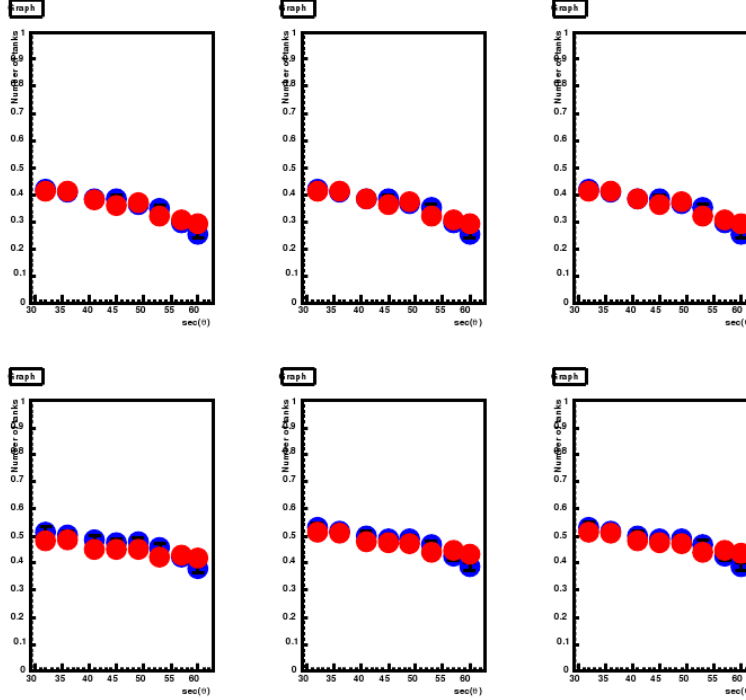


Figure 4.16: Efficiency (equation 4.11) versus zenith angle for different cuts in signal and core distance for 10^{19} eV primaries for both primaries proton (red) and iron (blue). Cuts in signal intensity are $S > 15$ VEM (top) and $S > 10$ VEM (bottom). The core distance intervals correspond to 500-1500 m, 500-1800 m and 500-2000 m (from left to right).

parameters on composition has been determined as follows. Firstly a two component composition (proton-iron) has been assumed. For each interval in energy, $\sec \theta$ and ζ , the risetime for proton and iron showers has been obtained following the procedure described in section 4.5. The corresponding risetime for a number of mixtures have been inferred as the average value for both components weighted with the corresponding composition factors.

The percentage of iron and proton events was changed in steps of 10%. As an example, the asymmetry longitudinal development curves at 10^{19} eV showers for different composition are shown in figure 4.18. A smooth transition between pure proton to pure iron compositions can be observed. This procedure allows to determine the dependence of the three parameters with composition, e. g. the iron fraction of the sample, named in the following X_{Fe} . The result for 10^{19} eV is shown in figure 4.19. These plots show that XAsymMax and AsymHeight present a strong correlation with X_{Fe} while there is

4.7 Extracting the mean primary mass

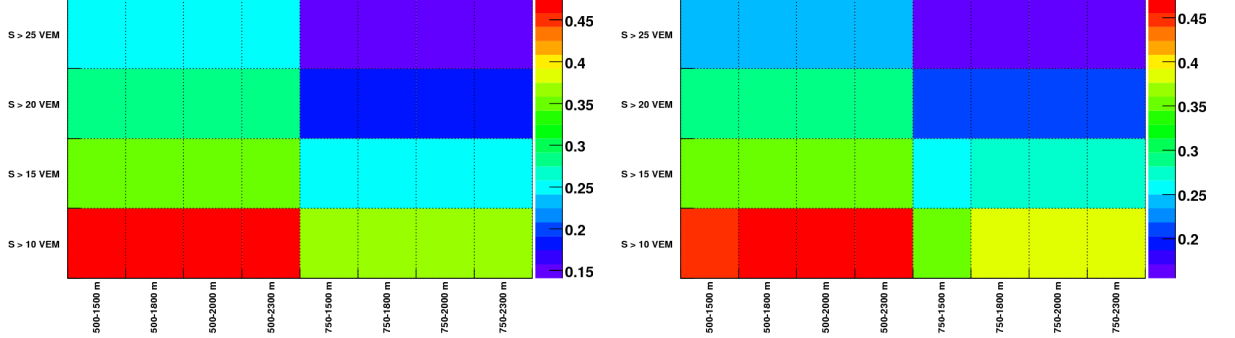


Figure 4.17: Efficiency (equation 4.11) for different cuts in signal and core distance for 10^{19} eV primaries for protons (left) and iron (right).

almost no correlation for XAsymWidth, as expected from the results of section 4.5. For all energy intervals studied, the three parameters follow a linear behavior, decreasing with the iron fraction. The result for a linear fit of these three separation parameters is also shown in 4.19 (solid line). The grey area in this figure represents the statistical uncertainties. The parameterisation of this dependence will be used later to extract the primary mass.

In principle the composition of a sample, i.e. the X_{Fe} value (either a Monte Carlo test sample or real data) can be determined by minimizing the following function

$$\Delta^2 = \frac{(m_s - f_m(X_{Fe}))^2}{\Delta m_s^2 + \Delta f_m^2} + \frac{(\sigma_s - f_\sigma(X_{Fe}))^2}{\Delta \sigma_s^2 + \Delta f_\sigma^2} + \frac{(h_s - f_h(X_{Fe}))^2}{\Delta h_s^2 + \Delta f_h^2} \quad (4.12)$$

where σ , h and m stand for the parameters XAsymWidth, AsymHeight and XAsymMax, respectively; the subindex s indicates the corresponding values of the sample to be studied. $\Delta\sigma$, Δh and Δm represent the standard deviation of the separation parameters obtained from the gaussian fit to the asymmetry longitudinal development. The functions $f_i(X_{Fe})$ stand for the parameterization of the dependence of each parameter with the iron fraction X_{Fe} obtained in figure 4.19. However as already mentioned the XAsymWidth parameter is not sensitive to primary mass. In addition in the next chapter it will be shown that XAsymMax is by far the most reliable parameter for mass composition. Therefore in practice only the first term in equation (4.12) will be used (see chapter 5).

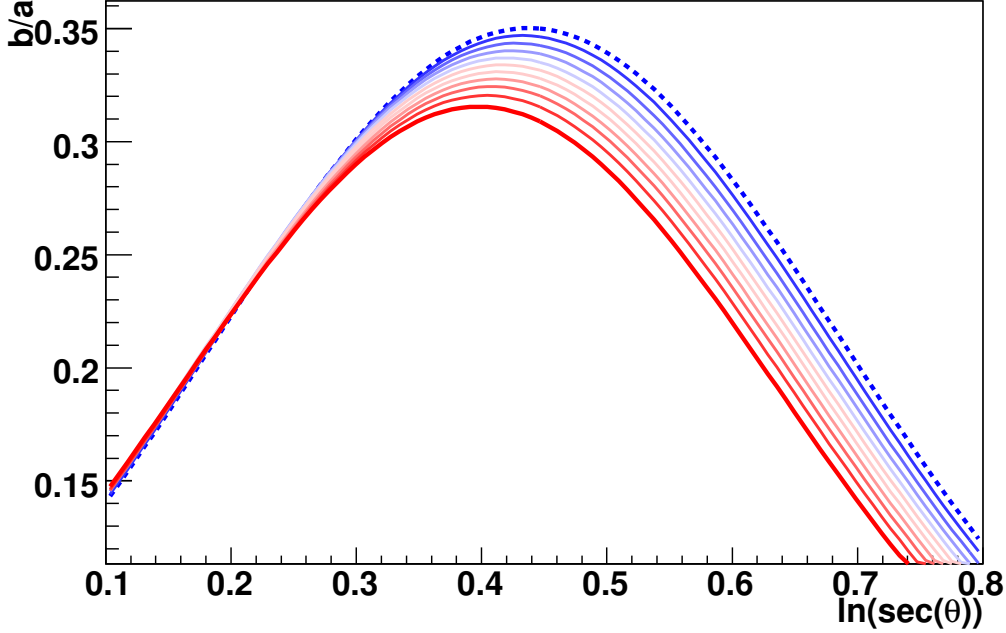


Figure 4.18: Asymmetry development for the different samples with mixed composition, from pure proton (dashed blue) to pure iron (continuous red) in steps of 10% for primary energy 10^{19} eV.

To check the reliability of this technique the X_{Fe} have been reconstructed for several samples containing a known fraction of proton and iron showers of the Monte Carlo sample by minimising Δ^2 . This test has been carried out using mixtures of showers containing the same events used previously to get the results of figure 4.19. The results are shown in figure 4.20 for three different samples: 100% iron, i.e. $X_{Fe} = 1$, (upper curve), 50% proton and 50% iron, i.e. $X_{Fe} = 0.5$, (middle curve) and 100% proton, i.e. $X_{Fe} = 0$, (lower curve). Deviations of reconstructed X_{Fe} values from the real ones are below 0.04 (i.e errors in the iron percentage below 4 units). The error bars represent the uncertainty due to the limited statistics and amount about $\Delta X_{Fe} = 0.1$.

4.7 Extracting the mean primary mass

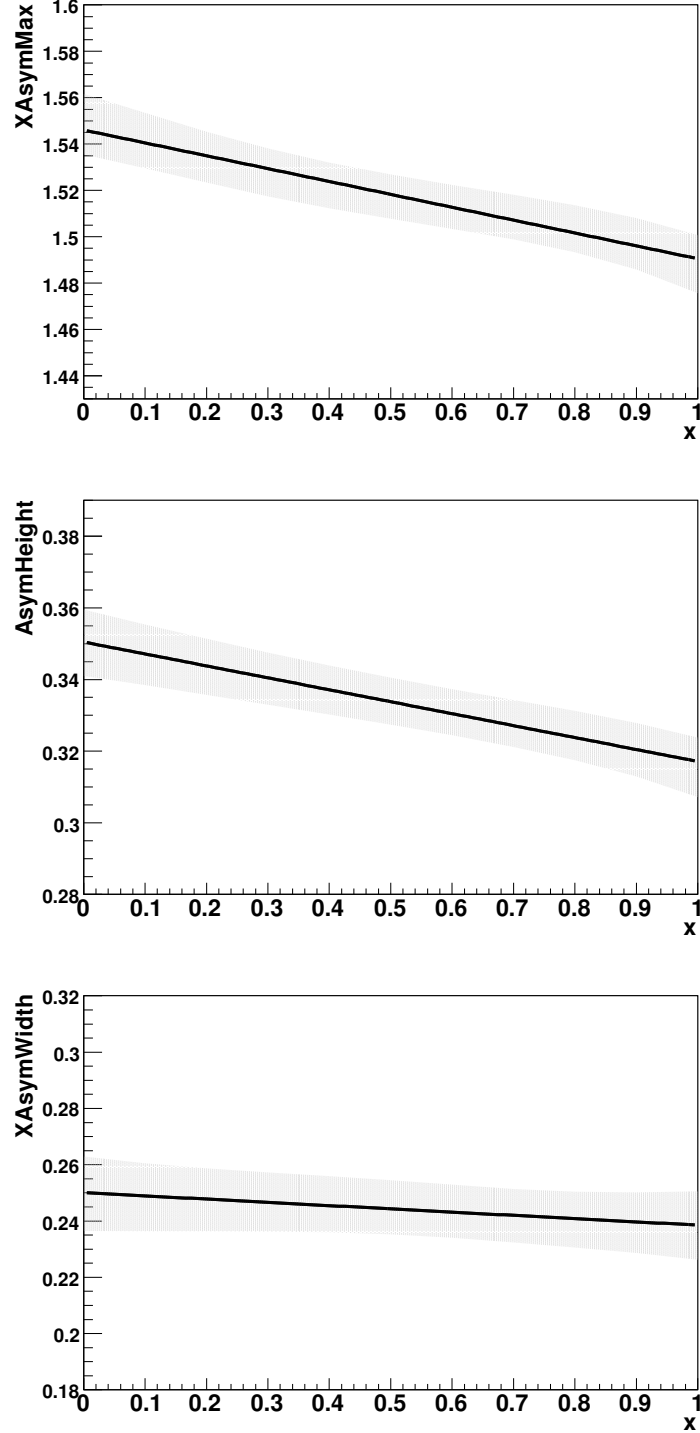


Figure 4.19: Position of the maximum, width and height of the longitudinal asymmetry development as a function of the mixture fraction X_{Fe} , for a primary energy of 10^{19} eV. $X_{Fe} = 1$ corresponds to 100 % iron and $X_{Fe} = 0$ to 100 % proton.



4. MASS DEPENDENCE OF THE TIME ASYMMETRY

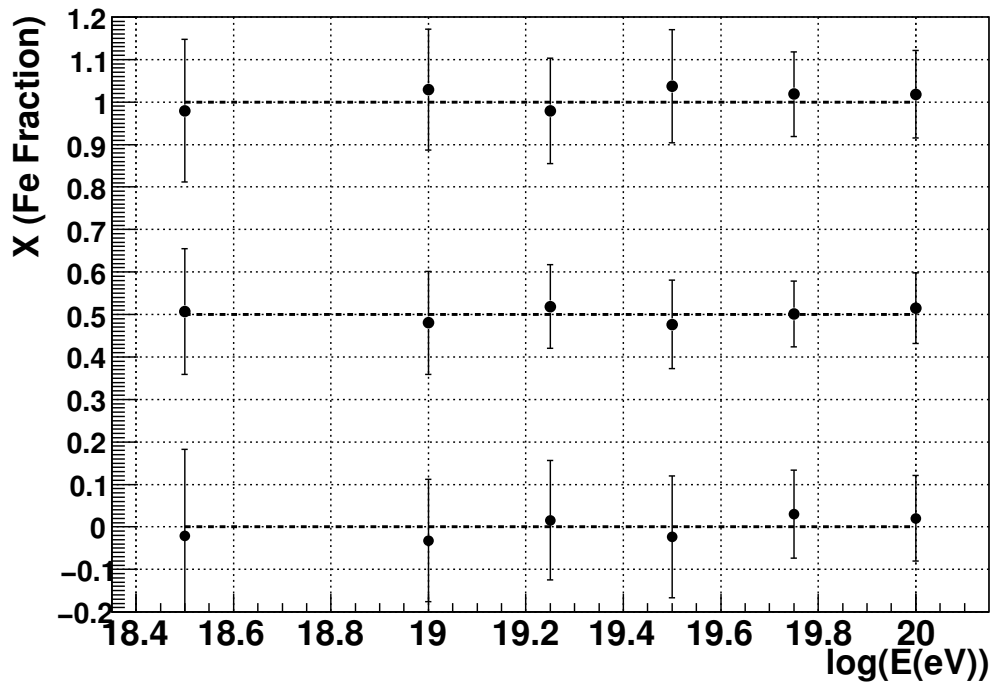


Figure 4.20: Consistency check of iron fraction with three different samples: 100% iron(upper curve), 50% iron and 50% proton(middle curve), 100% proton(lower curve).

Determination of Composition

5.1 Introduction

As already pointed out in 2.7 the knowledge of the mass composition of cosmic rays is deficient above 10^{17} eV. While the measure of the direction and energy of cosmic rays are relatively easy, the determination of the nature of primary particles from the features of the extensive air showers is very difficult. Primary photons and neutrinos produce showers markedly different from that of nuclei. However nuclear species, even with large mass difference (e.g. proton and iron), are not easy to distinguish. On average heavier-nuclei initiated showers develop higher in the atmosphere, due to their larger cross-section and the lower energy per nucleon, and they have higher muon-to-electron ratio at observation level. These differences are small and difficult to observe because of the uncertainty in the primary energy and shower to shower fluctuations.

As mentioned in section 2.7.1 the variation of X_{max} with the primary energy can be used to measure the mass composition of UHECRs. Hybrid events of the Pierre Auger Observatory provide direct measurement of the X_{max} value. Nevertheless, 90% of the total number of registered events have only information from the surface array and therefore, observables sensitive to the mass of the primary cosmic ray measured by the SD are very useful for composition analysis, in particular at the highest energies for which the statistics of the FD events is very poor.

In chapter 4 a mass composition method based on the $t_{1/2}$ asymmetry measured by the surface detector was presented. The main objective of this thesis is to obtain the mass composition of UHECRs from the data recorded by the Pierre Auger Observatory



5. DETERMINATION OF COMPOSITION

using this new technique. We will probe that this asymmetry method is a very powerful tool for mass composition.

This chapter is organized as follows. In section 5.2 the sample of real events used in the analysis is described and the steps involved in the analysis are detailed. To obtain the composition of UHECRs, a comparison with MC predictions, as will be done in section 5.3, is necessary. In section 5.4, the calibration of the asymmetry parameter X_{AsymMax} with X_{max} will be performed using hybrid events. The mass composition obtained with the asymmetry technique will be presented in section 5.5, also in this section several systematics studies will be shown.

5.2 Asymmetry analysis on data

5.2.1 Event Selection

In this work risetime data collected using the Pierre Auger surface detector from 1 January 2004 until March 2008 are used. The events included in the selection are those which have passed the T4 and T5 trigger levels (see section 3.4.2), with zenith angle in the range 30 to 62°. The array is 100% efficient for cosmic ray detection for energies above 3.16 EeV, as can be seen in figure 5.1. Although the array is not 100% efficient for energies below 3.16 EeV, the analysis was extended to 1 EeV.

The data sample is splitted in energy and zenith angle as follows:

- Zenith angle: From 30° to 62° in ten equal bins.
- Energy $\log(E/\text{eV})$:
 - From 18.0 to 18.4 in bins of 0.10.
 - From 18.4 to 19.2 in bins of 0.15.
 - 19.2 - 19.5 and > 19.5 .

As mentioned in section 3.5 the Offline framework allows to treat simulations and real data with the same software chain, so the code developed for the asymmetry analysis in simulated events can be used for the data recorded by The Pierre Auger Observatory. The Auger Software Offline RandomTask - v2r4p1 has been used for both the generation of ground signals for the tanks in the Auger Observatory and

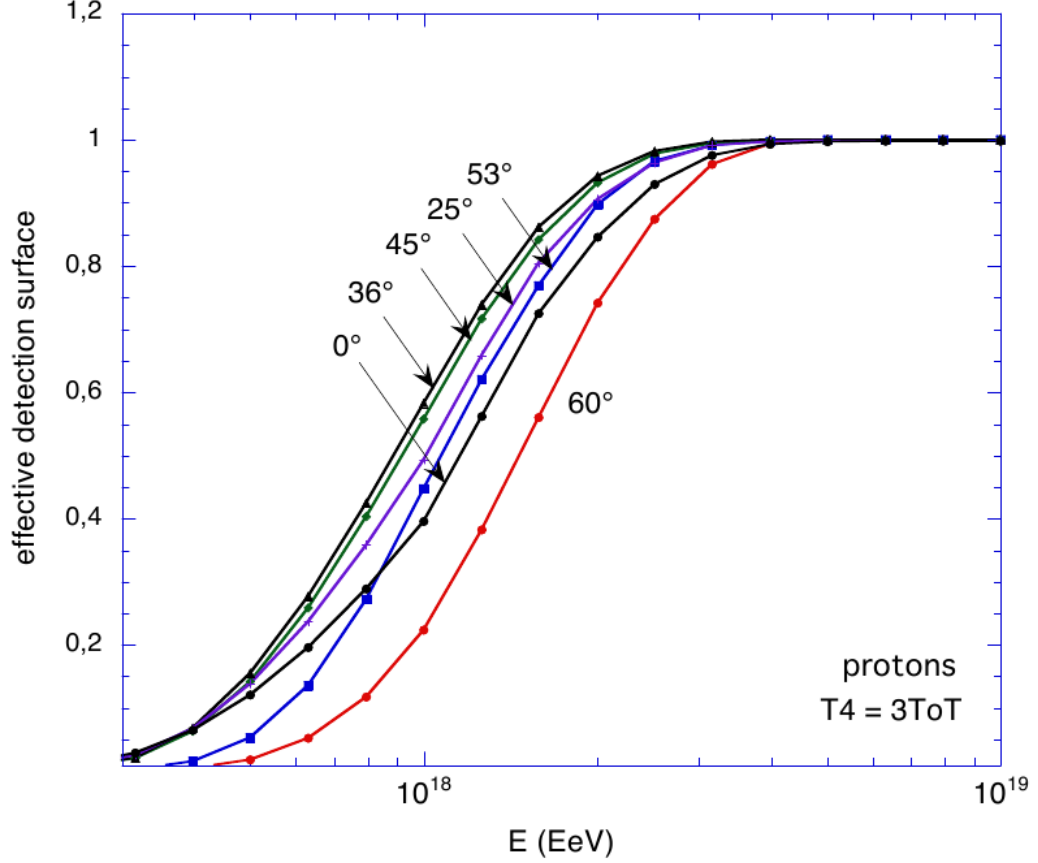


Figure 5.1: Detection efficiency of protons showers as a function of the energy for the Pierre Auger Observatory surface array using the T5 trigger condition. The effective detection area becomes saturated at 3.16 EeV ($\log(E/\text{eV}) = 18.5$).

for the reconstruction of the various shower parameters. Reconstructed values of the primary energy, arrival direction and core location of the shower have been used for this analysis. In addition the risetime has been determined with this version of the Offline as the average value of the three PMTs of the station.

5.2.2 Asymmetry longitudinal development and asymmetry parameters

As will be shown below the asymmetry features predicted by the simulations are fully consistent with those observed in real data. The modulation of $\langle t_{1/2}/r \rangle$ with ζ is present in real data and the dependence of the asymmetry factor b/a with $\sec\theta$ follows the



5. DETERMINATION OF COMPOSITION

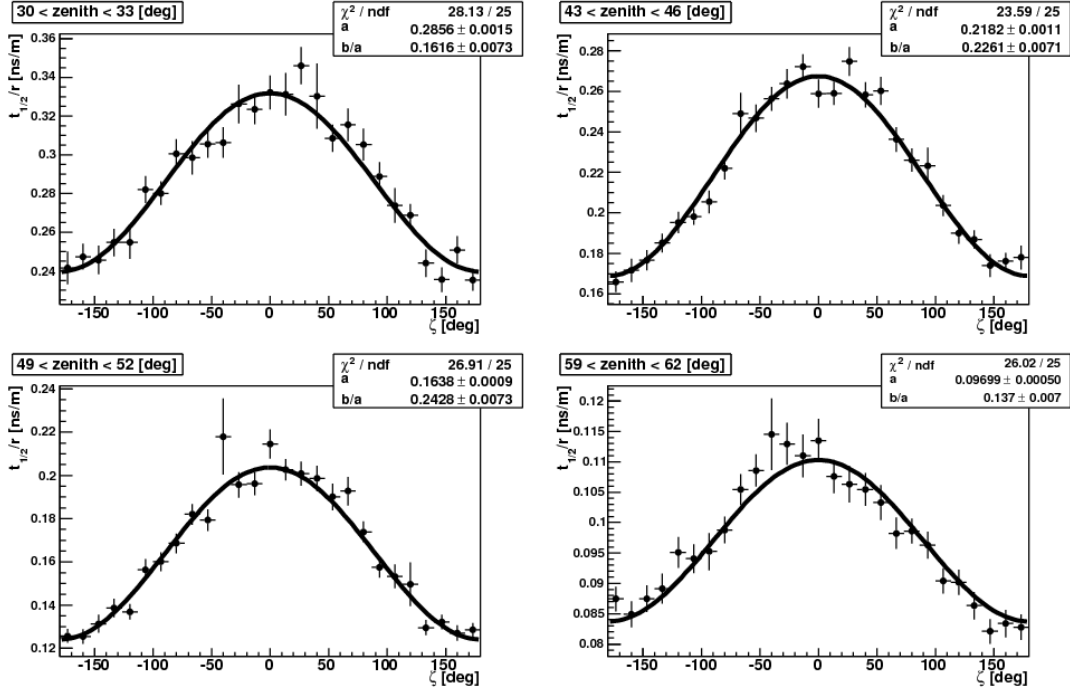


Figure 5.2: $\langle t_{1/2}/r \rangle$ vs ζ for events with reconstructed $\log(E/\text{eV})$ in the bin 18.4 - 18.6 and reconstructed θ in four different bins, 30° - 33°, 43° - 46°, 49° - 52° and 59° - 62°, from top to bottom and from left to right.

behavior expected from the simulations. Next the main features of the time asymmetry observed in real data are described and representative examples are shown.

- **$\langle t_{1/2}/r \rangle$ vs ζ .** Figures 5.2, 5.3 shows the $\langle t_{1/2}/r \rangle$ value as a function of ζ for two different energy bins and four zenith angle bins. In these plots the asymmetry between early and late risetimes is clearly observed. The corresponding fits to equation (4.7) are also shown in this figure, allowing the determination of the asymmetry factor b/a .
- **Asymmetry longitudinal development.** Once the asymmetry factor is obtained, the longitudinal development of the asymmetry parameter (see section 4.5.1) can be easily constructed. In figure 5.5 the result for all the energy bins used in this analysis is presented. These plots show the expected behavior, the asymmetry parameter increases with the zenith angle, reaches a maximum and then start decreasing. To characterise the asymmetry development, the Gaus-

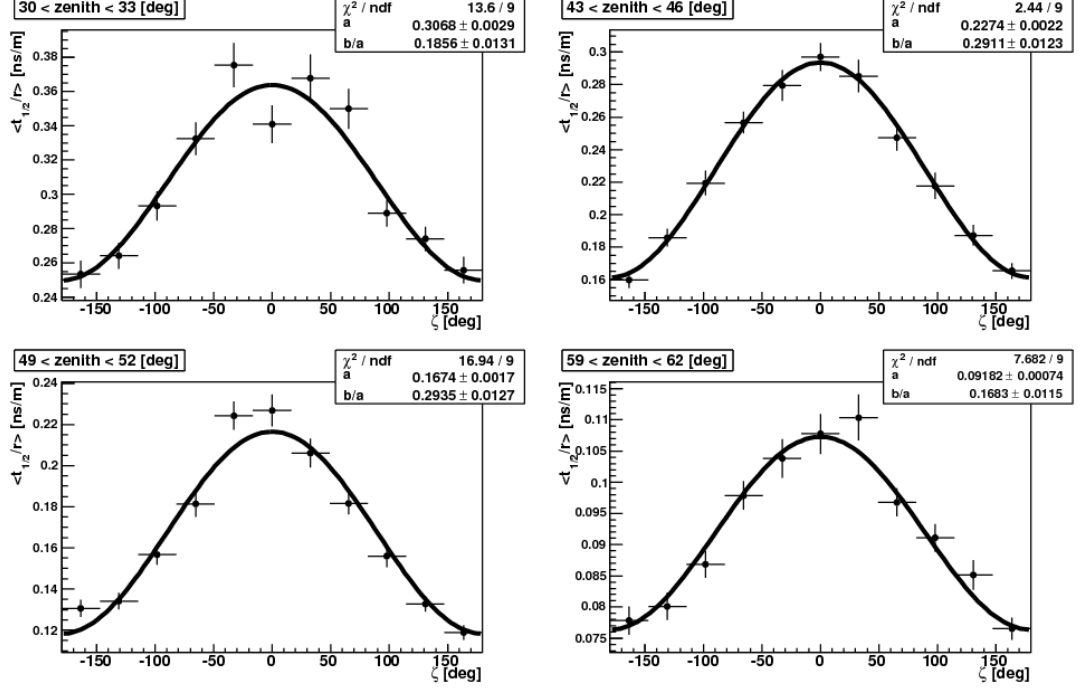


Figure 5.3: $\langle t_{1/2}/r \rangle$ vs ζ for events with reconstructed $\log(E/\text{eV})$ in the bin 18.8 - 19.0 and reconstructed θ in four different bins, 30° - 33°, 43° - 46°, 49° - 52° and 59° - 62°, from top to bottom and from left to right.

sian/normal fit proposed in 4.5 is performed and showed in figure 5.5.

- Asymmetry parameters vs Energy** The energy dependence of the three parameters describing the longitudinal development, XAsymMax, AsymHeight and XAsymWidth, are shown in figures 5.6, 5.7 and 5.8 respectively. The dependence of the composition-sensitive parameters, XAsymMax and AsymHeight, are similar to those observed in simulations. XAsymMax shows a weaker energy dependence than AsymHeight. As expected XAsymWidth shows no energy dependence (see section 4.5.2).

5.3 Comparison of Data and MC

In order to obtain the mass composition of Auger SD data it is mandatory to compare with MC simulations. For that reason this section is focused in a detailed comparison of the asymmetry parameters XAsymMax, AsymHeight and XAsymWidth of MC



5. DETERMINATION OF COMPOSITION

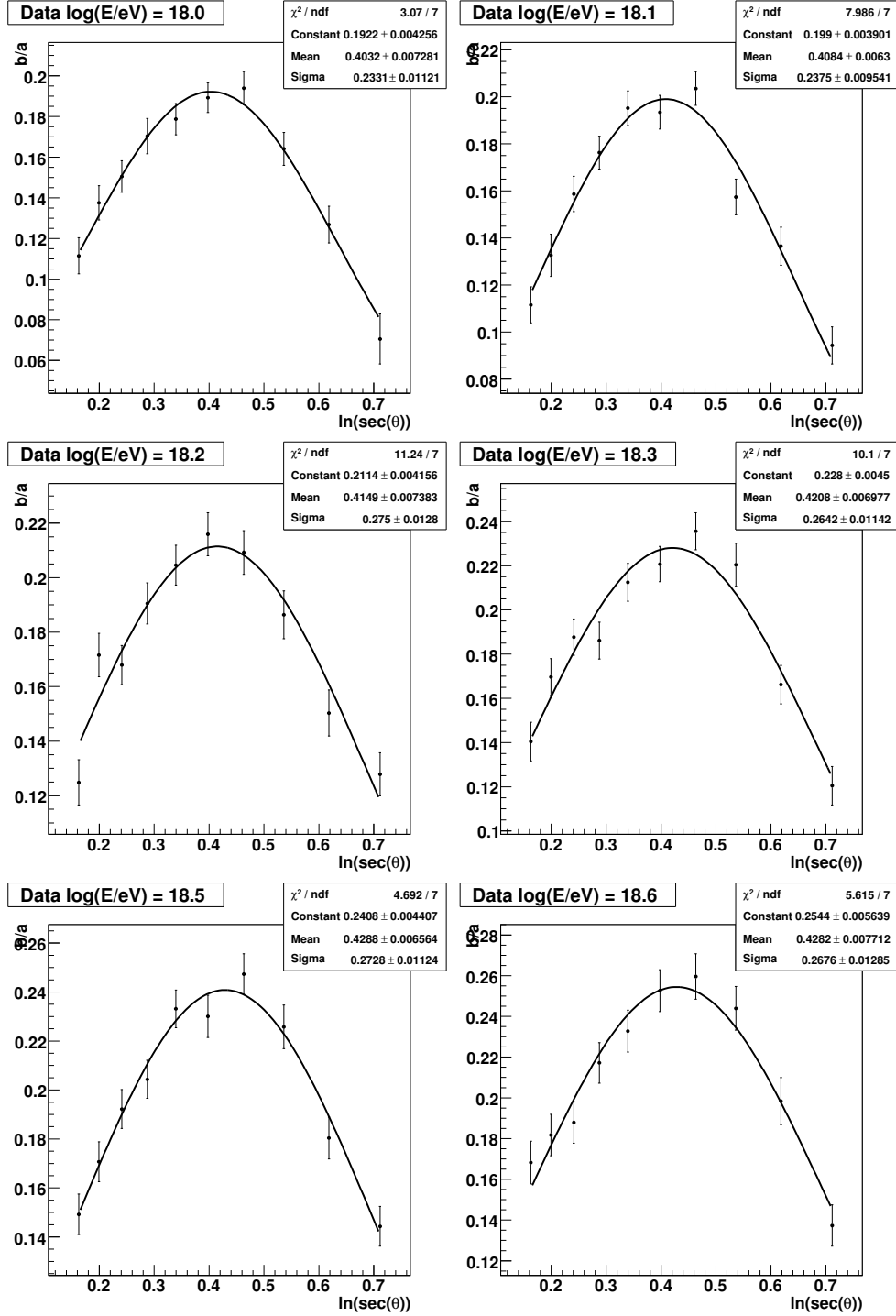


Figure 5.4: Asymmetry longitudinal development of real data for six $\log(E/eV)$ bins in the range 18.0 – 18.6.

5.3 Comparison of Data and MC

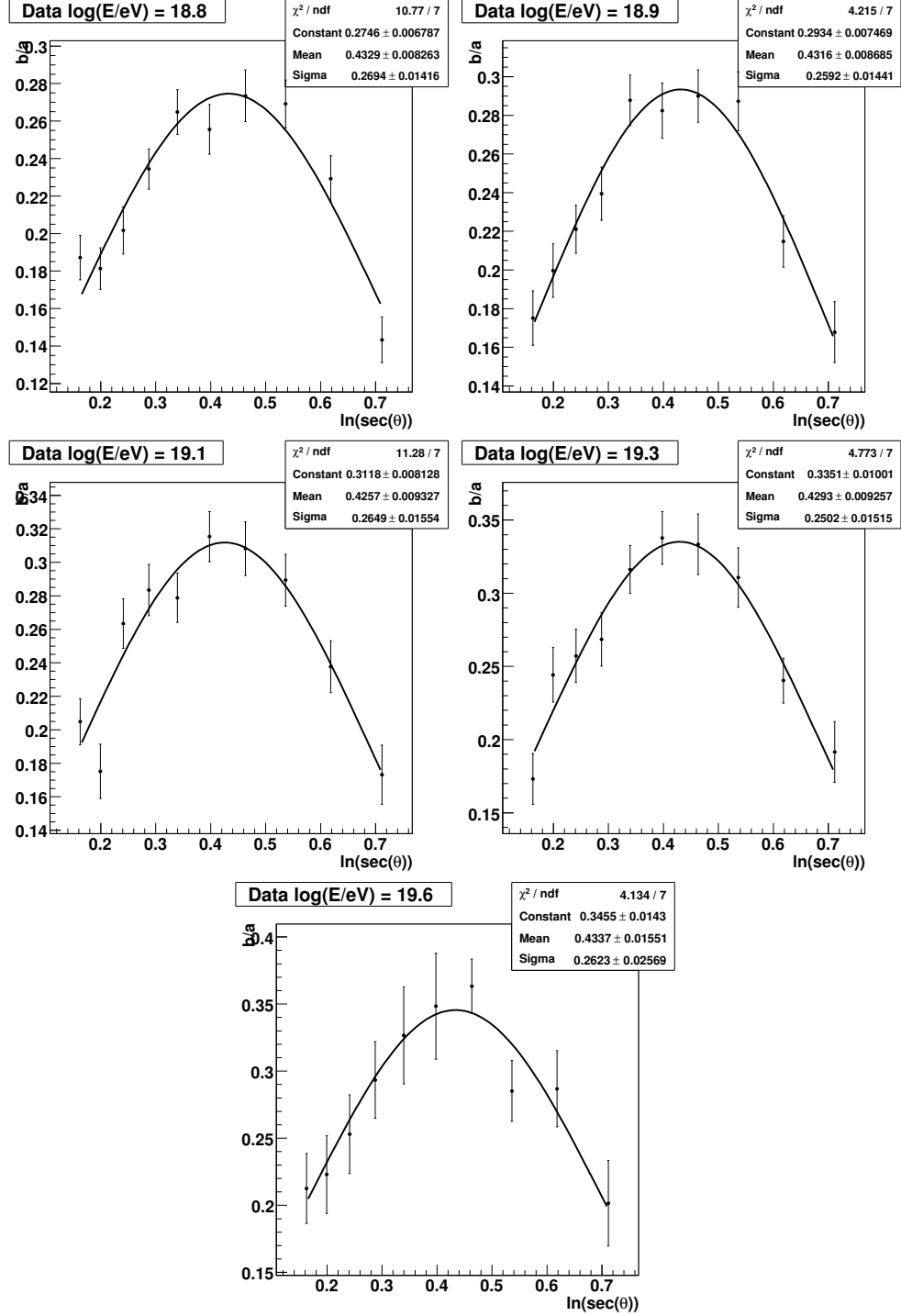


Figure 5.5: Asymmetry longitudinal development of real data for five $\log(E/eV)$ bins in the range > 18.6 .



5. DETERMINATION OF COMPOSITION

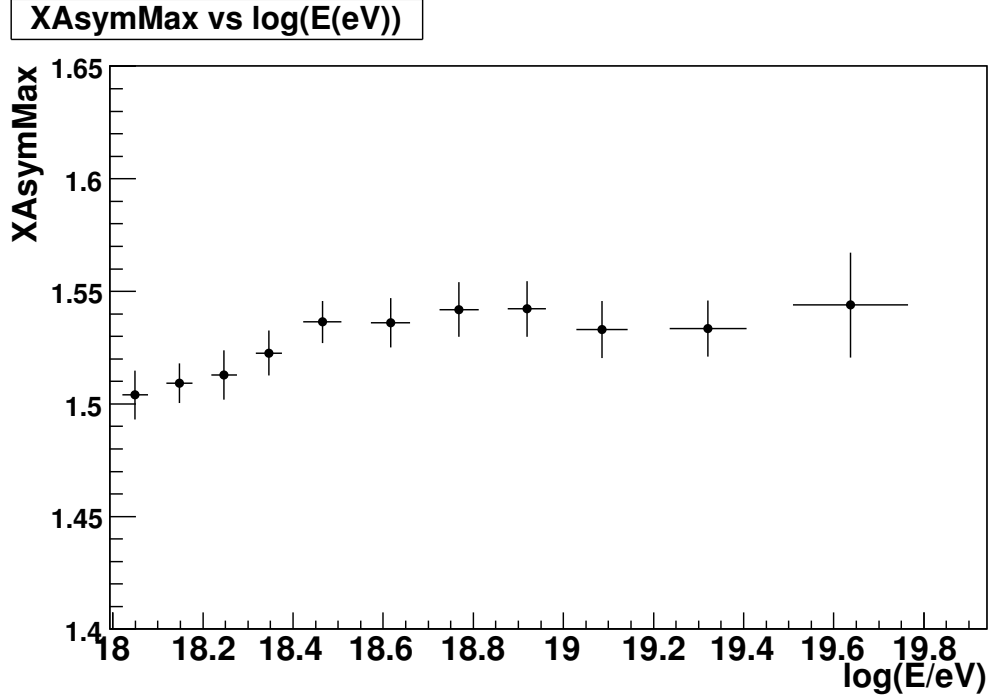


Figure 5.6: XAsymMax vs $\log(E/\text{eV})$ for real data.

simulations with data. Figure 5.9 shows the comparison of the energy dependence of XAsymMax. MC events are now represented by their fitted functions. Figures 5.10 and 5.11 shows the energy dependence of AsymHeight and XAsymWidth respectively.

The data points of XAsymWidth agree well with those obtained with MC simulations, this implies that the speed of the asymmetry evolution is well predicted by simulations. As mentioned before this parameter does not give any information of composition. The XAsymMax values of real data are contained between those for proton and iron primaries, which can be interpreted as a mixed composition. Possible changes in the composition can be observed.

However, there is a clear discrepancy between data and MC in the values of AsymHeight, simulations for both proton and iron primaries, predict higher values. This discrepancy can be interpreted as due to a difference in the number of muons at ground between data and simulation, since AsymHeight, that is the maximum b/a value, is strongly dependent of the number of muons at ground. Indication of a lack of muons in simulations was also found in other studies [173; 174; 175; 176], where comparison between data and simulations were performed. The AsymHeight disagreement between simulations and

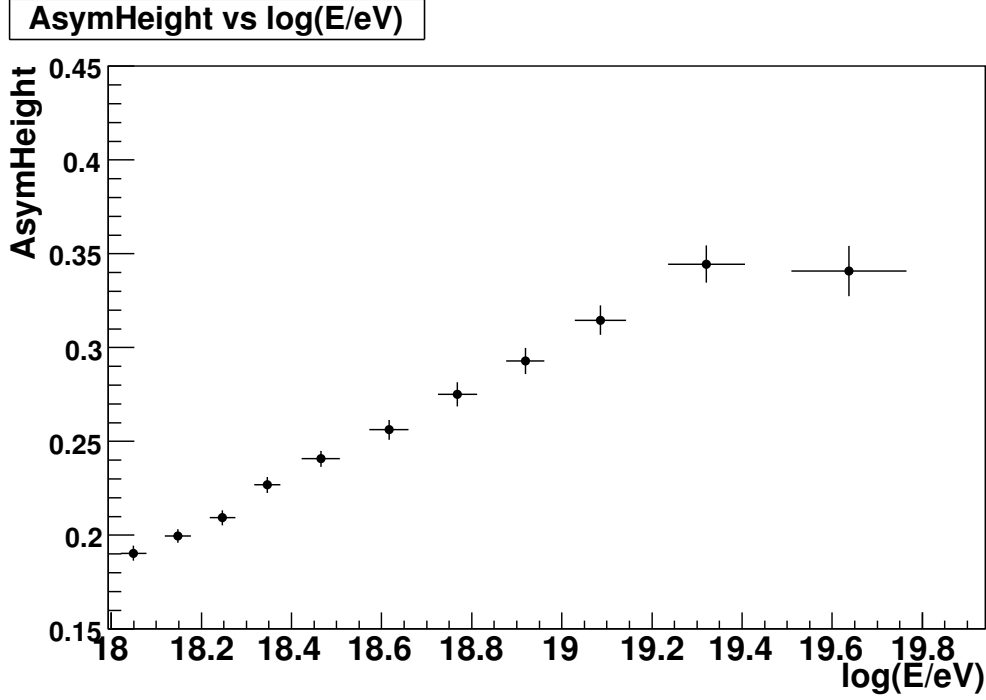


Figure 5.7: AsymHeight vs $\log(E/eV)$ for real data.

data hold for both hadronic interaction models QGSJETII and SIBYLL 2.1. Therefore the derived composition from the asymmetry analysis using the AsymHeight parameter would correspond to a primary cosmic ray even heavier than iron which is clearly at variance with the measured XAsymMax values¹ where data values are between proton and iron.

5.3.1 Muon Test

The muon component of EAS is closely related with the hadronic interaction model. At the highest energies models are based on extrapolations from much lower energies. Those extrapolations could not be able to infer the muon content accurately and therefore it does not seem unfeasible that the muon content could be underestimate. This might be the origin of the discrepancy between simulation and data.

¹And also with the X_{max} behavior.



5. DETERMINATION OF COMPOSITION

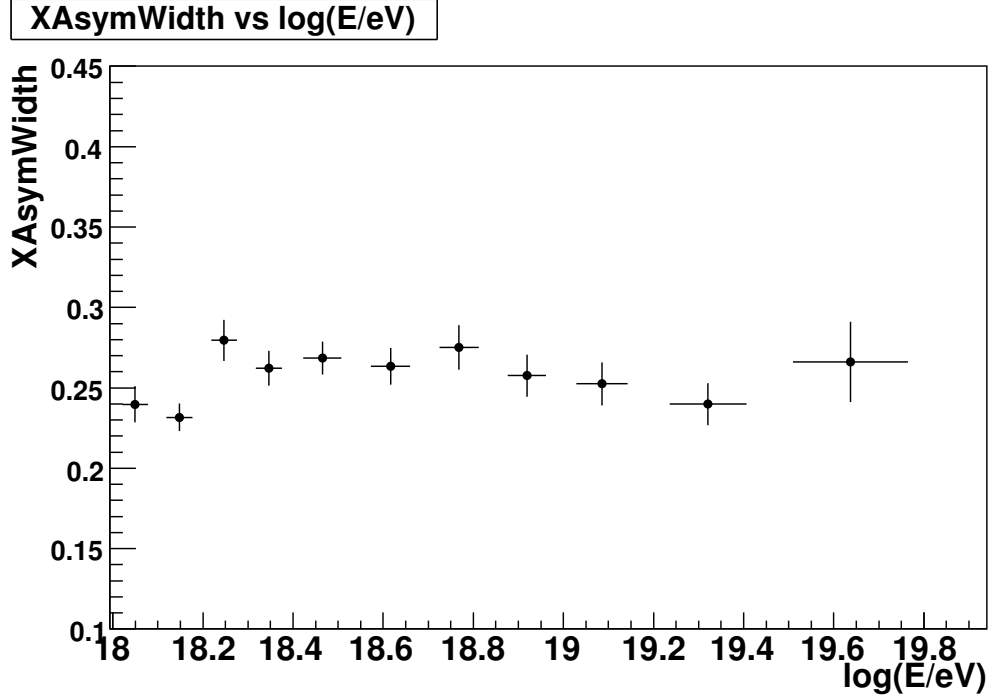


Figure 5.8: XAsymWidth vs log(E/eV) for real data.

5.3.1.1 Simulation with enhanced muon content

To study the dependency of the number of muons on the asymmetry analysis, simulations with enlarged number of muons at the detector level were carried out. These simulations did not introduce any modification on possible mechanisms responsible of muon generation but just a direct increase of the number of muons at ground level. For this purpose the same Offline simulation chain (see section 4.4) was used. The Offline module responsible of the unthinning was changed in such a way that the weight w_i of each muon read in the AIREs ground particle file [167] was multiplied by a factor f . Then the response of the detector to the $f \cdot w_i$ muons is performed as in the standard simulation.

The test with the enhanced number of muons, $f = 1.6$, was done in a subset of the AIREs showers used in this work, the ones obtained with the QGSJET-II model. The asymmetry analysis is then repeated for these new simulations and compared with the standard ones.

Figures 5.12, 5.13 and 5.14 shows the comparison of the energy dependency of the

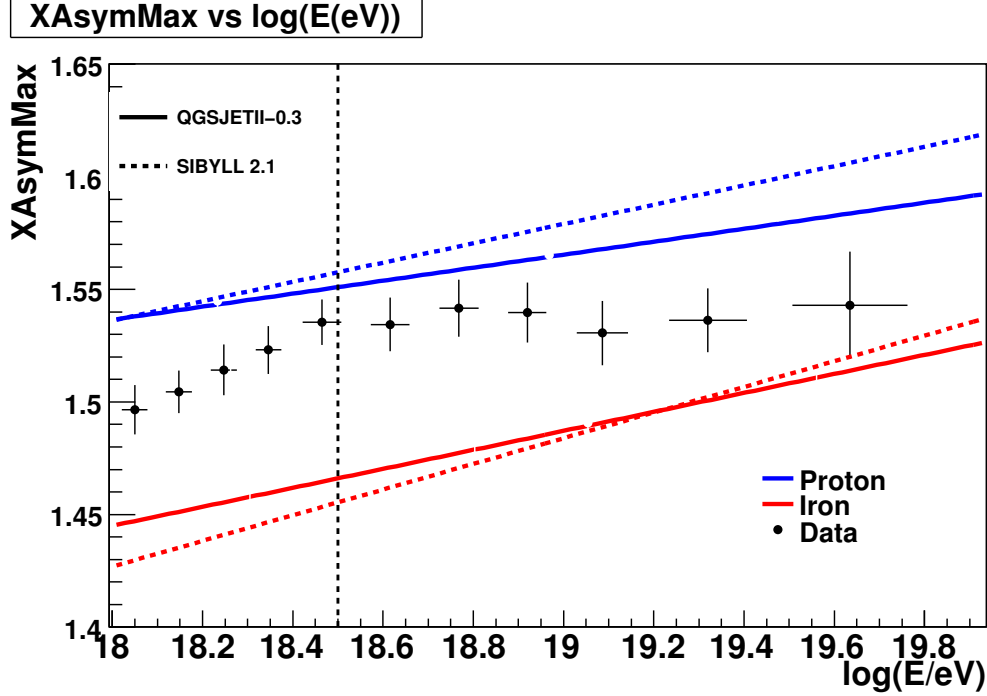


Figure 5.9: XAsymMax vs $\log(E/eV)$ for data (points) and Monte Carlo. Monte Carlo predictions are represented by their fitted functions. At $\log(E/eV) < 18.5$ (dashed vertical line) SD trigger efficiency depends on primary and therefore data can not provide reliable information on mass composition.

three parameters, XAsymMax, AsymHeight and XAsymWidth respectively. The position of the maximum (XAsymMax) and the width (XAsymWidth) of the asymmetry longitudinal development, are marginally modified. This was expected since these two parameters mainly depend on the attenuation of the electromagnetic component in the atmosphere which is insensitive to a modification of the number of muons. The minor differences in the XAsymMax parameter are due to statistical fluctuations. In contrast, the prediction of the maximum, AsymHeight is strongly modified. This effect is consistent with the fact that the asymmetry is reduced when the muon content of the signal is enhanced compared with the electromagnetic one.

In the figures 5.15, 5.16 and 5.17 the comparison of the asymmetry parameters from simulations with enhanced muon content and data are shown. It is noticeable that AsymHeight get closer to real data values for increased muon content. This may indicate a lack of muons in the simulations as shown in previous studies [173; 174; 175;

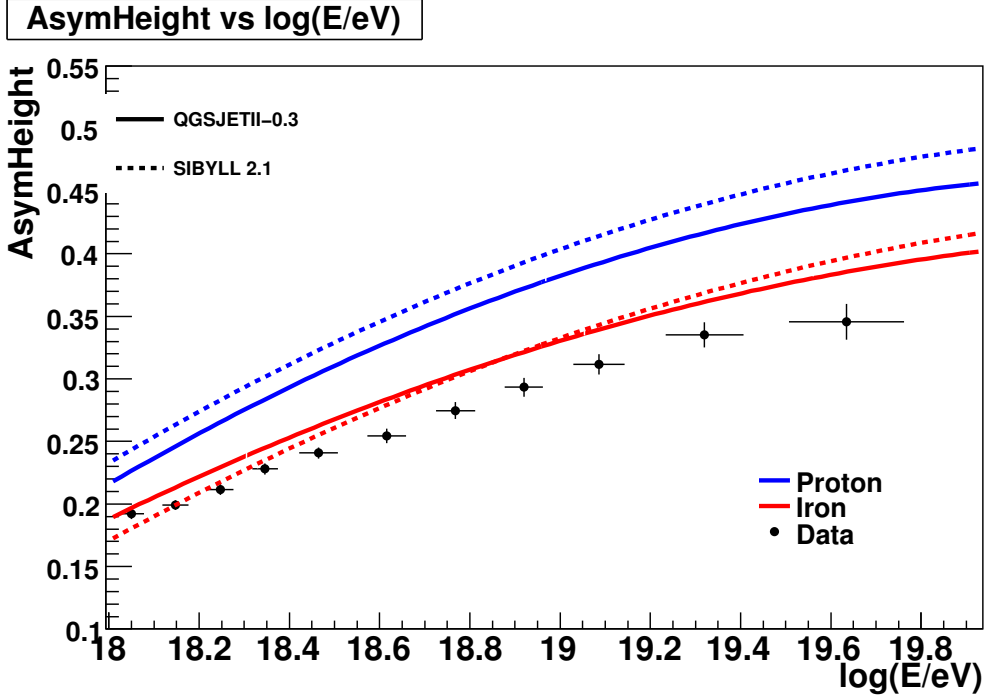


Figure 5.10: AsymHeight vs log(E/eV) for data (points) and Monte Carlo. Monte Carlo predictions are represented by their fitted functions.

176]. Assuming that XAsymMax parameter gives the correct composition of data, it could be possible in the future to use the AsymHeight parameter to constraint the hadronic models.

Taking into account this inconsistency between simulation and data in the AsymHeight parameter and the strong effect of total number of muons at ground, XAsymMax has been chosen as the main variable for composition analysis. Favoring this selection is also the weak energy dependence of this parameter which reduces possible effects from uncertainties in the energy reconstruction.

5.3.2 Effect of the measurement uncertainties in $t_{1/2}$

It is interesting to check the effect of the measurement uncertainties in $t_{1/2}$ in the asymmetry analysis. This uncertainty is the combination of the detector effects and data treatment [85]. Including in the detector effects are those caused by the detector electronics, such as the uncertainty introduced by the FADCs in digitising the signal and the finite size of the tanks, introducing statistical fluctuations in the fraction of

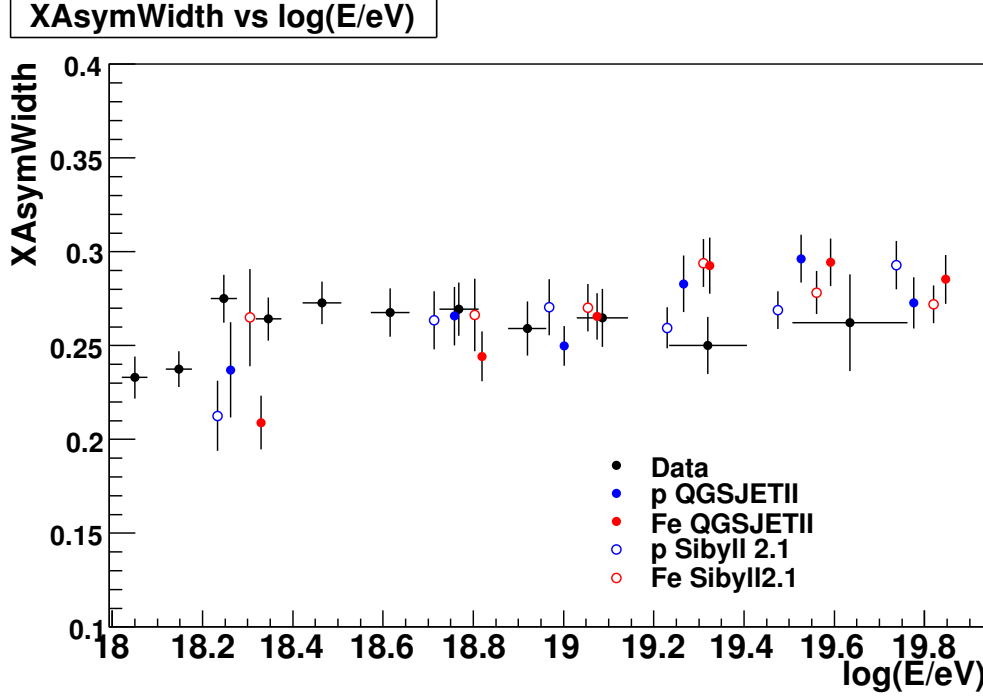


Figure 5.11: XAsymWidth vs $\log(E/eV)$ for data (black points) and Monte Carlo.

the shower observed.

The measurement uncertainties in the risetime recorded in the Pierre Auger Observatory have been parametrized by C. Wileman in [85], using an empirical determination. The method consist of using two measurements of the $t_{1/2}$ in the *twins*¹, then the measurement uncertainty can be calculated as:

$$\sigma_{t_{1/2}} = \frac{\sqrt{\pi}}{2} < \Delta t_{1/2} > \quad (5.1)$$

where $\Delta t_{1/2}$ is the difference between both $t_{1/2}$ measurements.

This uncertainty is parameterized as a function of core distance r , the total signal S and the incident angle θ of the event. Te data set containing twins is very small, then to overcome this problem, *pairs*² were used. After corrections of asymmetry effects and using the risetime parametrization with r (see section 4.3) (pairs does not lie at the

¹A twin is a set of two tanks deployed on an standard hexagonal point, with a separation of only 11 m.

²A pair is two tanks observing an event, with a difference of less than 100 m in r .



5. DETERMINATION OF COMPOSITION

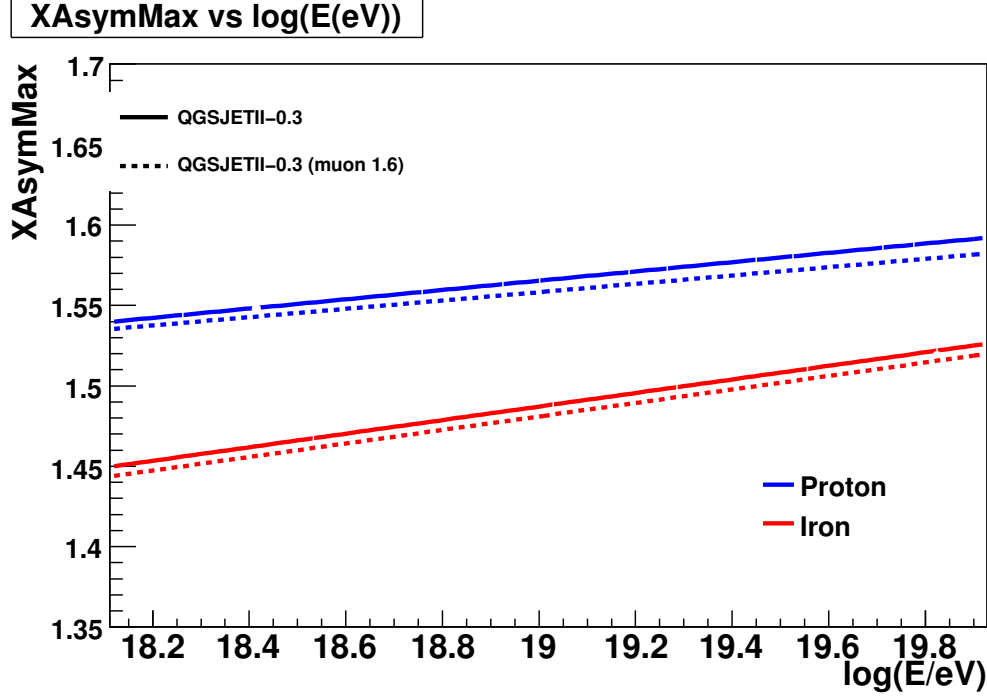


Figure 5.12: XAsymMax versus log(E/eV) for MC simulations using the QGSJETII interaction model for both standard and that with enhanced muon content at ground.

same core distance) the average uncertainty in the risetime is parameterized as:

$$\sigma_{t_{1/2}}(\theta, r, S) = \frac{j(r, \theta)}{S} + k(r, \theta) \quad (5.2)$$

Using this parameterization, the average value of the $t_{1/2}/r$ and its error for each bin of (E, θ, ζ, r) is calculated as:

$$\langle t_{1/2}/r \rangle_w = \frac{\sum w_i \cdot t_{1/2}/r_i}{\sum w_i} \quad (5.3)$$

and its standard deviation:

$$\sigma_w^2 = \frac{\sum w_i (t_{1/2}/r_i - \langle t_{1/2}/r \rangle_w)^2}{\sum w_i} \quad (5.4)$$

where the weights are:

$$w_i = \frac{1}{(\sigma_{t_{1/2}}/r)^2} \quad (5.5)$$

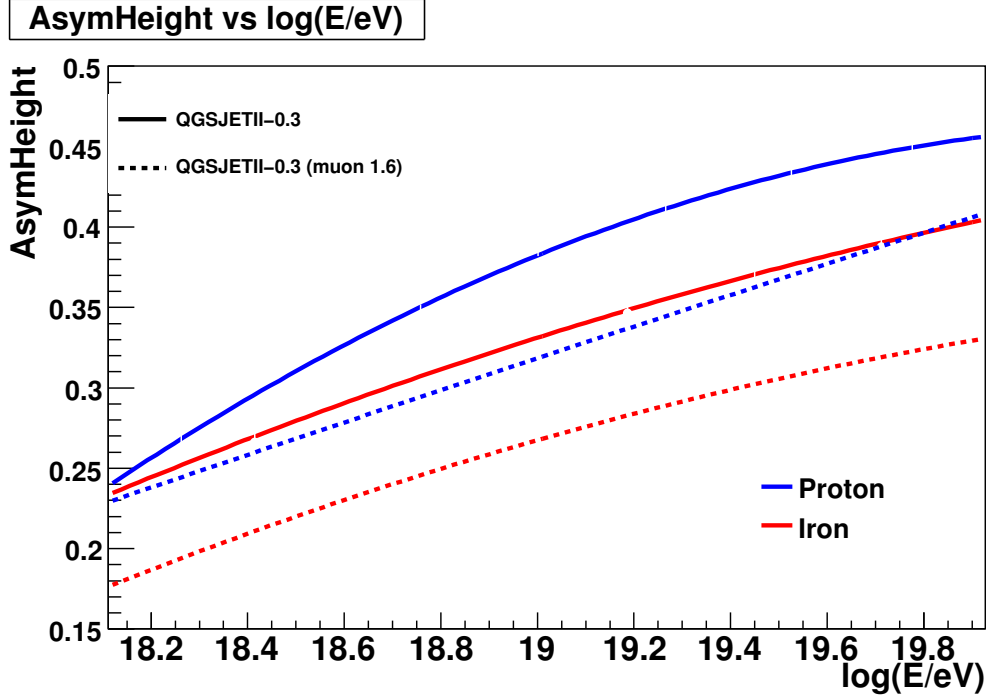


Figure 5.13: AsymHeight versus $\log(E/eV)$ for MC simulations using the QGSJETII interaction model for both standard and that with enhanced muon content at ground.

Once the average $t_{1/2}$ is calculated using the reciprocal measurement uncertainties as weights, the asymmetry analysis is repeated.

The usage of weighted means has been found very useful for low statistics samples. For instance in figure 5.19 the XAsymMax parameter calculated from a reduced data sample with (dots) and without weights (triangles) is compared. It is clearly seen that the analysis including weights reduce the variance for highest energies, where the statistics is more limited.

5.4 Inferring X_{max}

5.4.1 Data samples and event selection

For this study we use both Monte Carlo and real data samples in ADST format [178]. The MC sample consists of the same library of 24.000 proton and iron air showers used in chapter 4. The data sample contains events recorded from January 2004 to October 2008. The samples are binned in X_{max} and zenith angle θ . At present full hybrid



5. DETERMINATION OF COMPOSITION

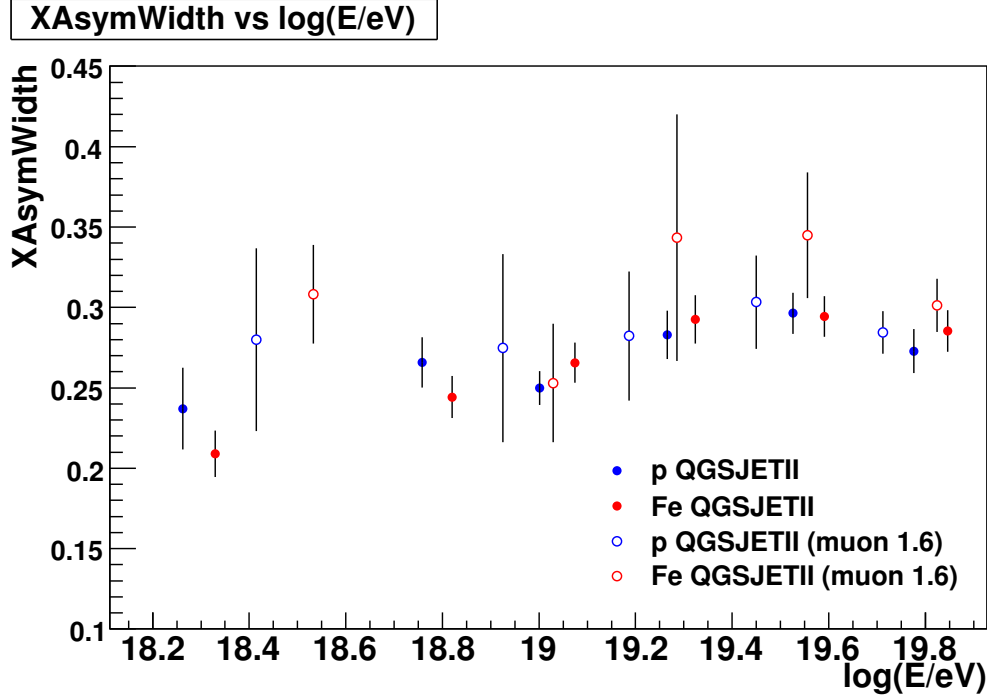


Figure 5.14: XAsymWidth versus $\log(E/eV)$ for MC simulations using the QGSJETII interaction model for both standard and that with enhanced muon content at ground.

simulation is not available and therefore in the MC sample the X_{max} value is taken from the shower simulation in the atmosphere instead from the detector reconstruction.

In regard with real data, golden hybrid events are selected following a similar criteria used for the elongation rate analysis [179]:

- Events are required to contain the reconstructed X_{max} parameter within the FoV.
- Showers must be well-fitted to a Gaisser-Hillas function.
- Errors in the reconstructed energy and X_{max} are constrained to a maximum value of 20% (relative) and 40 g/cm² respectively.
- The minimum viewing angle must be larger than 20°

Other quality cuts are related to the reduced χ^2 of the Gaisser-Hillas fit which is required to be smaller than 2.5 and to exceed in more than 4 the reduced χ^2 of a linear fit to the longitudinal profile. Atmospheric quality cuts have been relaxed in order to increase the statistics. Regarding the SD part of the event, shower candidates must

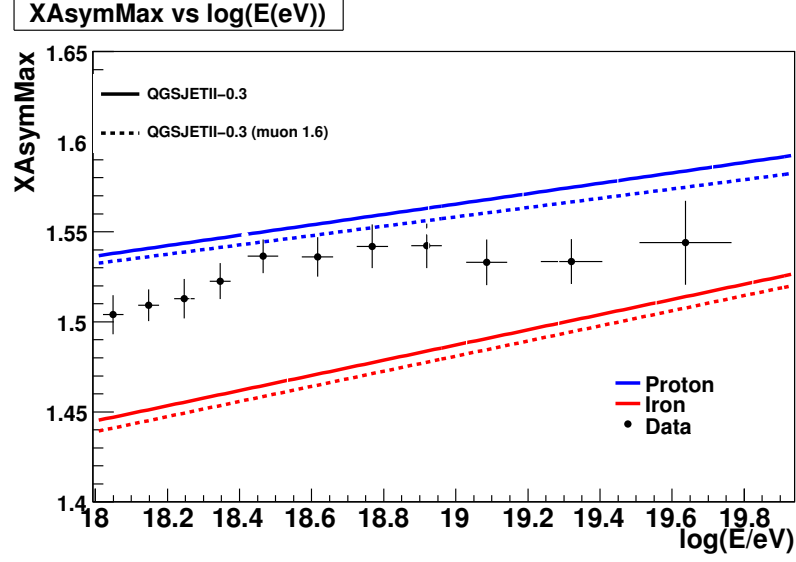


Figure 5.15: Comparison of XAsymMax between data and QGSJETII simulation with enhanced muon content at ground.

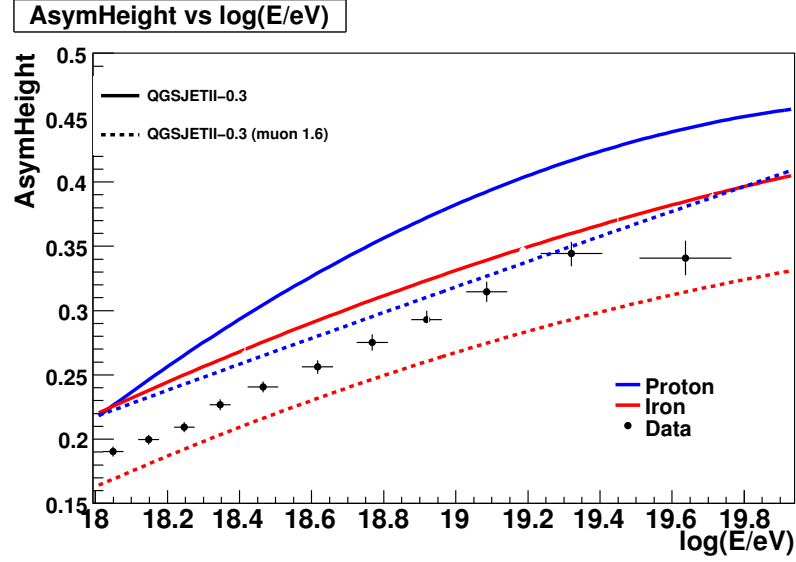


Figure 5.16: Comparison of AsymHeight between data and QGSJETII simulation with enhanced muon content at ground.



5. DETERMINATION OF COMPOSITION

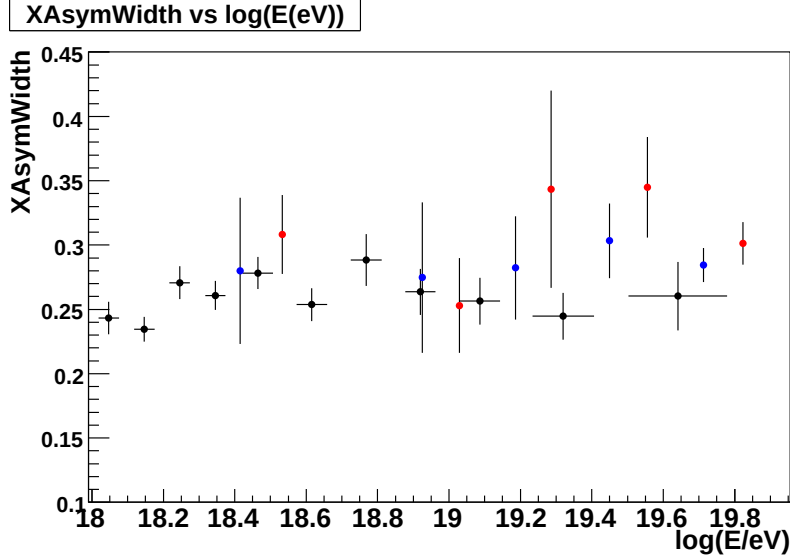


Figure 5.17: Comparison of XAsymWidth between data and QGSJETII simulation with enhanced muon content at ground.

fulfill the same requirements employed in the asymmetry analysis of SD-only events, and thus, only showers with zenith angles between 30° and 62° are selected, see section 5.2.1 for further details. For the SD stations, the distance to the shower core must be contained within 500 and 2000 m and only the risetime of stations with signals greater than 10 VEM are considered.

5.4.2 Relationship between X_{max} and the asymmetry parameters

The correlation of the asymmetry parameters and X_{max} has been studied for both MC events and real data. The first step is to group events within bins of X_{max} . For real data, bins were defined within the following limits: 600, 640, 680, 710, 740, 760, 790, 820 and 860 g/cm^2 . Bin widths (ranging from 20 to 40 g/cm^2) have been chosen in order to have enough statistics to apply accurately the asymmetry technique. For MC events the X_{max} limits were: 625, 675, 725, 775, 825, 875 and 925 g/cm^2 . For each sample corresponding to a given grammage interval in X_{max} , the asymmetry parameters XAsymMax, AsymHeight and XAsymWidth (see section 4.5) have been obtained as well as the corresponding mean value of the shower maximum depth $\langle X_{max} \rangle$ (weighted with individual errors). It is worth reminding that this calibration is not based on event by event relations as for example in [180], but on relations between

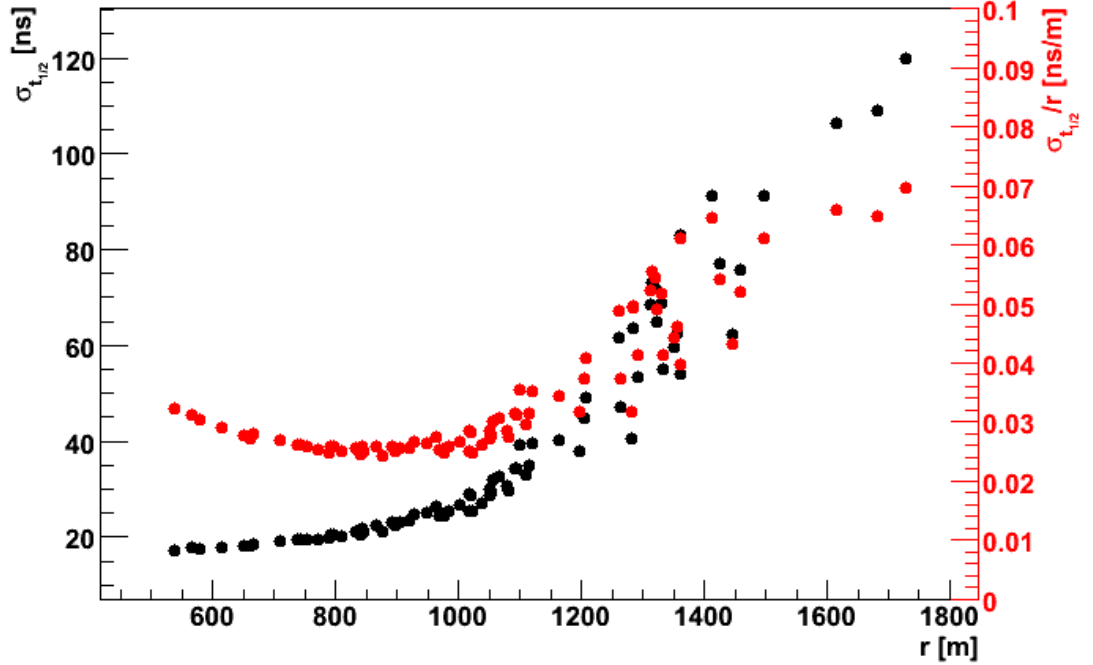


Figure 5.18: Example of $\sigma_{t_{1/2}}$ (black dots) and $\sigma_{t_{1/2}}/r$ (red dots) as a function of the core distance.

average values where a unique number of the asymmetry parameter is obtained for each $\langle X_{max} \rangle$ value.

The correlation of the three asymmetry parameters with $\langle X_{max} \rangle$ for the sample of real data has been obtained and results are shown in figure 5.20. As expected from section 4.5, both XAsymMax and AsymHeight show a strong correlation with X_{max} while XAsymWidth is found to be nearly independent on the shower maximum depth.

As explained in section 5.3 XAsymMax has turned out to be the most efficient asymmetry parameter for mass discrimination. In the following we will study in more detail the features of the correlation between this asymmetry parameter and X_{max} . In figure 5.21 several plots of XAsymMax versus X_{max} are shown to demonstrate the universality of this correlation as well as the consistency between MC and real data. In the upper plots results from MC proton samples are compared with that obtained for iron showers for two hadronic models QGSJETII-03 (green symbols) and SIBYLL 2.1 (pink symbols). Black dots represent the result for real data. Since the energy of



5. DETERMINATION OF COMPOSITION

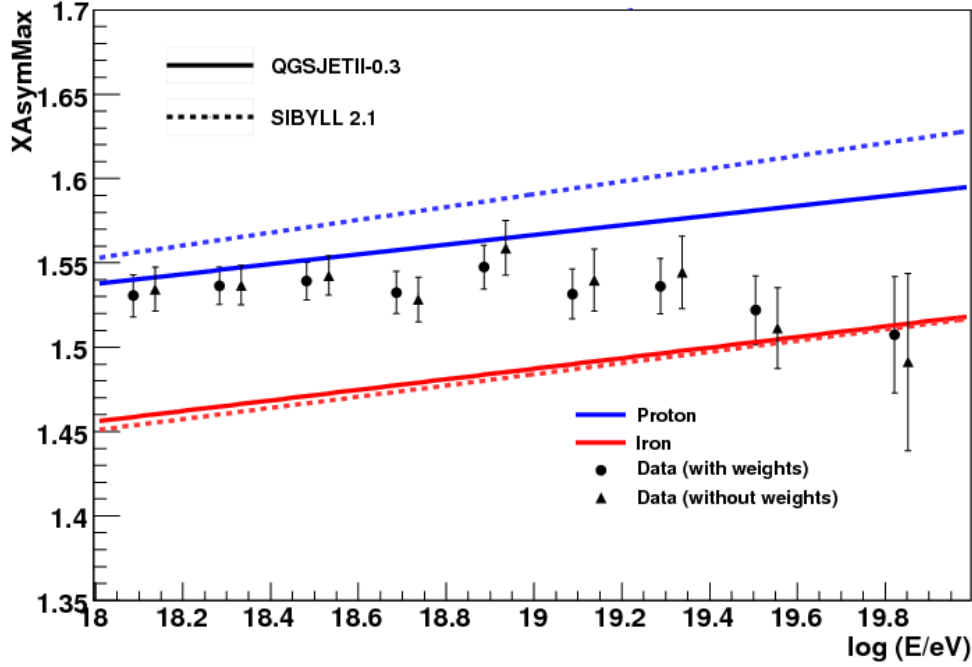


Figure 5.19: Comparison of the XAsymMax parameter versus energy obtained weighting rise-times with the measurement uncertainties (dots) and without weights, i.e. arithmetic mean values (triangles), obtained for a reduced data sample.

MC samples for different primaries extend within the same interval, the covered X_{max} range is different for proton and iron nuclei populating protons the upper region of the curve (highest X_{max} values) and iron the lower region (lowest X_{max} values). In consequence the overlap region in the X_{max} parameter is very small. In spite of this limitation upper plots of figure 5.21 show that the relationship between XAsymMax and X_{max} is independent on the primary particle. More precisely, the same correlation is found for proton and iron nuclei.

Real data results are also consistent with the correlation predicted by the MC sample as can be seen in all plots of figure 5.21. In the lower panel of this figure, linear fits to the samples containing real data (black line) and all MC events merging proton and iron showers simulated with QGSJETII-03 (green line) and SIBYLL2.1 (pink line) are shown. The three linear fits are very similar and therefore we can conclude that the relationship between XAsymMax and X_{max} is unique independently on the primary mass and the assumed hadronic model and consistent with real data. Obviously this

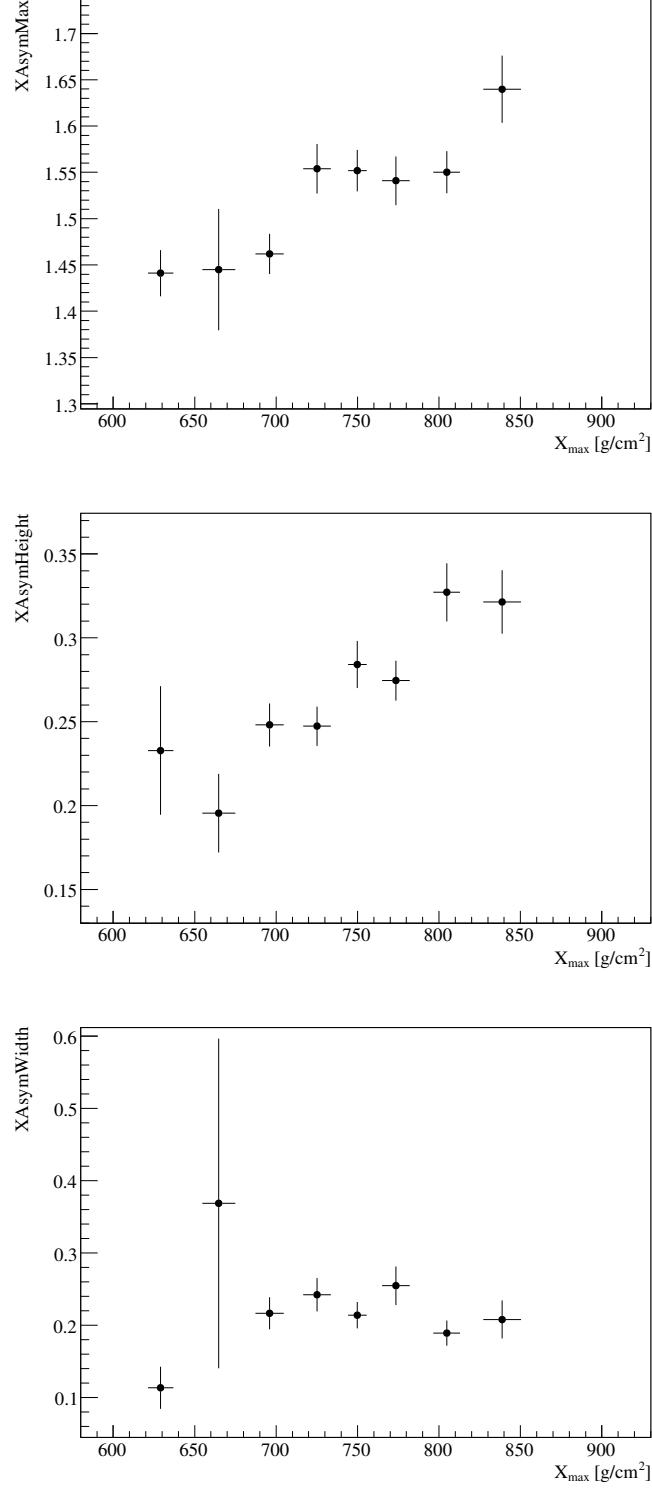


Figure 5.20: Correlation with $\langle X_{max} \rangle$ of the three parameters which describe the asymmetry in the shower development: XAsymMax (upper plot), XAsymHeight (middle plot) and XAsymWidth (lower plot).



5. DETERMINATION OF COMPOSITION

assertion is limited by the statistical fluctuations of both the MC and data samples.

For this analysis, events have been grouped in bins of X_{max} . Since the X_{max} parameter is mass dependent for a given energy, different primaries generating showers with a similar value of X_{max} will have different energies. As data suggest that mass composition changes with energy, the dependence of this calibration curve with composition should be studied, binning not only in X_{max} but also in energy. Although the present statistics is not yet high enough to perform this analysis no changes with composition are expected since, within the actual uncertainties, the observed correlation is the same even for extreme compositions (pure iron and pure proton compositions). Merging different compositions in a given X_{max} bin implies merging different energies. However the asymmetry is essentially determined by the position of the shower maximum, independently on energy.

The relationship between XAsymMax and X_{max} has been obtained using the real data sample. In figure 5.22 the result of a linear fit $X_{max} = a + b \cdot \text{XAsymMax}$ is shown (solid line). Results for the fitted parameters are $a = 0.92 \pm 0.12$ and $b = (8 \pm 2) \cdot 10^{-4}$. The predicted calibration curve obtained for the MC sample is fully consistent with that obtained with real data. The systematic uncertainty can be defined by the region within the linear functions with parameters $(a + \sigma_a, b - \sigma_b)$ and $(a - \sigma_a, b + \sigma_b)$ [181]. The uncertainty in the parameterization was propagated to estimate the systematic error in X_{max} as the half deviation of X_{max} within those two limit curves for a given XAsymMax value. A similar procedure to estimate the uncertainty of the calibration curve relies in the generation of possible linear fits from points randomly shifted around their mean value with a rms given by their error bars, the result is shown in 5.22. As pointed out in [180] this procedure predicts a similar uncertainty in the calibration curve as that defined above although a more realistic estimate is reached in the region where both curves intersect.

From the results shown in figure 5.22 a mean systematic uncertainty in the determination of X_{max} from a measurement of XAsymMax ranging between 10 and 20 g/cm² is estimated. The systematic uncertainty in the calibration curve will be naturally reduced with increasing statistics.

The dependence of AsymHeight with X_{max} has been also compared in figure 5.23 with predictions from simulations showing a general disagreement between both real (black symbols) and MC data (green and pink symbols). This effect was expected since

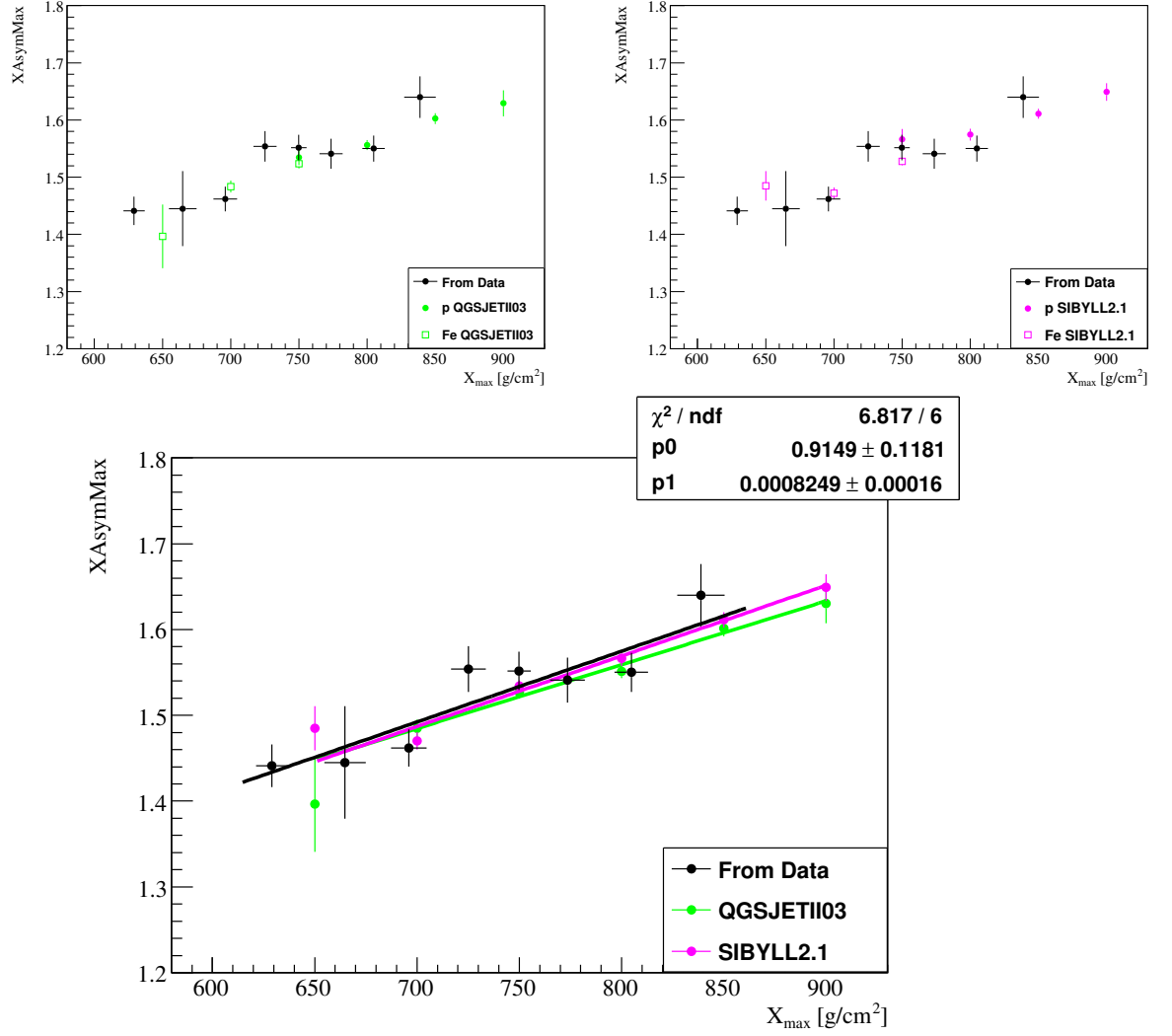


Figure 5.21: Correlation of $X_{AsymMax}$ with $\langle X_{max} \rangle$ for the different hadronic interaction models and different primaries along with data for comparison (upper plots). In the lower plot the calibration curves for data and two different hadronic interaction models are shown. Values of the parameters obtained from the linear fit to data are also presented.



5. DETERMINATION OF COMPOSITION

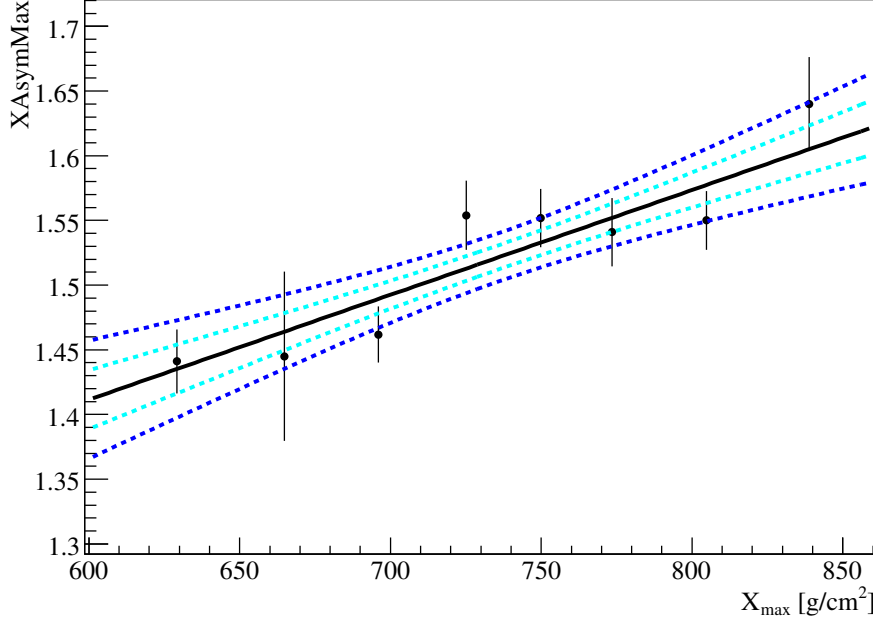


Figure 5.22: $X_{AsymMax}$ versus X_{max} (calibration curve) obtained for real data (solid line). Deviations of the calibration curve (dashed lines) obtained from the propagation of uncertainties are shown for 1σ region (cyan) and 2σ region (blue).

as shown in section 5.3, $AsymHeight$ values in real data were found to be smaller than those predicted for both iron and proton showers in the whole energy range. The result shown in figure 5.23 is consistent with the evidence of the lack of muons at ground level observed in simulations.

5.4.3 The elongation rate method using the $X_{AsymMax}$ parameter

Using the calibration curve shown in figure 5.22 $\langle X_{max} \rangle$ can be now determined for different energy bins from the asymmetry observed in inclined showers measured using the large statistics provided by the SD. Therefore the elongation rate method can be applied for the determination of mass composition using SD data.

The procedure consists of several steps. Firstly, $X_{AsymMax}$ is determined following the method described in 4.5 for samples of events grouped in energy bins. Then the calibration curve is applied to get the average X_{max} value for each energy bin. In figure 5.24 the result for our real data sample is shown. Error bars in figure accounts for

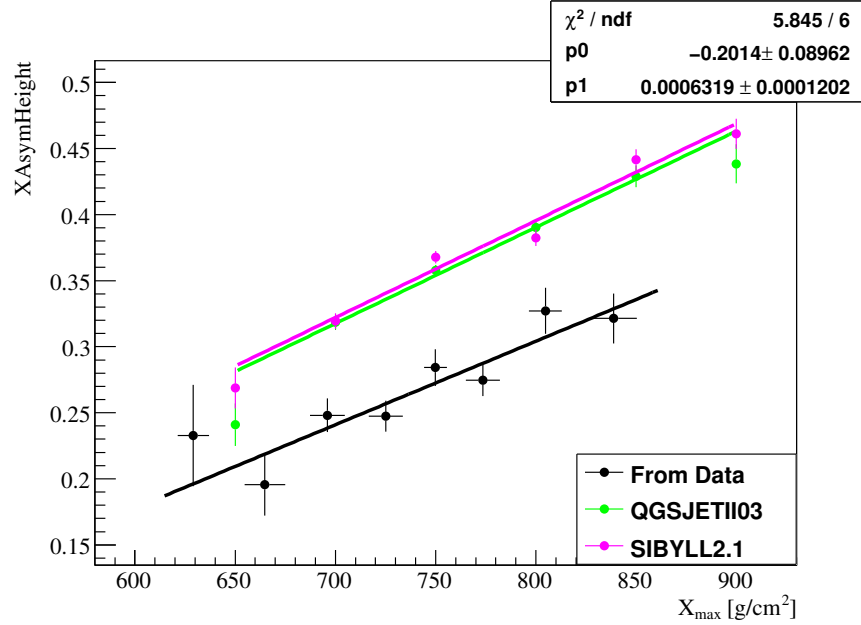


Figure 5.23: Correlation of AsymHeight with $\langle X_{max} \rangle$ for different hadronic interaction models. Linear fits to data are also shown.

statistical uncertainties due to our limited sample of real events. The actual systematic uncertainties of the direct FD measurement is of about ± 18 g/cm². In addition as mentioned above, a further systematic uncertainty due to the calibration procedure amounts on average to about ± 15 g/cm².

In figure 5.25 the $\log E$ dependence of the shower maximum depth for several theoretical and experimental data is shown. Notice that the result obtained by the asymmetry method (black circles) is consistent within uncertainties with that from the direct measurement of the shower maximum depth X_{max} with the FD.

5.5 Mass composition

The asymmetry analysis performed in previous sections allows to obtain the mass composition of UHECRs detected by the Pierre Auger Observatory. The general method presented in section 4.7, where the three asymmetry parameters were used can not be directly applied to data. As already explained, XAsymWidth is not sensitive to primary composition and thus this parameter does not provide any useful information.



5. DETERMINATION OF COMPOSITION

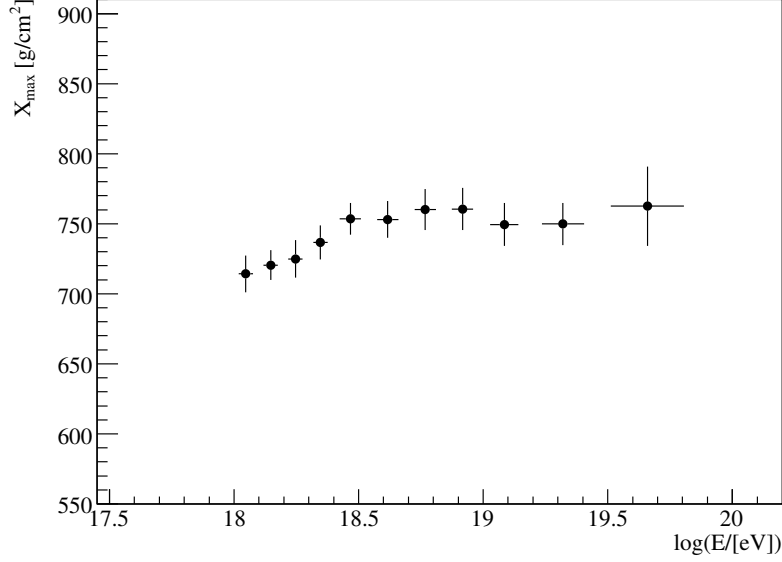


Figure 5.24: $\langle X_{max} \rangle$ vs $\log E$ as inferred from the XAsymMax parameter measured with the SD. Results for $\log E$ lower than 18.4 should be taken with caution since the surface detector is not fully efficient.

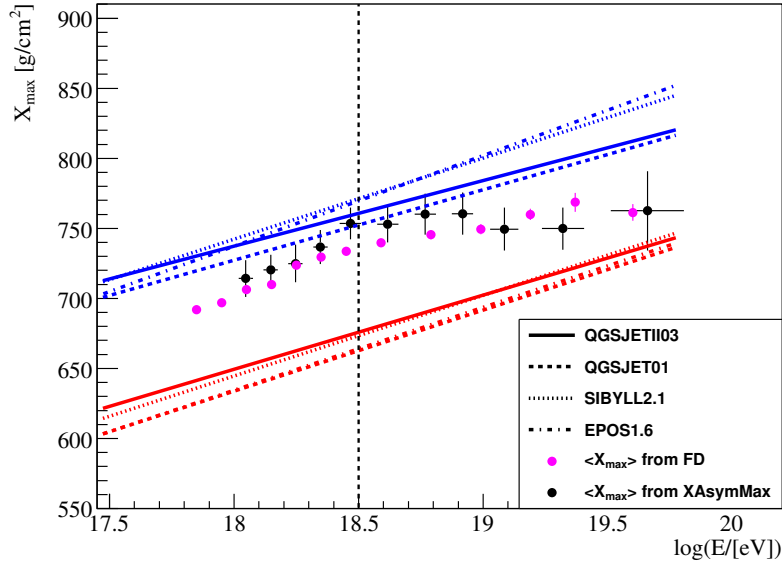


Figure 5.25: $\langle X_{max} \rangle$ vs $\log E$ inferred from the XAsymMax parameter measured with the SD. Predictions for a pure iron or proton composition according to several models as well as results from direct measurement of X_{max} using the FD are also shown for comparison.

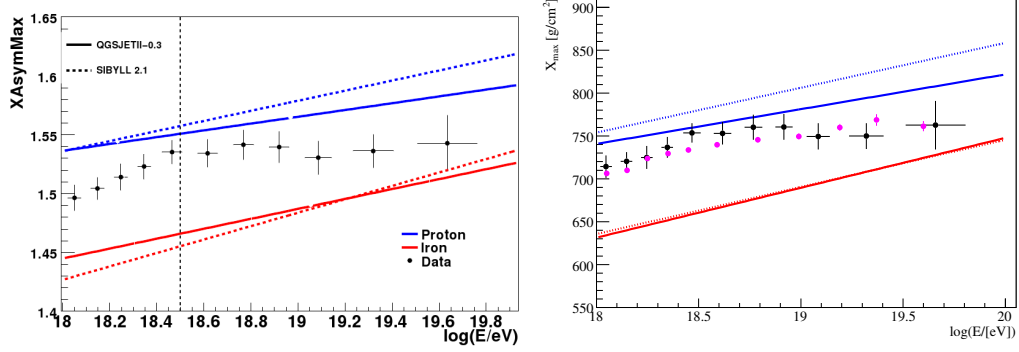


Figure 5.26: XAsymMax parameter vs log E (left panel) and $\langle X_{max} \rangle$ vs log E (right panel) inferred from XAsymMax using the presented calibration curve for both data (black symbols) and MC predictions (red lines: proton, blue lines: iron) from QGSJETII-03 (solid lines) and Sibyll 2.1 (dashed lines).

On the the other hand, AsymHeight, the maximum asymmetry in the corresponding energy bin, is not well reproduced by simulations.

However it has been demonstrated that XAsymMax is sensitive to the primary mass and the properties predicted in simulations are well reproduced in real data. In addition XAsymMax presents a strong correlation with the depth of the shower maximum. In fact this parameter was used in previous section to obtain the elongation rate from SD data. Also, it is marginally modified by an increase of the total number of muons at ground and its energy dependence is reasonably weak. These two features reduce the systematic uncertainties inherit from the interaction hadronic models and the ones introduced by the energy reconstruction procedure.

The asymmetry method developed in section 5.5 has been used although only the XAsymMax parameter, and thus only the first term of equation (4.12), has been employed to extract the iron fraction. To apply the method, the XAsymMax parametrization as a function of the iron fraction X_{Fe} , shown in section 4.7 has been used.

The resulting mass composition is shown in figure 5.27 (using QGSJETII model) and figure 5.28 (using Sibyll 2.1 model). This result indicates that within the still large systematic (5.5.1) and statistical errors, a mixed composition of $\sim 20\%$ ($\sim 30\%$) iron for QGSJETII (SYBILL 2.1) is obtained for the energy interval $10^{18.5} - 10^{19}$ eV¹.

¹ Although this behaviour is observed in the range $10^{18} - 10^{19}$ eV below $10^{18.5}$ the analysis is not fully reliable.



5. DETERMINATION OF COMPOSITION

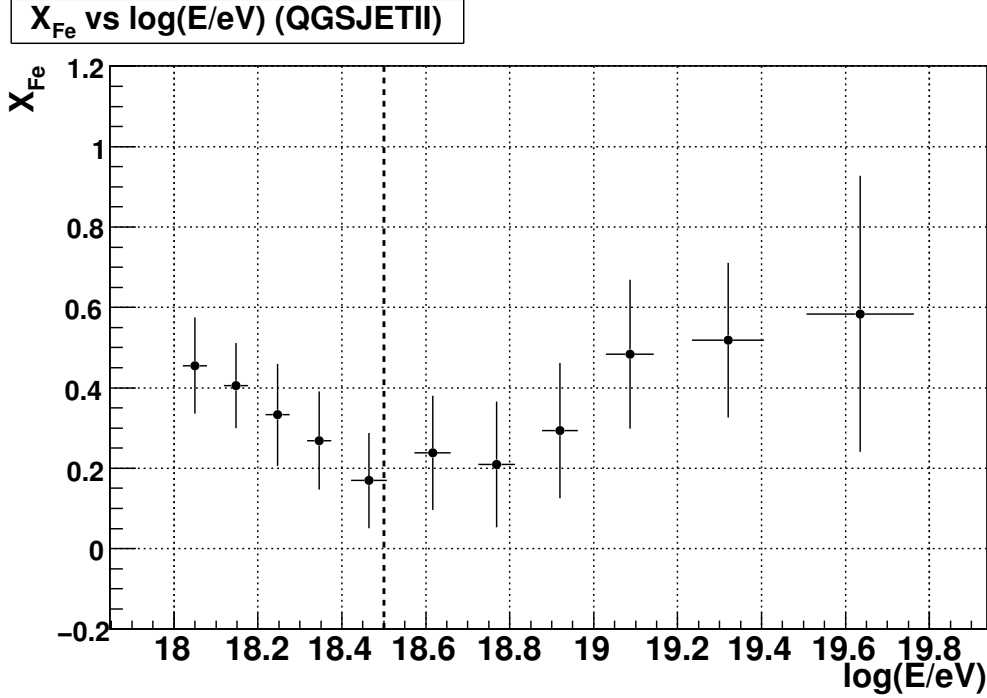


Figure 5.27: Mass composition X_{Fe} obtained by minimizing equation (4.12) using only the term corresponding of XAsymMax. Predictions from QGSJETII model are used.

At larger energies composition seems to become heavier for both hadronic interaction models.

5.5.1 Uncertainties in the determination of the mass composition

Several contributions to the uncertainty in the measure of the mass composition using the time asymmetry technique have been identified. Different errors of both statistical and systematic nature will be analyzed and their contribution to the mass composition will be evaluated. The effect of the uncertainties to the fraction of iron, ΔX_{Fe} ; i.e. the absolute deviation of the X_{Fe} parameter will be estimated.

In the first place, statistical fluctuation in the XAsymMax measurement (error bars in figure 5.9) due to the limited number of events in the current real data sample, give rise to a statistical uncertainty in X_{Fe} which varies between $\Delta X_{Fe} \approx 0.1$ and $\Delta X_{Fe} \approx 0.25$.

On the other hand a number of effects contribute to a total systematic uncertainty in X_{Fe} . The main ones are discussed next:

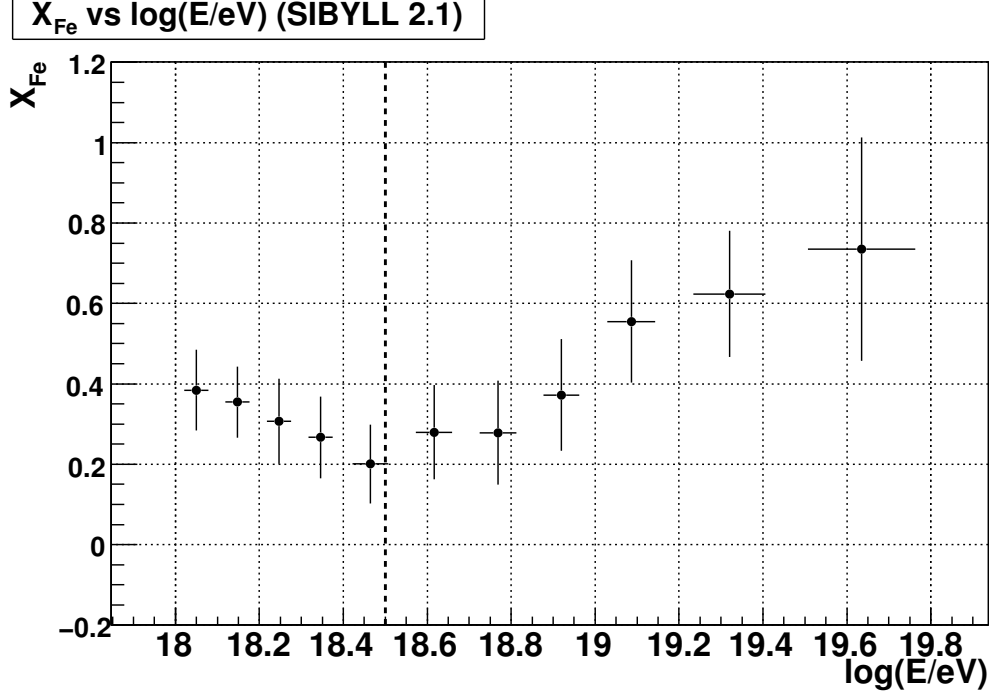


Figure 5.28: Mass composition X_{Fe} obtained by minimizing equation (4.12) using only the term corresponding of XAsymMax. Predictions from SIBYLL 2.1 model are used.

- **Energy reconstruction and muon content:** Presently a deviation (of about 30%) between the reconstructed energy in MC SD events and the input one is found. In addition a lack of muons have been observed in simulated showers (see section 5.3.1). These effects, likely induced by hadronic interaction models, can be treated as a systematic uncertainty. The total effect (dominated by the energy contribution) has been found to be of about $\Delta X_{Fe} = 0.1$ (see figure 5.29).
- **$t_{1/2}(r)$ parameterization:** As shown in section 2.7.4, several parameterizations of the core distance dependence of the risetime has been used. To check the possible systematic effect on X_{Fe} , the mass composition analysis has been repeated using a quadratic core distance dependence. For example $\langle t_{1/2}/(40 + a \cdot r + b \cdot r^2) \rangle$ has been used to obtain the asymmetry factor b/a and the corresponding X_{Fe} values were compared with the ones obtained by the standard procedure. Differences in the iron fraction, from the two analysis are at the level of $\Delta X_{Fe} = 0.02$.
- **Limited MC statistics:** Due to the limited statistics of the MC sample, the



5. DETERMINATION OF COMPOSITION

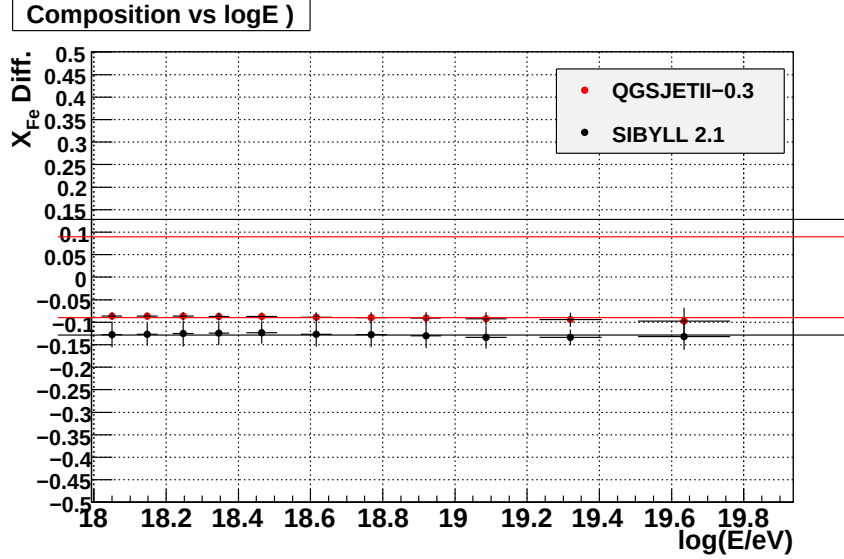


Figure 5.29: Effect of the assumed primary energy in MC on mass composition. Ordinate represents the difference in X_{Fe} when the MC input energy, instead of the reconstructed one, is used. The continuous line represents the average deviation in the full energy range.

parameterization of the XAsymMax as a function of X_{Fe} is subjected to an uncertainty. Although this error is of statistical nature in the evaluation of the asymmetry parameters of the MC sample (error bars in figure 4.11), it turns in a systematic error in the calibration function (see figure 4.19) and therefore it gives rise to a systematic error in the measure of the mass composition. With the actual MC sample, the impact of this systematic uncertainty on the mass composition amounts about $\Delta X_{Fe} = 0.15^1$.

- **Effects of cuts:** The effect of the selection criteria, i.e. the cuts, on the X_{Fe} values has been analyzed by changing in a certain amount the cuts in core distance and minimum signal size, for both MC simulations and real data. Core distance has been varied from the original 500 - 2000 m interval to 600 - 1800 m while the minimum signal size has been enlarged from 10 VEM, i.e. the usual cut, up to 15 VEM. The effect on XAsymMax for real data has been of about 0.05 units. Neglecting the effect of the cut variation on the MC samples, the contribution of this effect on real data would be of less than $\Delta X_{Fe} = 0.01$.

¹Notice that in the data analysis only the XAsymMax parameter can be used and therefore the uncertainty is slightly larger than that predicted in section 4.5.

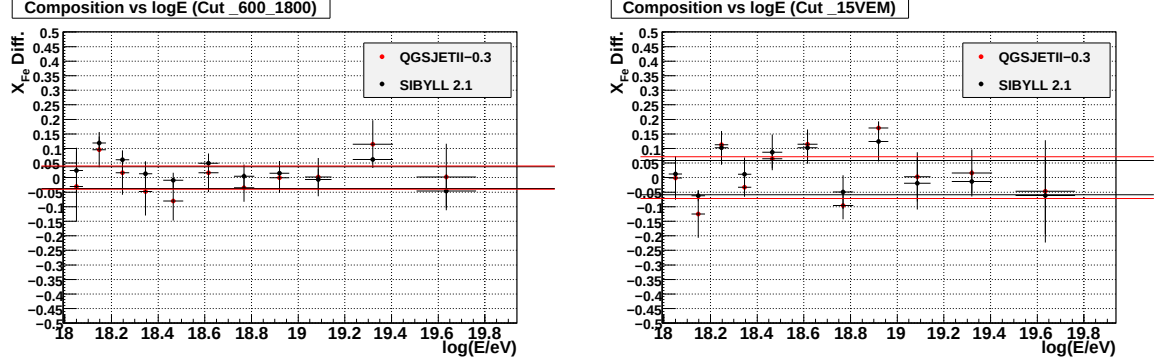


Figure 5.30: Effect of the cuts on mass composition. Ordinate represents the difference in X_{Fe} when the cut in core distance is restricted to the 600 - 1800 m interval (left) for both MC and real data samples. Effect for an intensity signal cut of 15 VEM (right). The continuous lines represents the average deviation in the full energy range.

The same analysis has been carried out for the MC samples. In this case, the variation of XAsymMax results of $\Delta X_{Fe} = 0.07$ (see figure 5.30). This different behaviour is very likely due to the much smaller statistics of the MC sample. The more strict cuts reduce the size of the sample in such a way that the new fitted parameters vary very likely not because of the effect of the new cut but due to the statistical properties of the reduced sample.

- **Hadronic interaction model:** The full analysis has been carried out using both hadronic models, using the parametrization from QGSJETII and then applying to SIBYLL 2.1 data set. The prediction in the X_{Fe} value for both models are different by an amount which ranges up to $\Delta X_{Fe} = 0.14$.

As mentioned above a very relevant contribution to the total uncertainty comes from the statistical fluctuations of our limited real data sample (ΔX_{Fe} of 0.1 - 0.25) and that of the MC sample (ΔX_{Fe} of about 0.15). Notice that both uncertainties will be reduced significantly in the near future when much larger data and MC samples are available.



5. DETERMINATION OF COMPOSITION

Chapter 6

Conclusions

The construction of the Pierre Auger Observatory has opened a new era in the physics of ultra-high energy cosmic rays. This detector offers the largest exposition ever achieved and, for the first time, with a hybrid design combining successfully surface detection with the fluorescence technique. Even before full completion of the detector, the Observatory has measured the energy spectrum of cosmic rays ($E > 10^{18}$ eV) with unprecedented accuracy confirming the expected GZK suppression. In addition the analysis of the angular distribution of cosmic rays has revealed anisotropic features for $E > 57$ EeV possibly correlated with close AGNs.

The mass of cosmic rays is a fundamental ingredient for the interpretation of the combined results on anisotropies and spectrum suppression. In this work a novel method for the measure of mass composition of UHECRs has been developed. For inclined cascades the late portion of the shower front at ground is subjected to a larger atmospheric attenuation, and as a result the reconstructed signals of the SD tanks in the shower plane exhibit asymmetry, not only in their integrated values but also in the time features. This time asymmetry is due to the variation of the ratio of the electron to muon component in the late and early regions. The dependence of this asymmetry with the incident angle θ is correlated with the mass of the primary cosmic rays.

The main milestones of this work have been the following:

- 1) By means of the official software of the Auger collaboration the response of extensive air showers, previously generated by the AIRES code, has been simulated. Around 2×10^4 MC events initiated by both protons and iron nuclei with energies in the



6. CONCLUSIONS

range $\log(E/eV) = 18.5$ to 20 and incident angle between 30 and 60 degrees have been used in this study. The reconstruction of the simulated showers has been performed using the Auger official reconstruction code. Dedicated tools for the determination of the time asymmetry have been developed in this work and implemented in the framework of the official software.

Simulations reveal the expected time asymmetry in the surface detectors. An asymmetry factor b/a has been defined which turns out to depend on the incident angle of the primary particle as expected, that is, it grows with $\sec \theta$ up to a maximum value, decreasing beyond it.

2) This longitudinal asymmetry development of the shower has been characterized by three parameters: XAsymMax, i.e. the $\sec \theta$ value for which b/a maximizes; AsymHeight, i.e. the maximum b/a value and XAsymWidth, i.e. the full width at half maximum of the longitudinal asymmetry development. It has been demonstrated that XAsymMax and AsymHeight are sensitive to the primary mass while XAsymWidth is nearly mass independent.

Using the simulated events, the dependence of the above parameters on composition has been obtained in several energy intervals, assuming a simple p-Fe model. Both XAsymMax and AsymHeight decreases linearly with the iron fraction. A relationship of these asymmetry parameters with the iron fraction allows a simple procedure for the measure of the mass which has been tested with the simulated sample.

3) The method developed in this work has been applied to the sample of real data of the Pierre Auger Observatory corresponding to the period January 1st 2004 to March 2009. The asymmetry features predicted by the simulations have been observed in real data. Asymmetry longitudinal developments for several energy intervals, have been measured. The dependence of the three asymmetry parameters with energy has been studied and compared with MC calculations. As predicted, XAsymWidth turns out to be nearly independent on energy. XAsymMax for real data takes values in between those predicted from simulation for protons and iron nuclei. However, the measured AsymHeight is in clear disagreement with those from simulations since values for real data result smaller than predictions for both proton and iron nuclei.

4) The discrepancy in AsymHeight has been studied assuming that it is due to an incorrect prediction of the muon content of the shower at ground. The number of muons of simulated showers has been artificially enlarged before being injected in

the detector simulator and the results of the asymmetry analysis have been compared. The effect of increasing the muon content at ground by a 60% is the following. The values of XAsymMax and XAsymWidth are basically insensitive to the muon content as expected. On the other hand, the AsymHeight value decreases and thus reaching a value closer to the data ones. Although a full agreement is not possible this result support other indications on a possible lack of muons at ground in simulations. The comparison of AsymHeight predictions with real data could be a useful tool to test hadronic interaction models in our energy range.

5) Comparison of real data and MC suggests that XAsymMax is the most convenient and reliable parameter for mass composition studies since it shows a weak energy dependence and seems to be nearly independent on possible uncertainties in the muon content at ground. The correlation between XAsymMax and X_{max} has been studied in detail for both simulated events and hybrid events of real data. It has been found that both parameters are very well correlated. A relevant result is that the relationship between both parameters is universal and independent on model, that is, the same relationship has been found for real data and simulations and for both hadronic models.

6) Finally, preliminary results has been found for the mass composition of UHECRs at energies over $10^{18.5}$ eV by comparing the XAsymMax values measured by the SD of the Pierre Auger Observatory with MC predictions. Under a two component (p-Fe) approximation the analysis presented in this work indicates a mixed composition at low energy $X_{Fe} \sim 0.25$, evolving to a heavier one above 10^{19} eV.

These predictions are subjected to uncertainties of about 10-25% (i.e ΔX_{Fe} in the range 0.1-0.25) due to limited statistics of real data, a contribution of about 10% ($\Delta X_{Fe} = 0.1$) due to the systematic uncertainty in the muon content and primary energy and a systematic uncertainty of about 15% ($\Delta X_{Fe} = 0.15$) because of the limited sample of MC events. The uncertainty of the hadronic interaction model contributes a 14% ($\Delta X_{Fe} = 0.14$) to the systematic error.

References

- [1] V. F. Hess, Phys. Z. **13** (1912) 1084. [6](#)
- [2] R. A. Millikan and G. Harvey Cameron, Phys. Rev. **28** (1926) 851. [6](#)
- [3] C. D. Anderson, Phys. Rev. **43** (1933) 491. [6](#)
- [4] P. A. M. Dirac, Proc. Roy. Soc. Lond. A**133** (1931) 60. [6](#)
- [5] J. C. Street and E. C. Stevenson, Phys. Rev. **52** (1937) 1003. [6](#)
- [6] H. Neddermeyer and C. D. Anderson, Phys. Rev. **51** (1937) 884. [6](#)
- [7] H. Yukawa, Proc. Phys. Math. Soc. Jap. **17** (1935) 48. [6](#)
- [8] C. M. G. Lattes, H. Muirhead, G. P. S. Occhialini and C. F. Powell, Nature **159** (1947) 694. [6](#)
- [9] B. Rossi, Ric. Sic. Suppl. **1** (1934) 579. [6](#)
- [10] P. Auger, P. Ehrenfest, R. Maze, J. Daudin and A. F. Robley, Rev. Mod. Phys. **11** (1939) 288. [6](#)
- [11] P. Bassi, G. Glark and B. Rossi, Phys. Rev. **92** (1953) 441. [6](#)
- [12] J. Linsley, Phys. Rev. Lett. **10** (1963) 146. [6](#), [39](#)
- [13] K. Suga, Proc. 5th Interamerican Seminar on Cosmic Ray, La Paz, Bolivia, vol. II (1962) XLIX. [7](#)



REFERENCES

- [14] A. E. Chudakov, Proc. 5th Interamerican Seminar on Cosmic Ray, La Paz, Bolivia, vol. II (1962) XLIX. [7](#)
- [15] T. Antoni *et al.* [The KASCADE Collaboration], Astropart. Phys. **24** (2005) 1. [7](#), [35](#)
- [16] R. Engel, talk presented in the International School on Astroparticle Physics 2005, Belgirate, Italy. [vii](#), [9](#), [15](#)
- [17] T. Wibig and A. W. Wolfendale, J. Phys. G **31** (2005) 255. [9](#)
- [18] A. M. Hillas, J. Phys. Conf. Ser. **47** (2006) 168. [10](#)
- [19] D. Allard, E. Parizot, E. Khan, S. Goriely and A. V. Olinto, Astron. Astrophys. **443** (2005) L29. [10](#)
- [20] R. Aloisio, V. Berezhinsky, P. Blasi, A. Gazizov, S. Grigorieva and B. Hnatyk, Astropart. Phys. **27** (2007) 76. [10](#)
- [21] R. Aloisio, V. Berezhinsky, P. Blasi and S. Ostapchenko, Phys. Rev. D **77** (2008) 025007. [10](#)
- [22] D. Allard, A. V. Olinto and E. Parizot, Astron. Astrophys. **473** (2007) 59. [10](#)
- [23] L. O. Drury, Rept. Prog. Phys. **46** (1983) 973. [10](#)
- [24] E. Fermi, Phys. Rev. **75** (1949) 1169. [10](#)
- [25] P. Bhattacharjee and G. Sigl, Phys. Rept. **327** (2000) 109. [vii](#), [11](#)
- [26] A. M. Hillas, arXiv:astro-ph/0607109. [11](#)
- [27] F. A. Aharonian *et al.* [The HESS Collaboration], Nature **432** (2004) 75. [11](#)
- [28] F. Aharonian *et al.* [HESS Collaboration], Astron. Astrophys. **449** (2006) 223. [11](#)
- [29] V. S. Ptuskin and V. N. Zirakashvili, Astron. Astrophys. **429** (2005) 755. [11](#)
- [30] A. M. Hillas, J. Phys. G **31** (2005) R95. [11](#)
- [31] C. T. Hill, Nucl. Phys. B **224** (1983) 469. [12](#)
M. B. Hindmarsh and T. W. B. Kibble, Rept. Prog. Phys. **58** (1995) 477.

-
- [32] V. Berezhinsky, M. Kachelriess and A. Vilenkin, Phys. Rev. Lett. **79** (1997) 4302. [12](#)
M. Birkel and S. Sarkar, Astropart. Phys. **9** (1998) 297.
- [33] R. Aloisio, V. Berezhinsky and M. Kachelriess, Phys. Rev. D **69** (2004) 094023. [12](#)
- [34] J. R. Ellis, V. E. Mayes and D. V. Nanopoulos, Phys. Rev. D **74** (2006) 115003. [12](#)
- [35] T. J. Weiler, Astropart. Phys. **11** (1999) 303. [12](#)
D. Fargion, B. Mele and A. Salis, Astrophys. J. **517** (1999) 725.
- [36] M. Risse and P. Homola, Mod. Phys. Lett. A **22** (2007) 749. [12](#)
- [37] J. Abraham *et al.* [Pierre Auger Collaboration], Astropart. Phys. **29** (2008) 243. [12](#), [36](#), [91](#)
- [38] J. Abraham *et al.* [The Pierre Auger Collaboration], Phys. Rev. Lett. **100** (2008) 211101. [12](#), [91](#)
- [39] K. Greisen, Phys. Rev. Lett. **16** (1966) 748. [12](#)
- [40] G. T. Zatsepin and V. A. Kuzmin, JETP Lett. **4** (1966) 78. [13](#)
- [41] F. A. Aharonian and J. W. Cronin, Phys. Rev. D **50** (1994) 1982. [vii](#), [14](#)
- [42] T. Stanev, R. Engel, A. Mucke, R. J. Protheroe and J. P. Rachen, Phys. Rev. D **62** (2000) 093005. [14](#)
- [43] L. N. Epele and E. Roulet, JHEP **9810** (1998) 009. [14](#)
- [44] F. W. Stecker and M. H. Salamon, Astrophys. J. **512** (1999) 521. [14](#)
- [45] J. Abraham *et al.* [Pierre Auger Collaboration], Phys. Rev. Lett. **101** (2008) 061101. [x](#), [14](#), [48](#), [61](#), [62](#), [63](#)
- [46] J. W. Cronin, Nucl. Phys. Proc. Suppl. **138** (2005) 465. [viii](#), [17](#)



REFERENCES

- [47] P. G. Tinyakov and I. I. Tkachev, JETP Lett. **74** (2001) 445.
P. G. Tinyakov and I. I. Tkachev, Astropart. Phys. **18** (2002) 165.
D. S. Gorbunov, P. G. Tinyakov, I. I. Tkachev and S. V. Troitsky, Astrophys. J. **577** (2002) L93.
R. U. Abbasi *et al.* [HiRes Collaboration], Astrophys. J. **636** (2006) 680.
D. S. Gorbunov, P. G. Tinyakov, I. I. Tkachev and S. V. Troitsky, JETP Lett. **80** (2004) 145. [15](#)
- [48] N. Hayashida *et al.* [AGASA Collaboration], Proc. 27th International Cosmic Ray Conference, Hamburg, Germany **1** (2001) 337. [15](#)
- [49] J. A. Bellido, R. W. Clay, B. R. Dawson and M. Johnston-Hollitt, Astropart. Phys. **15** (2001) 167. [15](#)
- [50] M. Teshima, *et al.* [AGASA Collaboration], Proc. 28th International Cosmic Ray Conference, Tsukuba, Japan HE**13**. (2003) 437. [15](#)
- [51] D. Harari [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.1715 [astro-ph] (2007). [16](#)
- [52] M. Aglietta *et al.* [Pierre Auger Collaboration], Astropart. Phys. **27** (2007) 244.
E. M. Santos [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.2669 [astro-ph] (2007). [16](#)
- [53] S. Mollerach [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.1749 [astro-ph] (2007). [16](#)
- [54] E. Armengaud [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.2640 [astro-ph] (2007). [16](#)
- [55] J. Abraham *et al.* [Pierre Auger Collaboration], Science **318** (2007) 939. [16](#)
J. Abraham *et al.* [Pierre Auger Collaboration], Astropart. Phys. **29** (2008) 188.
- [56] L. Anchordoqui, M. T. Dova, A. Mariazzi, T. McCauley, T. Paul, S. Reucroft and J. Swain, Annals Phys. **314** (2004) 145. [19](#), [21](#)
- [57] “LHC Design Report”, CERN-2004-003 (2004). [20](#)
- [58] G. Aad *et al.* [ATLAS Collaboration], JINST **3** (2008) S08003. [20](#)

-
- [59] G. L. Bayatian *et al.* [CMS Collaboration], J. Phys. G **34** (2007) 995. [20](#)
- [60] O. Adriani *et al.*, Czech. J. Phys. **56** (2006) A107. [20](#)
- [61] N. N. Kalmykov, S. S. Ostapchenko and A. I. Pavlov, Nucl. Phys. Proc. Suppl. **52B** (1997) 17. [20](#)
- [62] R. S. Fletcher, T. K. Gaisser, P. Lipari and T. Stanev, Phys. Rev. D **50** (1994) 5710. [20](#), [92](#)
- [63] M. T. Dova and S. Ferrari, Eur. Phys. J. C **52** (2007) 673. [20](#)
- [64] L. V. Gribov, E. M. Levin and M. G. Ryskin, Phys. Rept. **100** (1983) 1. [21](#)
- [65] K. Werner, Phys. Rept. **232** (1993) 87. [21](#)
- [66] B. Rossi and K. Greisen, Rev. Mod. Phys. **13** (1941) 240. [21](#)
- [67] W. Heitler, The Quantum Theory of Radiation, third ed., Oxford University Press London, 1954, p. 386 (Section 38). [23](#)
- [68] T. K. Gaisser, *Cambridge, UK: Univ. Pr. (1990)*. [23](#)
- [69] J. Matthews, Astropart. Phys. **22** (2005) 387. [viii](#), [24](#), [25](#), [26](#)
- [70] J. Linsley, Proc. 15th International Cosmic Ray Conference, Plovdiv, Bulgaria, **12** (1977) 89. [28](#)
- [71] J. Linsley and A. A. Watson, Phys. Rev. Lett. **46** (1981) 459. [28](#)
- [72] D. Heck, “The Influence of Hadronic Interaction Models on Simulated Air-Showers: A Phenomenological Comparison”, talk presented at the VIHOS CORSIKA School, Lauterbad, Germany (2005). [viii](#), [29](#)
- [73] M. Unger [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.2643 [astro-ph] 2007. [28](#)
- [74] M. Unger [Pierre Auger Collaboration], To appear in Proceedings of CRIS 2008, Malfa, Italy. [28](#)
- [75] R. U. Abbasi *et al.* [The High Resolution Fly’s Eye Collaboration], Astrophys. J. **622** (2005) 910. [28](#)



REFERENCES

- [76] B. Keilhauer *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 123. [28](#)
- [77] Ulrich, R. “Measurement of the proton-air cross section using hybrid data of the Pierre Auger Observatory”. Phd thesis, University of Karlsruhe, (2008). [viii](#), [31](#)
- [78] T. Abu-Zayyad *et al.*, Phys. Rev. Lett. **84** (2000) 4276. [30](#)
- [79] K. Shinozaki and M. Teshima [AGASA Collaboration], Nucl. Phys. Proc. Suppl. **136** (2004) 18. [30](#)
K. Shinozaki *et al.*, Proc. 28th International Cosmic Ray Conference, Tsukuba, Japan (2203) 401.
- [80] X. Garrido *et al.*, “Measurement of the number of muons in Auger tanks by the FADC jump counting method”, Auger Technical Note, GAP-07-060 (2007). [viii](#), [30](#), [32](#)
- [81] A. A. Watson and J. G. Wilson, J. Phys. A **7** (1974) 1199. [32](#)
- [82] R. Walker and A. A. Watson, J. Phys. G **7** (1981) 1297. [32](#), [84](#)
- [83] M. Ave, J. Knapp, M. Marchesini, M. Roth and A. A. Watson, Proc. 28th International Cosmic Ray Conference Tsukuba, Japan, **1** (2003) 349. [32](#)
- [84] M. Healy *et al.*, “A Study of Composition Trends Using Rise Time and Curvature Data” Auger Technical Note, GAP-2006-092 (2006). [viii](#), [32](#), [33](#)
- [85] C. Wileman, “The Spread in the Arrival Times of Particles in Air-Showers for Photon and Anisotropy Searches above 10 EeV.”, Ph.D. Thesis, University of Leeds, England (2008). [33](#), [91](#), [122](#), [123](#)
- [86] B. Smith, “The Mass Composition of Cosmic Rays Above 1 EeV inferred Using the Spread in Arrival Times of Air Shower Particles”, Ph.D. Thesis, University of Leeds, England (2008). [ix](#), [33](#), [34](#), [91](#)
- [87] M. Ave *et al.*, Astropart. Phys. **19** (2003) 61. [35](#)
- [88] M. T. Dova, M. E. Mancenido, A. G. Mariazzi, T. P. McCauley and A. A. Watson, Astropart. Phys. **21** (2004) 597. [35](#)

-
- [89] M. Bertaina *et al.*, Nucl. Instrum. Meth. A **588** (2008) 162. [36](#)
- [90] M. Healy *et al.* [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico (2007). [36](#)
- [91] Pierre Auger Collaboration, “Pierre Auger Project Design Report”, Fermi National Accelerator Laboratory (1997). [37](#), [49](#)
- [92] AMS Collaboration, Phys. Lett. **B490** (2000) 27. [38](#)
- [93] J. Linsley, Phys. Rev. Lett. **97** (1955) 1292. [39](#)
- [94] M. A. Lawrence, R. J. O. Reid and A. A. Watson, J. Phys. G **17** (1991) 733. [39](#), [91](#)
- [95] V. P. Artamonov *et al.*, Bull. Rus. Acad. Sci. Phys. **58** (1994) 2026. [39](#)
- [96] M. M. Winn, J. Ulrichs, L. S. Peak, C. B. A. Mccusker and L. Horton, J. Phys. G **12** (1986) 653. [39](#)
- [97] N. Chiba *et al.* Nucl. Instrum. Meth. A**311** (1992) 338. [39](#)
- [98] J. Delvaille *et al.*, J. Phys. Soc. Japan **17** Suppl. A-III (1962) 76. [39](#)
- [99] K. Suga, Proc. 5th Interamerican Seminar on Cosmic Ray La Paz, Bolivia vol.**II** (1962) XLIX. [39](#)
- [100] A. E. Chudakov, Proc. 5th Interamerican Seminar on Cosmic Ray La Paz Bolivia, vol.**II** (1962) XLIX. [39](#)
- [101] R. M. Baltrusaitis *et al.*, Nucl. Instrum. Meth. A**240** (1985) 410. [39](#)
- [102] S. C. Corbato *et al.*, Nucl. Phys. Proc. Suppl. **28B** (1992) 36. [39](#)
- [103] D. De Marco, P. Blasi and A. V. Olinto, Astropart. Phys. **20** (2003) 53. [39](#)
- [104] B. M. Connolly, S. Y. BenZvi, C. B. Finley, A. C. O'Neill and S. Westerhoff, Phys. Rev. **D74** (2006) 043001. [39](#)
- [105] R. Abbasi *et al.* [HiRes Collaboration], Phys. Rev. Lett. **100** (2008) 101101. [ix](#), [40](#)



REFERENCES

- [106] [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 13. [41](#)
- [107] R. Sato and C. Escobar [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India (2005). [41](#)
- [108] R. Sato, J. Bellido, H. Reis and C. Escobar, “Corrector plate with an annulus shape”, Auger Technical Note, GAP-99-014 (1999). [41](#)
- [109] G. Borreani *et al.*, Nucl. Instrum. Meth. A**461** (2001) 577. [41](#)
- [110] P. Facal and P. Privitera, “Measurement of the FD camera light collection efficiency and uniformity”, Auger Technical Note, GAP-2000-010 (2000). [41](#)
- [111] A. Schmidt *et al.*, arXiv:0807.4421 [astro-ph]. [43](#)
- [112] P. Bauleo *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **8** (2005) 5. [43](#)
- [113] C. Aramo *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **8** (2005) 101. [43](#)
- [114] R. Knapik *et al.* [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0708.1924 [astro-ph] (2007). [43](#)
- [115] M. Monasor, “Energy Calibration of the Pierre Auger Observatory. Measurement of the spectrum of Ultra-High Energy Cosmic Rays”, Ph.D. Thesis, Universidad de Alcalá, Spain (2008). [ix](#), [43](#), [45](#), [46](#)
- [116] B. Keilhauer *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 123. [44](#)
- [117] S. Y. BenZvi *et al.*, Nucl. Instrum. Meth. A **574** (2007) 171. [44](#)
- [118] P. Trávníček *et al.* [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.1710 [astro-ph] (2007). [44](#)
- [119] F. Arqueros *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **8** (2005) 335 arXiv: astro-ph/0507334. [44](#)

REFERENCES

- [120] R. Cester *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **8** (2005) 347. [44](#)
- [121] P. Sommers and B. Fick, “A Numerical Recipe for Fluorescence Detector Geometric Reconstruction”, Auger Technical Note, GAP-2003-045 (2003). [44](#)
- [122] C. Bonifazi [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference Pune, India, **7** (2005) 17. [45](#)
- [123] M. Unger, B. R. Dawson, R. Engel, F. Schussler and R. Ulrich, Nucl. Instrum. Meth. A **588** (2008) 433. [45](#), [48](#)
- [124] F. Nerling, J. Blumer, R. Engel and M. Risse, Astropart. Phys. **24** (2006) 421. [45](#)
- [125] F. Arqueros, J. R. Hoerandel and B. Keilhauer, Nucl. Instrum. Meth. A **597** (2008) 23. [46](#)
- [126] H. M. J. Barbosa, F. Catalani, J. A. Chinellato and C. Dobrigkeit, Astropart. Phys. **22** (2004) 159. [47](#)
- [127] [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 1. [48](#)
- [128] A. K. Tripathi, Nucl. Instrum. Meth. A **504** (2003) 1. [50](#)
- [129] T. Suomijärvi [Pierre Auger Collaboration], Nucl. Phys. Proc. Suppl. **136** (2004) 393. [50](#)
- [130] X. Bertou *et al.* [Pierre Auger Collaboration], Nucl. Instrum. Meth. A **568** (2006) 839. [ix](#), [51](#), [52](#)
- [131] M. Aglietta *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 279. [51](#)
- [132] M. Aglietta *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 83. [51](#)
- [133] D. Allard *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 287. [52](#)



REFERENCES

- [134] S. Dagoret-Campagne, “The Central Trigger, User Guide and Reference Manual”, Auger Technical Note, GAP-2004-020 (2004). [ix](#), [55](#)
- [135] D. Veberič and M. Roth, “Offline Reference Manual: SD Reconstruction”, Auger Technical Note, GAP-2005-035 (2005). [ix](#), [56](#), [59](#), [60](#)
- [136] Event Selection Group, “The Official Event Selection as of March 2005”, Auger Technical Note, GAP-2005-023 (2005). [57](#)
- [137] M. Horvat and D. Veberič, “On Shower-Front Start-Time Variance”, Auger Technical Note, GAP-2007-057 (2007). [58](#)
- [138] A. M. Hillas, Acta Phys. Acad. Sci. Hung. 29(Supl.3) (1970) 355.
A. M. Hillas, Proc. 12th International Cosmic Ray Conference, Tasmania, Australia, **3** (1971) 1001. [59](#)
- [139] D. Newton, J. Knapp and A. A. Watson, Astropart. Phys. **26** (2007) 414. [x](#), [59](#), [60](#)
- [140] D. Barnhill *et al.* [Pierre Auger Collaboration], Proc. 29th International Cosmic Ray Conference, Pune, India **7** (2005) 291 arXiv: astro-ph/0507590. [60](#)
- [141] K. Kamata and J. Nishimura, Prog. Theoret. Phys. Suppl. **6** (1956) 93.
- [142] K. Greisen, Progress in Cosmic Ray Physics **3** (1956). [60](#)
- [143] M. Ave, P. Bauleo, J. L. Harton, R. Knapik, A. Castellina, G. Navarra and A. Chou [Pierre Auger Collaboration], Nucl. Instrum. Meth. A **578** (2007) 180. [60](#)
- [144] F. Arneodo, S. Maldera and G. Navarra, “A procedure to derive the anode charges for saturated signals from the dynode and anode undershoots”, Auger Technical Note, GAP-2005-025 (2005). [61](#)
- [145] I. C. Maris, M. Roth and T. Schmidt, “A Phenomenological Method to Recover the Signal from Saturated Stations”, Auger Technical Note, GAP-2006-012 (2006). [61](#)
- [146] I. D. Mitri, S. Maglio and D. Martello, “A method for SD traces saturation recovery”, Auger Technical Note, GAP-2006-075 (2006). [61](#)

-
- [147] P. Billoir, “Reconstruction of showers with the Ground Array: status of the “prototype” program”, Auger Technical Note, GAP-2005-025 (2005). [61](#)
[65](#), [92](#)
- [148] J. Hersil *et al.*, Phys. Rev. Lett. **6** (1961) 22. [61](#)
- [149] S. Argirò *et al.* Nucl. Instrum. Meth. A**580** (2007) 1485. [63](#), [92](#)
- [150] C. D. England, “The Longitudinal Development of Extensive Air Showers at energies above 10¹⁷ eV”, PhD Thesis of the University of Leeds (1984).
- [151] M. Ave, R. A. Vazquez and E. Zas, Astropart. Phys. **14** (2000) 91. [66](#), [79](#)
- [152] X. Bertou and P. Billoir, “On the origin of the asymmetry of ground densities in inclined showers”, Auger Technical Note, GAP-2000-017 (2000). [68](#)
- [153] P. Da Silva, P. Billoir and X. Bertou, “Checking the origin of the Asymmetry of the Surface Detector Signals”, Auger Technical Note, GAP-2002-074 (2002). [x](#), [68](#), [69](#), [71](#)
- [154] M. T. Dova, L. N. Epele and A. G. Mariazzi, Astropart. Phys. **18**, 351 (2003) [arXiv:astro-ph/0110237]. [x](#), [68](#), [70](#), [73](#)
- [155] A. G. Mariazzi, PhD thesis UNLP (2004), “Asimetrías espacio-temporales en las cascadas de rayos cósmicos ultra energéticos y composición de la partícula primaria”. [74](#), [90](#)
- [156] C. Pryke, “Asymmetry of Air Showers at Ground Level”, Auger Technical Note, GAP-1998-034 (1998). [74](#), [76](#)
- [157] M. T. Dova, L. N. Epele and A. Mariazzi, Nuovo Cim. **24C** (2001) 745. [75](#)
- [158] Talk presented by P. Da Silva in Auger Collaboration Meeting, Malargüe, Nov 2004. [75](#)
- [159] D. Allard *et al.*, “A guide-line to the Auger Surface Detector analysis”, Auger Technical Note, GAP-2006-024 (2006). [78](#)



REFERENCES

- [160] M. T. Dova, F. G. Monticelli and H. Wahlberg “Influence of the reconstructed core position on $S(1000)$ for inclined showers”, Auger Technical Note, GAP-2006-035 (2006). [xi](#), [79](#), [80](#)
[79](#)
- [161] P. Billoir and P. Da Silva “Towards a Parametrization of the Lateral Distribution Function and its Asymmetries in the Surface Detector”, Auger Technical Note, GAP-2002-073 (2002). [xi](#), [79](#), [81](#)
- [162] M. T. Dova, M. E. Mancenido, A. G. Mariazzi, H. Wahlberg, F. Arqueros and D. Garcia-Pinto, *Astropart. Phys.* **31** (2009) 312. [83](#)
- [163] J. Abraham *et al.* [Pierre Auger Collaboration], *Nucl. Instrum. Meth.* **A523** (2004) 50. [83](#)
- [164] M. Ave, J. Knapp, M. Marchesini, M. Roth and A. Watson, “A Preliminary Analysis of the 11-fold Event 184599”, Auger Technical Note, GAP-2002-020 (2002). [84](#)
- [165] AUGER Collaboration (M. T. Dova for the Collaboration), *Proc. 28th International Cosmic Ray Conference*, Tsukuba, **HE1.3** 369 (2003). [84](#)
- [166] F. Gómez Albarracín, M. T. Dova and H. Wahlberg, “Study of timing parameters $t_y - t_x$ using SD data” Auger Technical Note, GAP-2008-017 (2008). [91](#)
- [167] S. J. Sciutto, *Proc. 27th International Cosmic Ray Conference Hamburg, Germany*, **1** (2001) 237. [91](#), [92](#), [120](#)
- [168] A. M. Hillas, *Nucl. Phys. Proc. Suppl.* **52B** (1997) 29. [91](#)
- [169] S. Ostapchenko, *Nucl. Phys. Proc. Suppl.* **151** (2006) 143. [92](#)
- [170] R. Engel, T. K. Gaisser, T. Stanev and P. Lipari, *Proc. 26th International Cosmic Ray Conference Salt Lake City, Utah*, **1** (1999) 415. [92](#)
- [171] P. Billoir, *Astropart. Phys.* **30** (2008) 270. [92](#)
- [172] M. Unger [The Pierre Auger Collaboration], arXiv:0706.1495 [astro-ph]. [102](#)

- [173] R. Engel [Pierre Auger Collaboration], Proc. 30th International Cosmic Ray Conference, Merida, Mexico, arXiv:0706.1749 [astro-ph] (2007). [118](#), [121](#)
- [174] P. Billoir and O. Blanch-Bigas, “About a possible muon excess in real data compared to simulations”, Auger Technical Note, GAP-2006-055 (2006). [118](#), [121](#)
- [175] M. Ave *et al.*, “Determining the SD Energy Scale and the Number of Muons using Air Shower Universality”, Auger Technical Note, GAP-2007-021 (2007). [118](#), [121](#)
- [176] M. Ave *et al.*, “Number of Muons and Fluctuations from Golden Hybrid Events”, Auger Technical Note, GAP-2007-051 (2007). [118](#), [122](#)
- [177] A. A. Watson, Nucl. Phys. Proc. Suppl. **136** (2004) 290.
- [178] I. C. Maris, F. Schüssler, R. Ulrich, M. Unger, “ADST and EventBrowser Reference Manual”. Auger Thechnical Note, GAP 2006-081. [125](#)
- [179] M. Unger personal web page, <http://www-ik.fzk.de/~munger/Xmax/>. [126](#)
- [180] B. Smith, C. Wileman and A. Watson, “Can the risetime be used to infer the position of shower maximum?”, Auger Technical Note, GAP 2007-0902. [128](#), [132](#)
- [181] F. Arqueros, D. Garcia-Pinto, M. Monasor, M. T. Dova, A. G. Mariazzi, H. P. Wahlberg, “Inferring Xmax from asymmetry in risetime”, Auger Technical Note, GAP-2008-179 (2008). [132](#)